Keldysh Path Integral Quantization of Nonlinear Open Nanophotonic Structures via Quasi-Normal Modes

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Abstract

We derive a first-principles Keldysh (closed-time-path) path integral for an open nonlinear nanophotonic structure, using quasi-normal modes (QNMs) as the fundamental degrees of freedom. Starting from the Maxwell action in the Coulomb gauge, we (i) decompose the electromagnetic field into QNM amplitudes and a free-space radiation continuum, (ii) perform the exact Gaussian integral over the radiation, and (iii) apply the Schwinger-Keldysh doubling to handle the non-unitary open dynamics. Five technical challenges that appear in this programme are addressed in full:

1. **Functional measure.** The Jacobian from the change of variables $\mathbf{A}(\mathbf{r}, t) \to \{q_\lambda(t)\}$ equals the Gram determinant of the QNM mode functions, is field-independent, and cancels exactly in every physical observable.
2. **Non-orthogonal overlapping modes.** The Hermitian overlap matrix $\mathcal{O}_{\lambda\lambda'}$ is positive definite; its Cholesky decomposition $\mathcal{O} = LL^\dagger$ defines dressed amplitudes $\Xi_\nu = (U^\dagger L^\dagger)q$ in which the kinetic term is diagonal and the resulting bosonic operators satisfy $[\hat{a}_\nu, \hat{a}^\dagger_{\nu'}] = \delta_{\nu\nu'}$ exactly.
3. **Exceptional points.** When two QNMs coalesce the eigenspace is defective. We derive the Jordan-block Keldysh action, the Jordan operator algebra $[\hat{a}_0, \hat{a}^\dagger_0] = 0$, $[\hat{a}_0, \hat{a}^\dagger_1] = 1$, and the double-pole retarded propagator $G^R_{\mathrm{EP}}(\omega) \sim (\omega - \tilde{\omega}_{\mathrm{EP}})^{-2}$, which generates a linear-in-time impulse response and a squared-Lorentzian transmission lineshape distinguishable from a standard Lorentzian.
4. **Rotating-wave approximation.** Counter-rotating terms are suppressed by $g^{(n)}/\omega_0 \sim 10^{-8}$–$10^{-5}$ for all known nanophotonic platforms; the leading correction is the Bloch-Siegert shift $\delta\omega_{\mathrm{BSS}} \approx (g^{(3)})^2\kappa/(8\omega_0^2)$ is negligible for all optical platforms ($\ll 10^{-8}\,\mathrm{Hz}$).
5. **Closed-system limit.** As $\kappa_\lambda \to 0$ the Keldysh component $D^{\mathrm{K}} \to 0$, the forward and backward paths decouple, and $Z_K \to |Z_{\text{in-out}}|^2$, recovering the standard coherent-state path integral and canonical commutators exactly.

A central analytic result establishes the symmetry structure of the nonlinear coupling constants in the QNM bilinear basis: for real Kleinman-symmetric $\chi^{(3)}$, the cross-Kerr coupling $g^{(3)}_{\lambda\mu\mu\lambda} \in \mathbb{R}$ exactly, even though the QNM mode functions are complex, so **passive cross-mode dephasing vanishes identically regardless of cavity Q or mode overlap**. In sharp contrast, the $\chi^{(2)}$ downconversion coupling is genuinely complex ($g^{(2)}_{\lambda\mu\nu} \in \mathbb{C}$, Im/Re $\sim (1/Q_\lambda + 1/Q_\mu + 1/Q_\nu)/2$), because the index pattern $(\lambda^*\mu\nu)$ admits no Kleinman cancellation. The primary falsifiable prediction of the passive-cavity framework is therefore a $\chi^{(2)}$ *squeezing-phase offset* $\delta\theta_{\rm sq} \sim 1/Q$ relative to the pump phase, measurable via balanced homodyne detection. For active structures, gain saturation contributes a purely imaginary $\chi^{(3)}_{\rm sat}$, driving a cross-mode dephasing $\delta\kappa_\lambda = 4|g^{(3),I}_{{\rm sat},\lambda\mu\mu\lambda}|\bar{n}_\mu$ that diverges in relative magnitude near threshold. For a GaAs photonic-crystal nanocavity ($g^{(3)}/2\pi \approx 235\,$kHz, $Q \approx 3\times 10^4$), the order-of-magnitude estimate for the passive squeezing-phase offset is $\sim 10^{-4}\,$rad (specific value requires a doubly-resonant Maxwell FDTD of the geometry), and the active dephasing is within reach of current experimental techniques.

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1 A Guided Tour for Quantum Optics Practitioners

This section explains the Feynman path integral and Keldysh formalism in plain language for readers who know the operator formalism and Heisenberg-Langevin equations. No new physics is introduced; the goal is to show that everything in this paper is something you already know, written in a different language.

1.1 The coherent state: the bridge between operators and c-numbers

Start with something you know. A coherent state $|\alpha\rangle$ is the eigenstate of the annihilation operator:

$$\hat{a}|\alpha\rangle = \alpha|\alpha\rangle, \quad \alpha \in \mathbb{C} \tag{1}$$

In this state, the operator $\hat{a}$ is replaced by a *complex number* α. The quantum mechanics of the state is encoded in the distribution of values that α can take.

The coherent states are **overcomplete**: they resolve the identity as $\int d^2\alpha/\pi\, |\alpha\rangle\langle\alpha| = \mathbf{1}$. This means any quantum expectation value can be written as an integral over coherent-state eigenvalues:

$$\langle\hat{O}\rangle = \mathrm{Tr}[\rho\hat{O}] = \int \frac{d^2\alpha}{\pi}\, \langle\alpha|\rho\hat{O}|\alpha\rangle \tag{2}$$

Inserting the identity repeatedly at successive time steps $t_0, t_1 = t_0 + \delta t, \ldots, t_N = t$ turns this integral into a sum over all *trajectories* $\{\alpha(t)\}$ — this is the **path integral**.

1.2 The path integral: a sum over c-number trajectories

For a cavity mode with Hamiltonian $\hat{H} = \hbar\omega\hat{a}^\dagger\hat{a}$ (no coupling to environment yet), the coherent-state path integral is:

$$\langle\hat{O}(t)\rangle = \int \mathcal{D}[\alpha^*(t'), \alpha(t')]\, O[\alpha^*, \alpha]\, e^{iS[\alpha^*,\alpha]/\hbar} \tag{3}$$

where the **action** is:

$$S = \int_{t_0}^{t} dt'\, \hbar\big[i\alpha^*(t')\partial_{t'}\alpha(t') - \omega|\alpha(t')|^2\big] \tag{4}$$

Two terms. The second, $-\omega|\alpha|^2$, is just the energy $-\hbar\omega n$ — nothing surprising. The first, $i\alpha^*\partial_t\alpha$, is the **Berry phase** term. This first-order kinetic term is the path-integral encoding of the commutation relation $[\hat{a}, \hat{a}^\dagger] = 1$: it is what makes the theory quantum rather than classical. Without it, you would have α evolving like a classical variable with no quantum noise.

Remark

Plain-language summary so far. A path integral is just a fancy way of computing $\langle\hat{O}\rangle$ by treating the quantum state as a probability distribution over complex-number trajectories $\alpha(t)$. The c-number field $\alpha(t)$ is the coherent-state eigenvalue, not the

quantum operator. The quantum nature lives in the Berry phase term $i\alpha^*\partial_t\alpha$ and in the integration measure.

1.3 The problem with open systems

You know the HL equation for a damped cavity:

$$\frac{d\hat{a}}{dt} = -i\omega\hat{a} - \frac{\kappa}{2}\hat{a} + \sqrt{\kappa}\,\hat{a}_{\text{in}}(t) \tag{5}$$

This involves both $\hat{a}$ (forward in time) and $\hat{a}_{\text{in}}$ (the environment). The corresponding density matrix evolves as:

$$\langle\hat{O}(t)\rangle = \text{Tr}\big[U(t,t_0)\rho_0 U^\dagger(t,t_0)\,\hat{O}\big] \tag{6}$$

The issue: $U(t,t_0)$ propagates *forward* in time, while $U^\dagger(t,t_0)$ propagates *backward*. A single path integral with a single time contour can only represent one direction.

The **Keldysh trick**: run the time contour forward and then backward — a closed loop. Introduce two copies of the field:

- $a^+(t)$: the c-number field on the *forward* branch (corresponding to the U factor)
- $a^-(t)$: the c-number field on the *backward* branch (corresponding to the $U^\dagger$ factor)

The Keldysh path integral is then:

$$\langle\hat{O}(t)\rangle = \int \mathcal{D}[a^\pm]\,O[a^+]\,e^{iS[a^+]-iS^*[a^-]} \tag{7}$$

The forward action $iS[a^+]$ and the backward action $-iS^*[a^-]$ combine into the Keldysh action $S_K[a^+,a^-]$.

Remark

Why two fields? In quantum optics terms: computing $\langle\hat{O}(t)\rangle$ requires both the "ket" (forward evolution $U\rho_0$) and the "bra" (backward evolution $\rho_0 U^\dagger$). The a^+ field carries the ket information; the a^- field carries the bra information. The path integral integrates over all consistent pairs of forward/backward trajectories, weighted by e^{iS_K}.

1.4 The classical-quantum rotation: what you actually compute

Instead of working with $a^\pm$, it is more transparent to rotate to:

$$a^{\text{cl}} = \tfrac{1}{2}(a^+ + a^-), \quad a^{\text{q}} = a^+ - a^- \tag{8}$$

The classical field a^{cl} is the average of forward and backward trajectories. Its expectation value is the quantum mean field: $\langle a^{\text{cl}}(t)\rangle = \langle\hat{a}(t)\rangle$. When you solve for a^{cl} at the saddle point of the path integral, you get the classical equation of motion — the same as the HL equation for $\langle\hat{a}\rangle$.

The quantum field a^{q} is the difference between forward and backward trajectories. It measures how much the "ket" and "bra" paths deviate from each other. In classical physics, both paths are identical and $a^{\mathrm{q}} = 0$. In quantum physics, $a^{\mathrm{q}} \neq 0$ and it carries the quantum fluctuation information.

1.5 The three propagators you already know

The heart of the Keldysh formalism is the 2×2 matrix of Green's functions. In the cl-q basis:

$$\begin{pmatrix} \langle a^{\mathrm{cl}} a^{\mathrm{cl}*} \rangle & \langle a^{\mathrm{cl}} a^{\mathrm{q}*} \rangle \\ \langle a^{\mathrm{q}} a^{\mathrm{cl}*} \rangle & \langle a^{\mathrm{q}} a^{\mathrm{q}*} \rangle \end{pmatrix} = \begin{pmatrix} G^{\mathrm{K}} & G^R \\ G^A & 0 \end{pmatrix} \tag{9}$$

Each entry is something you already know in quantum optics:

- $G^R(\omega) = \langle a^{\mathrm{cl}} a^{\mathrm{q}*} \rangle$ is the **retarded Green's function** — the cavity *response function*. If you drive the cavity with a probe at frequency ω, the intracavity field amplitude responds as $G^R(\omega)$. For a single mode: $G^R(\omega) = 1/(\omega - \omega_0 + i\kappa/2)$. You know this as the Lorentzian susceptibility.
- $G^A(\omega) = [G^R(\omega)]^*$ is the **advanced Green's function** — just the complex conjugate. No new information.
- $G^{\mathrm{K}}(\omega) = \langle a^{\mathrm{cl}} a^{\mathrm{cl}*} \rangle$ is the **noise spectrum** of the cavity field. You know this as the output power spectral density measured by a spectrum analyser. For a cavity at temperature T: $G^{\mathrm{K}}(\omega) = -i\kappa(2\bar{n}+1)/|\omega - \omega_0|^2$ where $\bar{n} = 1/(e^{\hbar\omega/k_B T} - 1)$. At $T = 0$: $G^{\mathrm{K}} = -i\kappa/|\omega - \omega_0|^2$ — this is the *vacuum noise* output spectrum.
- $\langle a^{\mathrm{q}} a^{\mathrm{q}*} \rangle = 0$ always. This is the causality condition: the quantum field cannot correlate with itself. It is the path-integral expression of $\mathrm{Tr}[\rho U^\dagger U] = 1$.

The **fluctuation-dissipation theorem** is the relation between these:

$$G^{\mathrm{K}}(\omega) = \left(G^R(\omega) - G^A(\omega)\right)(2\bar{n} + 1) \tag{10}$$

In plain language: the noise spectrum is the response function times the thermal occupation. You know this from the quantum regression theorem. The Keldysh formalism *derives* it from the path integral structure rather than assuming it.

1.6 The Keldysh action: where the physics lives

The Keldysh action for a single open cavity mode, in the cl-q basis:

$$S_K = \int \frac{d\omega}{2\pi} \begin{pmatrix} a^{\mathrm{cl}*} & a^{\mathrm{q}*} \end{pmatrix} \begin{pmatrix} 0 & (\omega - \omega_0 - i\kappa/2) \\ (\omega - \omega_0 + i\kappa/2) & -i\kappa(2\bar{n}+1) \end{pmatrix} \begin{pmatrix} a^{\mathrm{cl}} \\ a^{\mathrm{q}} \end{pmatrix} \tag{11}$$

Read off this matrix directly:

- The (2,1) entry $\omega - \omega_0 + i\kappa/2$ is $[G^R]^{-1}$ — the inverse retarded propagator. Setting this to zero gives the cavity resonance condition.

- The (1,2) entry $\omega - \omega_0 - i\kappa/2$ is $[G^A]^{-1}$.
- The (2,2) entry $-i\kappa(2\bar{n}+1)$ is D^{K} — the *noise kernel*. This encodes how much noise the environment injects.
- The (1,1) entry is *zero* — the causality constraint.

1.7 The saddle point = the HL equation, derived

The equation of motion for a^{cl} follows from varying S_K with respect to $a^{\mathrm{q}*}$:

$$\frac{\delta S_K}{\delta a^{\mathrm{q}*}} = 0 \quad \Rightarrow \quad (\omega - \omega_0 + i\kappa/2)\, a^{\mathrm{cl}}(\omega) = 0 \tag{12}$$

In the time domain: $\dot{a}^{\mathrm{cl}} = (-i\omega_0 - \kappa/2)a^{\mathrm{cl}}$.

Including the noise from integrating out a^{q}:

$$\dot{a}^{\mathrm{cl}} = \left(-i\omega_0 - \frac{\kappa}{2}\right) a^{\mathrm{cl}} + \xi(t), \quad \langle \xi^*(t)\xi(t')\rangle = \kappa\bar{n}\,\delta(t-t') \tag{13}$$

This is exactly your HL equation for $\langle \hat{a}(t)\rangle$! It emerged from varying the Keldysh action — no additional input needed. The noise statistics are fixed by the (2,2) entry of the Keldysh matrix: $D^{\mathrm{K}} = -i\kappa(2\bar{n}+1)$ gives $\langle \xi^*\xi\rangle = \kappa\bar{n}$ and $\langle \xi\xi^*\rangle = \kappa(\bar{n}+1)$.

1.8 Nonlinear interactions: vertices in the Keldysh action

Adding a Kerr interaction $\hat{H}_{\mathrm{Kerr}} = g^{(3)}\hat{a}^\dagger\hat{a}^\dagger\hat{a}\hat{a}$:

In the $a^\pm$ basis, the action gets the quartic terms $g^{(3)}[(a^+)^{*2}(a^+)^2 - (a^-)^{*2}(a^-)^2]$. After rotating to cl-q basis, the dominant terms are:

$$S_{\chi^{(3)}} = 2g^{(3)} \int dt\, |a^{\mathrm{cl}}|^2 (a^{\mathrm{cl}*}a^{\mathrm{q}} + a^{\mathrm{q}*}a^{\mathrm{cl}}) + (\text{subleading terms}) \tag{14}$$

Physical meaning:

- The term $2g^{(3)}|a^{\mathrm{cl}}|^2 a^{\mathrm{cl}*}a^{\mathrm{q}}$ means: the Kerr shift of the resonance frequency is $2g^{(3)}|a^{\mathrm{cl}}|^2$ (the mean-field Kerr shift), which appears as a modification of the (2,1) entry of the Keldysh matrix.
- The same vertex, at one loop (one internal propagator loop), modifies the noise kernel D^{K} — it changes the photon number statistics.
- The all-classical term $|a^{\mathrm{cl}}|^4$ is absent (the causality constraint forbids it): this says that classical fields alone cannot drive quantum fluctuations.

For $\chi^{(2)}$: the cubic interaction generates the three-field vertices listed in App. B. The key vertex for squeezing is $g^{(2)}a^{\mathrm{cl}*}a^{\mathrm{cl}}a^{\mathrm{q}}$ (a signal and idler photon pair generated from the pump).

1.9 The loop expansion: how quantum corrections are organized

Every calculation in the path integral can be organized by the number of **loops** in the Feynman diagrams:

Tree level (zero loops) = mean field = saddle point of S_K. This gives the classical equations of motion: the HL equation for $\langle \hat{a} \rangle$, the laser threshold condition, the comb mean field. No quantum noise at this level.

One loop = Gaussian fluctuations around the mean field. This is the level where the Keldysh propagators G^R, G^A, G^K live. One-loop gives:

- The cavity linewidth (κ and $\Delta\nu_{\mathrm{ST}}$)
- The photon number $\bar{n} = \langle \hat{a}^\dagger \hat{a} \rangle$
- The output noise spectrum $S_{\mathrm{out}}(\omega) \propto G^{\mathrm{K}}$
- Two-mode squeezing $F(\omega) = \langle a^{\mathrm{cl}} a^{\mathrm{cl}} \rangle$

This is where most of the paper lives. One-loop = Gaussian = the calculation you would do using the quantum Langevin equations and the quantum regression theorem.

Two loops = first correction beyond Gaussian. This is where the cubic and quartic vertices contribute to the first time. Two loops give:

- $g^{(2)}(0)$ corrections from $\chi^{(3)}$ (amplitude squeezing)
- $g^{(2)}(0)$ corrections from $\chi^{(2)}$ (parametric squeezing)
- First correction to the linewidth from nonlinear interactions

Higher loops = photon blockade regime, non-Gaussian states, $g^{(n)}(0)$ for $n \geq 3$. Requires resummation of the loop expansion (or exact numerical methods like the quantum master equation).

[1] *Loop-counting convention.* Throughout this paper, "one loop" and "two loops" count the number of *interaction vertices* in the diagram, not the number of independent loop momenta (which is the standard field-theory convention). Under the standard convention, the $g^{(3)}$ anomalous self-energy (quartic vertex contracted against the mean field twice) counts as *one* loop (one independent frequency integration). The two conventions agree on the loop order of the leading-order terms ($g^{(2)}(0)$ at Gaussian level, blockade at higher loops) and the relative ordering of corrections; only the naming differs.

1.10 Connecting to what you already know: a dictionary

Quantum optics concept	Keldysh language
Heisenberg operator $\hat{a}(t)$	c-number field $a^{\mathrm{cl}}(t) \approx \langle \hat{a}(t) \rangle$ (saddle point)
HL equation for $\langle \hat{a} \rangle$	Saddle-point equation $\delta S_K / \delta a^{\mathrm{q}*} = 0$
Cavity susceptibility $\chi(\omega)$	Retarded propagator $G^R(\omega)$
Output noise spectrum $S_{\mathrm{out}}(\omega)$	Keldysh propagator $G^{\mathrm{K}}(\omega)$
Quantum regression theorem	Keldysh two-time Green's function
Fluctuation-dissipation theorem	$G^{\mathrm{K}} = (G^R - G^A)(2\bar{n} + 1)$
Lindblad jump operator $\sqrt{\kappa}\hat{a}$	Noise kernel $D^{\mathrm{K}} = -i\kappa(2\bar{n} + 1)$
Coherent state mean field α_0	Saddle-point value of a^{cl}
Two-mode squeezing $\langle \hat{a}_s \hat{a}_i \rangle$	Anomalous propagator $F(\omega) = \langle a_s^{\mathrm{cl}}(\omega) a_i^{\mathrm{cl}}(-\omega) \rangle$
$g^{(2)}(0)$	Four-point Keldysh correlator, one-loop (Gaussian)
Photon blockade, non-Gaussian statistics	Multi-loop Keldysh expansion or resummation
Input-output relation $\hat{a}_{\mathrm{out}} = \hat{a}_{\mathrm{in}} + \sqrt{\kappa}\hat{a}$	Transfer function $G^R(\omega)\sqrt{\kappa}$ mapping source to output

1.11 Why use the Keldysh path integral instead of the master equation?

For a single mode, the Lindblad master equation and the Keldysh path integral are entirely equivalent: they produce the same G^R, G^K, and photon statistics. The path integral is more cumbersome for simple problems.

The path integral becomes advantageous when:

1. **Many modes with nonlinear interactions.** The master equation lives in Hilbert space; its dimension scales exponentially with the number of modes. The Keldysh path integral scales polynomially (one field per mode, plus vertices connecting them). A 50-mode comb is intractable with master equations but natural in the Keldysh language.

2. **Connections to mode functions.** The Keldysh path integral is written in terms of the QNM amplitudes $q_\lambda(t)$ with known spatial mode functions $\tilde{\mathbf{f}}_\lambda(\mathbf{r})$. The nonlinear coupling constants $g^{(2,3)}_{\lambda\mu\nu}$ are explicit spatial overlap integrals (Eqs. 75–76). The master equation does not naturally connect to the spatial structure of the modes.

3. **Perturbative corrections from nonlinearity.** The loop expansion organizes the corrections from $\chi^{(2)}$, $\chi^{(3)}$, and gain saturation systematically at each order. The master equation requires truncating the Fock space, which is less transparent.

4. **Topology optimization.** The gradient of the QFI with respect to the structure $\rho(\mathbf{r})$ uses the bilinear form $\tilde{\mathbf{f}} \cdot \tilde{\mathbf{f}}$ (Secs. 16–20). These gradients are natural in the path integral language because the mode functions are explicit.

1.12 Summary: the minimum you need to read this paper

1. The **Keldysh action** $S_K[a^{\mathrm{cl}}, a^{\mathrm{q}}]$ is a compact encoding of all dynamics. It contains $G^{R,A}$ in its off-diagonal entries and D^{K} in its diagonal.

2. The **saddle point** $\delta S_K/\delta a^{\mathrm{q}*} = 0$ gives the mean-field equations — your HL equations.

3. The **Keldysh propagator** $G^{\mathrm{K}}(\omega)$ is the output noise spectrum. It encodes photon number, linewidth, and squeezing.

4. **Nonlinear vertices** ($g^{(2,3)}$ terms in S_K) enter as Feynman diagram insertions. At tree level they shift the eigenfrequency; at one loop they modify the noise.

5. **Gain** contributes a *positive* $D^{\mathrm{K}}_{\mathrm{gain}} = +ig(2N_{\mathrm{sp}} + 1)$, opposite in sign to the radiation loss $D^{\mathrm{K}}_{\mathrm{rad}} = -i\kappa(2\bar{n} + 1)$. This is the quantum statement that gain amplifies vacuum fluctuations while loss damps them.

6. The **c-number fields** $a^{\pm}(t)$ are not classical fields — they are coherent-state eigenvalues. The theory is quantum because of the Berry phase term $i\int a^*\partial_t a\, dt$ and the Keldysh structure D^{K}, not because the fields are operators.

2 Introduction

The quantization of electromagnetic fields in open, spatially inhomogeneous, and nonlinear nanophotonic structures is a problem of both fundamental and practical importance. Canonical quantization, well established for closed Hermitian cavities [2, 3], encounters serious difficulties when applied to open photonic structures whose natural modes — *quasi-normal modes* (QNMs) — carry complex eigenfrequencies, non-orthogonal and exponentially growing mode functions, and no obvious Hilbert-space inner product [4, 5].

A QNM $\tilde{\mathbf{f}}_\lambda$ satisfies the Helmholtz equation with outgoing (Silver-Müller) radiation boundary conditions, yielding $\tilde{\omega}_\lambda = \omega_\lambda - i\kappa_\lambda/2$ with $\kappa_\lambda > 0$ the radiative linewidth. Substantial progress has been made on the quantization of a *single* QNM coupled to a radiation bath using canonical methods: Fano diagonalization [6], Lindblad master equations [7], and extensions to nonlinear processes [8]. However, a complete treatment that

1. derives rather than postulates the QNM dynamics from the Maxwell action,

2. handles multiple non-orthogonal modes and exceptional points,

3. includes nonlinear $\chi^{(2)}$ and $\chi^{(3)}$ interactions with physically correct, first-principles coupling coefficients,

4. is consistent with canonical quantization in the closed-cavity limit,

5. and comes with explicit Feynman rules for perturbative calculations,

has been absent.

In this paper we provide such a treatment using the Schwinger-Keldysh (closed-time-path) path integral formalism [11, 12, 13]. The Keldysh framework is the natural language for open quantum systems away from equilibrium [14, 15]: it maintains unitarity of the full system (field plus radiation bath) while generating an effective action for the QNM subsystem alone after the bath is integrated out.

Main contributions. Our principal results are:

- **Complete effective action.** Starting from the transverse Maxwell action (Sec. 3), we integrate out the free-space radiation continuum exactly (Sec. 5) to obtain a Keldysh effective action for the QNM amplitudes with complex poles at $\tilde{\omega}_\lambda$ and a Keldysh component determined by the fluctuation-dissipation theorem (Sec. 8).

- **Functional Jacobian.** The change-of-variables Jacobian is the Gram determinant of the QNM functions; it is field-independent and cancels in all observables (Sec. 4).

- **Non-orthogonal modes.** Cholesky decomposition $\mathcal{O} = LL^\dagger$ of the Hermitian overlap matrix diagonalizes the kinetic term and yields bosonic operators with exact commutator algebra $[\hat{a}_\nu, \hat{a}^\dagger_{\nu'}] = \delta_{\nu\nu'}$ *within the chosen N-mode subspace* (Sec. 6). Truncation to N QNMs introduces the same kind of $O(\sum_{\text{discarded}} |\mathcal{O}_{\lambda\lambda'}|^2)$ error as the Franke–Hughes Fano construction; the Cholesky improvement is in the operator algebra of the kept modes, not in the basis-completeness side of the problem.

- **Exceptional points.** The Jordan-chain structure yields a Jordan-block Keldysh action and a double-pole propagator $G^R_{\text{EP}} \sim (\omega - \tilde{\omega}_{\text{EP}})^{-2}$. The on-resonance Purcell factor scales as Q/V (same as a normal mode of equal Q), but the time-domain impulse response grows linearly before decaying, giving a squared-Lorentzian power transmission $|T(\omega)|^2 \propto [(\omega - \omega_{\text{EP}})^2 + (\kappa_{\text{EP}}/2)^2]^{-2}$ that is distinguishable from a Lorentzian (Sec. 7).

- **Nonlinear coupling structure.** We prove: (a) the single-mode self-Kerr coupling is real ($\propto \int |\tilde{f}|^4$); (b) the cross-Kerr coupling is *also* real for real Kleinman-symmetric $\chi^{(3)}$ — proved by an exact index-relabeling argument, confirmed numerically to $|\text{Im}/\text{Re}| < 10^{-15}$ (Sec. 9.4); (c) the $\chi^{(2)}$ downconversion coupling *is* generically complex with $\text{Im}/\text{Re} \sim 1/Q$ because the index pattern $(\lambda^* \mu \nu)$ lacks the self-conjugacy property (Sec. 9.5); (d) gain-saturation $\chi^{(3)}_{\text{sat}}$ is purely imaginary at resonance, driving cross-mode dephasing in active structures.

- **RWA and Bloch-Siegert shift.** The rotating-wave approximation is controlled by $g^{(3)}/\omega_0$; the leading correction is calculated explicitly (Sec. 10).

- **Canonical limit.** Taking $\kappa \to 0$ reproduces the standard coherent-state path integral and canonical commutators (Sec. 12).

- **Numerical estimates.** The self-Kerr coupling for a GaAs H1 PhC nanocavity (Q=3×10^4, $V_{\text{eff}} \approx 0.023\,\mu\text{m}^3$) is $g^{(3)}/2\pi \approx 235\,\text{kHz}$ (Sec. 13).

We use SI units throughout. Operator hats are dropped in path integrals where fields are classical variables.

3 Classical Maxwell Action and QNM Decomposition

3.1 Maxwell action in the Coulomb gauge

Consider a lossless nanophotonic structure with spatially varying permittivity $\varepsilon(\mathbf{r}) \in \mathbb{R}^+$ embedded in free space. In the Coulomb gauge $\nabla \cdot \mathbf{A} = 0$, with $\mathbf{E} = -\partial_t \mathbf{A}$ and $\mathbf{B} = \nabla \times \mathbf{A}$, the action is:

$$S[\mathbf{A}] = \int dt \int d^3r \left[\frac{\varepsilon_0}{2} \varepsilon(\mathbf{r}) |\dot{\mathbf{A}}|^2 - \frac{1}{2\mu_0} |\nabla \times \mathbf{A}|^2 + \mathcal{L}_{\mathrm{NL}}[\dot{\mathbf{A}}, \mathbf{r}] \right] \tag{15}$$

with nonlinear Lagrangian density:

$$\mathcal{L}_{\mathrm{NL}} = \frac{\varepsilon_0}{3} \chi^{(2)}_{ijk}(\mathbf{r}) \dot{A}_i \dot{A}_j \dot{A}_k + \frac{\varepsilon_0}{4} \chi^{(3)}_{ijkl}(\mathbf{r}) \dot{A}_i \dot{A}_j \dot{A}_k \dot{A}_l + \cdots \tag{16}$$

Repeated Cartesian indices i, j, k, l are summed. The susceptibility tensors are real (lossless medium) and satisfy Kleinman symmetry.

3.2 Quasi-normal modes

QNMs satisfy the generalized eigenvalue problem:

$$\nabla \times \nabla \times \tilde{\mathbf{f}}_\lambda(\mathbf{r}) = \frac{\tilde{\omega}_\lambda^2}{c^2} \varepsilon(\mathbf{r}) \tilde{\mathbf{f}}_\lambda(\mathbf{r}) \tag{17}$$

with outgoing radiation boundary conditions $\hat{\mathbf{r}} \times (\nabla \times \tilde{\mathbf{f}}_\lambda) - ik\tilde{\mathbf{f}}_\lambda \to 0$ as $r \to \infty$. The eigenfrequencies are complex:

$$\tilde{\omega}_\lambda = \omega_\lambda - i\frac{\kappa_\lambda}{2}, \qquad \omega_\lambda, \kappa_\lambda > 0, \qquad Q_\lambda = \omega_\lambda / \kappa_\lambda \tag{18}$$

Two inner products

Two distinct inner products appear and must not be confused:

Bilinear form (for QNM normalization, involves $\tilde{\mathbf{f}} \cdot \tilde{\mathbf{f}}$, not $\tilde{\mathbf{f}}^* \cdot \tilde{\mathbf{f}}$):

$$\mathcal{B}_{\lambda\lambda'} \equiv \int_V d^3r \, \varepsilon(\mathbf{r}) \, \tilde{\mathbf{f}}_\lambda \cdot \tilde{\mathbf{f}}_{\lambda'} + (\text{surface terms}) = \delta_{\lambda\lambda'} \tag{19}$$

Hermitian inner product (positive-definite, for path integral measure):

$$\mathcal{O}_{\lambda\lambda'} \equiv \int d^3r \, \varepsilon(\mathbf{r}) \, \tilde{\mathbf{f}}_\lambda^* \cdot \tilde{\mathbf{f}}_{\lambda'} \tag{20}$$

$\mathcal{O}$ is Hermitian positive-definite (Gram matrix of the linearly independent QNM functions); $\mathcal{O}_{\lambda\lambda'} \neq \delta_{\lambda\lambda'}$ in general, encoding non-orthogonality.

3.3 Field decomposition

We write the transverse vector potential as:

$$\mathbf{A}(\mathbf{r},t) = \sum_\lambda \left[q_\lambda(t)\tilde{\mathbf{u}}_\lambda(\mathbf{r}) + q_\lambda^*(t)\tilde{\mathbf{u}}_\lambda^*(\mathbf{r})\right] + \int \frac{d^3k}{(2\pi)^3} \sum_s \left[\phi_{\mathbf{k}s}(t)\, \boldsymbol{\epsilon}_{\mathbf{k}s} e^{i\mathbf{k}\cdot\mathbf{r}} + \text{c.c.}\right] \tag{21}$$

where $\tilde{\mathbf{u}}_\lambda$ are the QNM vector potentials ($\tilde{\mathbf{f}}_\lambda = i\tilde{\omega}_\lambda \tilde{\mathbf{u}}_\lambda / c^2$ up to normalization), and $\boldsymbol{\epsilon}_{\mathbf{k}s}$ ($s = 1, 2$) are free-space transverse polarization vectors. The path integral is:

$$Z = \int \mathcal{D}\mathbf{A}\, e^{iS[\mathbf{A}]/\hbar} \tag{22}$$

4 The Functional Jacobian

4.1 The transformation and its Jacobian

The map from mode amplitudes to the field at each time t is:

$$A_i(\mathbf{r}) = \sum_\lambda \left[q_\lambda[\tilde{\mathbf{u}}_\lambda]_i(\mathbf{r}) + q_\lambda^*[\tilde{\mathbf{u}}_\lambda^*]_i(\mathbf{r})\right] + \sum_{\mathbf{k}s} \left[\phi_{\mathbf{k}s}[\boldsymbol{\epsilon}_{\mathbf{k}s}]_i e^{i\mathbf{k}\cdot\mathbf{r}} + \text{c.c.}\right] \tag{23}$$

This is a linear, *time-local* map M from mode amplitudes to field values. Crucially, M depends only on the fixed mode functions $\tilde{\mathbf{u}}_\lambda(\mathbf{r})$ and the plane waves $\boldsymbol{\epsilon}_{\mathbf{k}s} e^{i\mathbf{k}\cdot\mathbf{r}}$ — it is independent of all dynamical variables. The functional measure transforms as:

$$\mathcal{D}\mathbf{A} = \mathcal{J} \prod_\lambda \mathcal{D}q_\lambda \mathcal{D}q_\lambda^* \cdot \prod_{\mathbf{k}s} \mathcal{D}\phi_{\mathbf{k}s} \mathcal{D}\phi_{\mathbf{k}s}^* \tag{24}$$

4.2 Explicit evaluation

Since M is time-independent, the functional determinant factorizes over time slices ($N_t = T/\delta t$ slices):

$$\mathcal{J} = (\det M)^{N_t} \tag{25}$$

The Gram determinant theorem gives:

$$\det(M^\dagger M) = \det \begin{pmatrix} \mathcal{O}^{\text{QNM}} & C \\ C^\dagger & \mathcal{O}^{\text{rad}} \end{pmatrix} \tag{26}$$

where $\mathcal{O}^{\text{QNM}}_{\lambda\lambda'} = \mathcal{O}_{\lambda\lambda'}$ (Eq. 20), $\mathcal{O}^{\text{rad}}_{\mathbf{k}s;\mathbf{k}'s'} = (2\pi)^3 \delta^{(3)}(\mathbf{k} - \mathbf{k}')\delta_{ss'}$, and $C_{\lambda;\mathbf{k}s} = V_{\lambda,\mathbf{k}s}$ is the QNM-radiation coupling element.

4.3 Cancellation in physical observables

Two properties guarantee the Jacobian is irrelevant:

(i) Field independence. $\det(M^\dagger M)$ depends only on the fixed mode functions, not on $q_\lambda(t)$ or $\phi_{\mathbf{k}s}(t)$. Therefore:

$$\boxed{\mathcal{J} = \left[\det \mathcal{O}^{\rm QNM} \cdot \det \mathcal{O}^{\rm rad}\right]^{N_t/2} = \text{const}} \tag{27}$$

(ii) Ratio cancellation. Every physical observable is computed as $\langle \mathcal{O} \rangle = (\int \mathcal{D}q\, \mathcal{O}\, e^{iS/\hbar})/(\int \mathcal{D}q\, e^{iS/\hbar})$. The same constant $\mathcal{J}$ appears in numerator and denominator and cancels exactly.

The divergence $\mathcal{J} \propto e^{T \log\det M/\delta t}$ as $\delta t \to 0$ is the standard vacuum-energy (cosmological-constant) divergence of QFT, absorbable into a constant additive renormalization of the action density that does not affect any S-matrix element.

5 Integrating Out the Radiation Continuum

5.1 Action in decomposed variables

Substituting Eq. (21) into the linear part of Eq. (15) (nonlinear terms are treated in Sec. 9):

$$S_{\rm QNM} = \frac{\varepsilon_0}{2} \sum_{\lambda\lambda'} \int dt\, \mathcal{O}_{\lambda\lambda'} \left[\dot{q}^*_\lambda \dot{q}_{\lambda'} - \tilde{\omega}^2_{\lambda'}\, q^*_\lambda q_{\lambda'}\right] \tag{28}$$

$$S_{\rm rad} = \frac{\varepsilon_0}{2} \int \frac{d^3k}{(2\pi)^3} \sum_s \int dt \left[|\dot{\phi}_{\mathbf{k}s}|^2 - c^2k^2|\phi_{\mathbf{k}s}|^2\right] \tag{29}$$

$$S_{\rm coup} = \varepsilon_0 \sum_\lambda \int \frac{d^3k}{(2\pi)^3} \sum_s \int dt \left[V_{\lambda,\mathbf{k}s}\dot{q}^*_\lambda \dot{\phi}_{\mathbf{k}s} + \text{c.c.}\right] \tag{30}$$

with coupling matrix elements $V_{\lambda,\mathbf{k}s} = \int d^3r\, \varepsilon(\mathbf{r}) \tilde{\mathbf{u}}^*_\lambda \cdot \boldsymbol{\epsilon}_{\mathbf{k}s} e^{i\mathbf{k}\cdot\mathbf{r}}$.

5.2 Exact Gaussian integration over radiation modes

The radiation action $S_{\rm rad} + S_{\rm coup}$ is quadratic in $\phi_{\mathbf{k}s}$ and can be integrated exactly. In frequency space, completing the square in $\tilde{\phi}_{\mathbf{k}s}(\omega)$ gives the saddle point:

$$\tilde{\phi}^{\rm sp}_{\mathbf{k}s}(\omega) = \frac{\omega^2 \sum_\lambda V^*_{\lambda,\mathbf{k}s} \tilde{q}_\lambda(\omega)}{\omega^2 - c^2k^2 + i0^+} \tag{31}$$

where $+i0^+$ enforces outgoing-wave (retarded) boundary conditions. Substituting back gives the effective action:

$$S^{\rm lin}_{\rm eff} = S_{\rm QNM} + \frac{\varepsilon_0^2}{2} \sum_{\lambda\lambda'} \int \frac{d\omega}{2\pi}\, \omega^4\, \tilde{q}^*_\lambda(\omega) \Sigma_{\lambda\lambda'}(\omega) \tilde{q}_{\lambda'}(\omega) \tag{32}$$

with self-energy kernel:

$$\Sigma_{\lambda\lambda'}(\omega) = \int \frac{d^3k}{(2\pi)^3} \sum_s \frac{V^*_{\lambda,\mathbf{k}s} V_{\lambda',\mathbf{k}s}}{\omega^2 - c^2k^2 + i0^+} \tag{33}$$

5.3 Decay rate and Lamb shift

By the Sokhotski-Plemelj theorem, the imaginary part of the diagonal self-energy is:

$$\operatorname{Im}\Sigma_{\lambda\lambda}(\omega_\lambda) = -\frac{\kappa_\lambda}{2\omega_\lambda^2\varepsilon_0^2} \tag{34}$$

This is the path-integral derivation of Fermi's golden rule: κ_λ is determined by the density of radiation modes at ω_λ weighted by $|V_{\lambda,\mathbf{k}s}|^2$. The real part gives a Lamb shift $\delta\omega_\lambda$ absorbed into the renormalized frequency.

5.4 Conditions for neglecting off-diagonal self-energy

The off-diagonal self-energy $\Sigma_{\lambda\mu}(\omega)$ ($\lambda \neq \mu$) couples the equations of motion for different modes. By the Cauchy-Schwarz inequality applied to Eq. (33):

$$|\Sigma_{\lambda\mu}(\omega)|^2 \leq \Sigma_{\lambda\lambda}(\omega)\,\Sigma_{\mu\mu}(\omega) \tag{35}$$

so $|\Sigma_{\lambda\mu}| \leq \sqrt{\kappa_\lambda\kappa_\mu}/(2\omega^2\varepsilon_0^2)$.

Near the resonance $\omega \approx \omega_\lambda$, the ratio of the off-diagonal self-energy shift to the resonance width is:

$$\frac{|\Sigma_{\lambda\mu}(\omega_\lambda)|}{|\Sigma_{\lambda\lambda}(\omega_\lambda)|} \leq \sqrt{\frac{\kappa_\mu}{\kappa_\lambda}} \cdot \frac{\sqrt{\kappa_\lambda\kappa_\mu}/2}{|\omega_\lambda - \omega_\mu|} \sim \frac{\sqrt{\kappa_\lambda\kappa_\mu}}{2|\omega_\lambda - \omega_\mu|} \tag{36}$$

The off-diagonal contribution is therefore negligible when:

$$\boxed{|\omega_\lambda - \omega_\mu| \gg \tfrac{1}{2}\sqrt{\kappa_\lambda\kappa_\mu}} \tag{37}$$

This is the **spectrally resolved** condition: modes must be separated by more than the geometric mean of their half-linewidths.

When Eq. (37) holds, each mode decays independently with rate κ_λ, and the effective action Eq. (32) separates mode by mode. This is the assumption used throughout this paper.

When the condition fails. If $|\omega_\lambda - \omega_\mu| \lesssim \sqrt{\kappa_\lambda\kappa_\mu}/2$, the off-diagonal self-energy generates a cross-coupling between modes at the *linear* level:

$$\delta S_{\text{cross}} = \varepsilon_0^2 \int \frac{d\omega}{2\pi}\,\omega^4\left[\tilde{q}_\lambda^*\Sigma_{\lambda\mu}(\omega)\tilde{q}_\mu + \text{c.c.}\right] \tag{38}$$

This becomes an additional off-diagonal entry in the linear Keldysh action Eq. (68), describing coherent hybridization of the two modes through their shared radiation bath — the photonic analogue of superradiance. This regime approaches the EP as $|\omega_\lambda - \omega_\mu| \to 0$ and is treated separately in Sec. 7.

Application to the GaAs H1 cavity. For the two dipole modes of the H1 cavity used in Sec. 13, $Q \approx 3 \times 10^4$ gives $\kappa/2\pi \approx 10\,\text{GHz}$. The condition Eq. (37) requires $|\omega_\lambda - \omega_\mu|/2\pi \gg 5\,\text{GHz}$. In a perfect triangular lattice the two dipole modes are exactly degenerate; symmetry breaking by the waveguide coupling or a slight ellipticity of the holes lifts the degeneracy. A splitting of $|\Delta\omega|/2\pi \gtrsim 50\,\text{GHz}$ (achievable by 1–2% asymmetry in hole radii) satisfies Eq. (37) with a factor of 10 margin, validating the independent-mode treatment.

5.5 Markov approximation and mode amplitudes

For high-Q resonators ($\kappa_\lambda \ll \omega_\lambda$), evaluating Σ at $\omega = \omega_\lambda$ (Markov approximation) and defining complex mode amplitudes:

$$a_\lambda(t) = \sqrt{\frac{\varepsilon_0 \mathcal{O}_{\lambda\lambda}}{2\hbar|\tilde{\omega}_\lambda|}}\,(\tilde{\omega}_\lambda q_\lambda + i\dot{q}_\lambda) \tag{39}$$

the single-mode effective action becomes:

$$S^{(1)}_{\text{eff}} = \int dt\,\hbar\left[ia^*_\lambda \dot{a}_\lambda - \omega_\lambda |a_\lambda|^2 + i\frac{\kappa_\lambda}{2}|a_\lambda|^2\right] \tag{40}$$

The imaginary term $+i\hbar\kappa_\lambda|a_\lambda|^2/2$ renders the action complex, signalling an open system and requiring the Keldysh doubling of Sec. 8.

6 Non-Orthogonal Multi-Mode QNMs

6.1 The overlap problem

When multiple QNMs are present, the QNM action Eq. (28) has a non-diagonal kinetic matrix $\mathcal{O}_{\lambda\lambda'}$. The mode amplitudes q_λ are coupled and the action cannot be factorized.

6.2 Cholesky diagonalization

Since $\mathcal{O}$ is Hermitian positive-definite, it has a unique Cholesky decomposition:

$$\mathcal{O} = LL^\dagger \tag{41}$$

with L lower triangular and positive diagonal. Define **Cholesky amplitudes**:

$$Q_\nu \equiv (L^\dagger)_{\nu\lambda} q_\lambda, \qquad q_\lambda = (L^{-\dagger})_{\lambda\nu} Q_\nu \tag{42}$$

The kinetic term diagonalizes exactly:

$$\sum_{\lambda\lambda'} \mathcal{O}_{\lambda\lambda'} \dot{q}^*_\lambda \dot{q}_{\lambda'} = \sum_\nu |\dot{Q}_\nu|^2 \tag{43}$$

The potential term transforms to $\sum_{\mu\nu} W_{\mu\nu} Q^*_\mu Q_\nu$ with $\mathbf{W} = L^{-1}\tilde{\omega}^2 L$ ($\tilde{\omega}^2$ diagonal).

6.3 Dressed eigenfrequencies

$\mathbf{W}$ is a similarity transform of $\tilde{\omega}^2$, so eigenvalues($\mathbf{W}$) $= \{\tilde{\omega}^2_\lambda\}$: the dressed eigenfrequencies equal the bare QNM eigenfrequencies. Diagonalize $\mathbf{W} = U\tilde{\Omega}^2 U^\dagger$ (unitary U) and define the final **dressed amplitudes**:

$$\Xi_\nu = (U^\dagger L^\dagger)_{\nu\lambda} q_\lambda \equiv (T^\dagger)_{\nu\lambda} q_\lambda \tag{44}$$

The fully diagonalized QNM action is:

$$S_{\text{QNM}} = \frac{\varepsilon_0}{2}\sum_\nu \int dt\left[|\dot{\Xi}_\nu|^2 - \tilde{\omega}^2_\nu |\Xi_\nu|^2\right] \tag{45}$$

6.4 Nearly degenerate modes: two-mode example

For two modes with $|\omega_1 - \omega_2| \lesssim \kappa$, $\mathcal{O} = \left(\begin{smallmatrix} 1 & \epsilon \\ \epsilon^* & 1 \end{smallmatrix}\right)$, $|\epsilon| \ll 1$. The dressed-frequency matrix:

$$\mathbf{W} = \begin{pmatrix} \tilde{\omega}_1^2 & 0 \\ \epsilon^*(\tilde{\omega}_2^2 - \tilde{\omega}_1^2) & \tilde{\omega}_2^2 \end{pmatrix} + O(\epsilon^2) \tag{46}$$

The off-diagonal entry $\epsilon^*(\tilde{\omega}_2^2 - \tilde{\omega}_1^2) \approx 2\epsilon^*\omega_0\delta\omega$ vanishes *both* as $\epsilon \to 0$ (orthogonal modes) *and* as $\delta\omega \to 0$ (degenerate modes), the latter signalling the exceptional-point limit of Sec. 7.

6.5 Nonlinear couplings in the dressed basis

Under the transformation T the $\chi^{(3)}$ coupling transforms as:

$$g^{(3),\mathrm{dress}}_{\nu_1\nu_2\nu_3\nu_4} = \sum_{\lambda_1\cdots\lambda_4} T^*_{\lambda_1\nu_1} T^*_{\lambda_2\nu_2} T_{\lambda_3\nu_3} T_{\lambda_4\nu_4}\, g^{(3)}_{\lambda_1\lambda_2\lambda_3\lambda_4} \tag{47}$$

Off-diagonal dressed couplings are of order $|\epsilon| g^{(3)}$: even modes uncoupled in the bare basis acquire cross-coupling through non-orthogonality.

6.6 Exact bosonic operators in the dressed basis

Each dressed mode Ξ_ν defines annihilation and creation operators:

$$\hat{a}_\nu = \sqrt{\frac{\varepsilon_0}{2\hbar|\tilde{\omega}_\nu|}} \left(\tilde{\omega}_\nu \Xi_\nu + i\dot{\Xi}_\nu\right), \qquad [\hat{a}_\nu, \hat{a}^\dagger_{\nu'}] = \delta_{\nu\nu'} \tag{48}$$

The commutation relations are *exact* by construction.

7 Exceptional Points: Jordan-Block Keldysh Action

7.1 The Jordan chain

At an exceptional point (EP) [17, 18], $\tilde{\omega}_1 = \tilde{\omega}_2 = \tilde{\omega}_{\mathrm{EP}}$ and the eigenspace is defective: there is only one eigenvector $\tilde{\mathbf{u}}_0$ satisfying Eq. (17). A second, *generalized eigenvector* $\tilde{\mathbf{u}}_1$ satisfies the Jordan chain:

$$(\mathbf{K} - \tilde{\omega}^2_{\mathrm{EP}}\mathcal{M})\tilde{\mathbf{u}}_0 = 0 \tag{49}$$

$$(\mathbf{K} - \tilde{\omega}^2_{\mathrm{EP}}\mathcal{M})\tilde{\mathbf{u}}_1 = \mathcal{M}\tilde{\mathbf{u}}_0 \tag{50}$$

where $\mathbf{K} = c^2\nabla \times \nabla\times$ and $\mathcal{M} = \varepsilon(\mathbf{r})$.

7.2 Jordan bilinear normalization

The bilinear form has a characteristic degeneracy at the EP:

$$\mathcal{B}_{00} = \int d^3r\, \varepsilon\, \tilde{\mathbf{u}}_0 \cdot \tilde{\mathbf{u}}_0 = 0 \quad \text{(self-orthogonality)} \tag{51}$$

$$\mathcal{B}_{01} = \int d^3r\, \varepsilon\, \tilde{\mathbf{u}}_0 \cdot \tilde{\mathbf{u}}_1 = 1 \quad \text{(Jordan normalization)} \tag{52}$$

$$\mathcal{B}_{11} = \int d^3r\, \varepsilon\, \tilde{\mathbf{u}}_1 \cdot \tilde{\mathbf{u}}_1 = \gamma \in \mathbb{C} \tag{53}$$

Self-orthogonality Eq. (51) is the defining signature of an EP: it replaces the standard bilinear normalization of isolated modes.

7.3 Jordan-block action

Expanding $\mathbf{A}_{\text{EP}} = q_0\tilde{\mathbf{u}}_0 + q_1\tilde{\mathbf{u}}_1 + \text{c.c.} +$ radiation, and using the Jordan chain conditions with the stiffness matrix elements $K_{00} = 0$, $K_{01} = \tilde{\omega}_{\text{EP}}^2$, $K_{11} = \tilde{\omega}_{\text{EP}}^2\gamma - 1$, the EP action takes the matrix form:

$$S_{\text{EP}} = \varepsilon_0 \int dt \begin{pmatrix} \dot{q}_0^* & \dot{q}_1^* \end{pmatrix} \mathcal{J}_{\text{EP}} \begin{pmatrix} \dot{q}_0 \\ \dot{q}_1 \end{pmatrix} - \tilde{\omega}_{\text{EP}}^2 \begin{pmatrix} q_0^* & q_1^* \end{pmatrix} \mathcal{J}_{\text{EP}} \begin{pmatrix} q_0 \\ q_1 \end{pmatrix} \tag{54}$$

with the **Jordan metric**:

$$\mathcal{J}_{\text{EP}} = \begin{pmatrix} 0 & 1 \\ 1 & \gamma \end{pmatrix}, \quad \det \mathcal{J}_{\text{EP}} = -1 \tag{55}$$

The negative determinant reflects the indefinite character of the bilinear form at the EP.

7.4 Jordan operator algebra

Defining Jordan-adapted mode operators:

$$\hat{a}_0 = \sqrt{\frac{\varepsilon_0}{\hbar|\tilde{\omega}_{\text{EP}}|}} \left(\tilde{\omega}_{\text{EP}} q_1 + i\dot{q}_1\right) \tag{56}$$

$$\hat{a}_1 = \sqrt{\frac{\varepsilon_0}{\hbar|\tilde{\omega}_{\text{EP}}|}} \left(\tilde{\omega}_{\text{EP}} q_0 + i\dot{q}_0\right) - \frac{\gamma}{2}\hat{a}_0 \tag{57}$$

these satisfy the **Jordan operator algebra**:

$$[\hat{a}_0, \hat{a}_0^\dagger] = 0, \quad [\hat{a}_0, \hat{a}_1^\dagger] = 1, \quad [\hat{a}_1, \hat{a}_1^\dagger] = -\gamma \tag{58}$$

The singular commutator $[\hat{a}_0, \hat{a}_0^\dagger] = 0$ is the operator manifestation of self-orthogonality; $\hat{a}_0$ is not a standard bosonic annihilator.

7.5 Double-pole retarded propagator

The Keldysh action at the EP is a 4×4 block matrix (two modes, two Keldysh branches). The retarded inverse propagator contains the Jordan block:

$$[\mathbf{G}^R_{\mathrm{EP}}(\omega)]^{-1} = (\omega - \tilde{\omega}_{\mathrm{EP}})\mathbf{1} - \begin{pmatrix} 0 & -1 \\ 0 & 0 \end{pmatrix} \tag{59}$$

Inverting:

$$\mathbf{G}^R_{\mathrm{EP}}(\omega) = \frac{\mathbf{1}}{\omega - \tilde{\omega}_{\mathrm{EP}}} + \frac{1}{(\omega - \tilde{\omega}_{\mathrm{EP}})^2} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \tag{60}$$

In the time domain, the double pole generates a linear-in-t response:

$$\mathbf{G}^R_{\mathrm{EP}}(t) = -i\,\theta(t)\,e^{-i\tilde{\omega}_{\mathrm{EP}}t} \left[\mathbf{1} - it \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}\right] \tag{61}$$

The diagonal $-i\mathbf{1}$ is the standard retarded propagator for each Jordan mode; the off-diagonal $-it\,e^{-i\tilde{\omega}_{\mathrm{EP}}t}$ grows linearly before the exponential decay (with rate $\kappa_{\mathrm{EP}}/2 = -\mathrm{Im}(\tilde{\omega}_{\mathrm{EP}})$) takes over. This linear growth is absent for a normal mode and constitutes the time-domain signature of the EP.

7.6 Transmission lineshape and time-domain response

The EP propagator has two physically distinct contributions (Eq. 60): a standard single-pole term and a double-pole term. For the (0,1) off-diagonal transmission amplitude from mode 0 to mode 1:

$$T_{01}(\omega) \propto \frac{1}{(\omega - \tilde{\omega}_{\mathrm{EP}})^2} \tag{62}$$

The power transmission $|T_{01}(\omega)|^2$ is a **squared Lorentzian**:

$$|T_{01}(\omega)|^2 \propto \frac{1}{\left[(\omega - \omega_{\mathrm{EP}})^2 + (\kappa_{\mathrm{EP}}/2)^2\right]^2} \tag{63}$$

This is narrower and has steeper wings than an ordinary Lorentzian $\sim [(\omega - \omega_{\mathrm{EP}})^2 + (\kappa_{\mathrm{EP}}/2)^2]^{-1}$, making it sensitive to small frequency perturbations.

Purcell factor at the EP. The spontaneous emission rate of a dipole emitter at $\mathbf{r}_0$ is proportional to $\mathrm{Im}\, G^R(\mathbf{r}_0, \mathbf{r}_0, \omega)$. The diagonal ($\lambda = \mu$) components of $\mathbf{G}^R_{\mathrm{EP}}(\omega)$ in Eq. (60) are single-pole terms: $G^{R,\lambda\lambda}_{\mathrm{EP}} = (\omega - \tilde{\omega}_{\mathrm{EP}})^{-1}$, identical to a standard QNM propagator. The double-pole term is purely off-diagonal. Therefore the *on-resonance* Purcell factor for an emitter coupling to a single Jordan eigenstate is:

$$F_P^{\mathrm{EP}}\big|_{\text{on-resonance}} \sim \frac{Q_{\mathrm{EP}}}{V_{\mathrm{EP}}} \quad \text{(same scaling as a normal mode)} \tag{64}$$

The EP does not enhance the steady-state spontaneous emission rate. Its distinct signature is in the **time-domain**: in the impulse response Eq. (61), the double pole contributes a term $\propto t\,e^{-\kappa_{\mathrm{EP}}t/2}$ that initially *grows* linearly before decaying. For $t \ll 2/\kappa_{\mathrm{EP}}$, the field amplitude from the EP is larger by a factor of $t\omega_{\mathrm{EP}}$ compared to a normal mode. This gives enhanced sensitivity in *pulsed* excitation schemes and in non-Hermitian sensing applications [27, 28].

7.7 Experimental realization: PT-symmetric coupled microrings

A concrete experimental system for the Keldysh EP theory is a pair of coupled microring resonators with balanced gain (ring 1) and loss (ring 2) — the PT-symmetric configuration realized by Peng et al. [29] and Hodaei et al. [27]. The coupling Hamiltonian:

$$H_{\mathrm{PT}} = \begin{pmatrix} \omega_0 + ig/2 & J \\ J & \omega_0 - ig/2 \end{pmatrix} \tag{65}$$

has eigenvalues $\omega_\pm = \omega_0 \pm \sqrt{J^2 - g^2/4}$. An EP occurs at $J = g/2$: the two eigenfrequencies coalesce and the eigenvectors become parallel.

In the Keldysh framework, the gain ring contributes a positive Keldysh component $D_1^{\mathrm{K}} = +ig_1(2N_{\mathrm{sp}}+1)$ (Eq. 194: gain amplifies vacuum fluctuations), while the loss ring contributes the usual $D_2^{\mathrm{K}} = -i\kappa_2(2n+1)$. The Jordan-block action Eq. (54) is recovered exactly at $J = g/2$.

Prediction. Driving the loss ring (mode 0) with a weak probe and measuring the gain ring (mode 1), the power transfer is:

$$|S_{21}(\omega)|^2 = \left| \frac{J}{(\omega-\omega_0)^2 + g^2/4 - J^2} \right|^2 \xrightarrow{J \to g/2} \frac{(g/2)^2}{[(\omega-\omega_{\mathrm{EP}})^2]^2} \tag{66}$$

The denominator $(\omega-\omega_0)^2 + g^2/4 - J^2$ is positive and non-singular for all real ω when $J < g/2$ (PT-broken phase, complex eigenfrequencies), and acquires real poles at $\omega = \omega_0 \pm \sqrt{J^2 - g^2/4}$ when $J > g/2$ (PT-unbroken phase, real eigenfrequencies). At the EP $J \to g/2$ the denominator becomes $(\omega - \omega_{\mathrm{EP}})^2$: exactly a squared Lorentzian with peak value $(g/2)^{-2}$. This is directly observable in microwave transmission spectroscopy and distinguishes the EP from two overlapping Lorentzians.

8 Keldysh Action for Open QNMs

The complex effective action Eq. (40) signals a non-unitary open system. The Schwinger-Keldysh (closed-time-path) formalism restores a well-defined path integral by doubling the fields: $a_\lambda^\pm(t)$ on forward/backward time branches. Rotating to the classical-quantum basis:

$$a_\lambda^{\mathrm{cl}} = \tfrac{1}{2}(a_\lambda^+ + a_\lambda^-), \quad a_\lambda^{\mathrm{q}} = a_\lambda^+ - a_\lambda^- \tag{67}$$

the linear Keldysh action for non-degenerate, well-separated QNMs is:

$$S_{\mathrm{K}}^{\mathrm{lin}} = \sum_\lambda \int \frac{d\omega}{2\pi} \begin{pmatrix} a_\lambda^{\mathrm{cl}*} & a_\lambda^{\mathrm{q}*} \end{pmatrix} \begin{pmatrix} 0 & [G_\lambda^A]^{-1} \\ [G_\lambda^R]^{-1} & D_\lambda^{\mathrm{K}} \end{pmatrix} \begin{pmatrix} a_\lambda^{\mathrm{cl}} \\ a_\lambda^{\mathrm{q}} \end{pmatrix} \tag{68}$$

with:

$$G_\lambda^R(\omega) = \frac{1}{\omega - \tilde{\omega}_\lambda} = \frac{1}{\omega - \omega_\lambda + i\kappa_\lambda/2} \tag{69}$$

$$G_\lambda^A(\omega) = [G_\lambda^R(\omega)]^\dagger \tag{70}$$

$$D_\lambda^{\mathrm{K}}(\omega) = -i\kappa_\lambda(2n_\lambda(\omega)+1) \tag{71}$$

At zero temperature, $n_\lambda = 0$ and $D^{\mathrm{K}} = -i\kappa_\lambda$. The Keldysh propagator:

$$G^{\mathrm{K}}_\lambda(\omega) = G^R_\lambda(\omega)\, D^{\mathrm{K}}_\lambda(\omega)\, G^A_\lambda(\omega) = \frac{-i\kappa_\lambda(2n_\lambda + 1)}{|\omega - \tilde{\omega}_\lambda|^2} \tag{72}$$

satisfies the fluctuation-dissipation relation, derived here from first principles rather than imposed phenomenologically.

9 Nonlinear Interactions: Real Self-Kerr and Complex Cross-Kerr

9.1 Electric field and zero-point amplitude

The electric field operator in the QNM basis:

$$\hat{\mathbf{E}}(\mathbf{r}, t) = i \sum_\lambda \mathcal{E}^{\mathrm{zpf}}_\lambda \left[a_\lambda(t) \tilde{\mathbf{f}}_\lambda(\mathbf{r}) - a^*_\lambda(t) \tilde{\mathbf{f}}^*_\lambda(\mathbf{r}) \right] \tag{73}$$

with zero-point field amplitude:

$$\mathcal{E}^{\mathrm{zpf}}_\lambda = \sqrt{\frac{\hbar|\tilde{\omega}_\lambda|}{2\varepsilon_0 \mathcal{O}_{\lambda\lambda}}} \in \mathbb{R}^+ \tag{74}$$

$\mathcal{E}^{\mathrm{zpf}}_\lambda$ is *real and positive* because $|\tilde{\omega}_\lambda| = |\tilde{\omega}^*_\lambda|$ and $\mathcal{O}_{\lambda\lambda} \in \mathbb{R}^+$.

9.2 Coupling coefficient formulas

After the rotating-wave approximation (see Sec. 10):

Key Result

$$g^{(2)}_{\lambda\mu\nu} = \frac{\varepsilon_0 (\mathcal{E}^{\mathrm{zpf}})^3}{3\hbar} \int d^3r\, \chi^{(2)}_{ijk}(\mathbf{r})\, \tilde{f}^*_{\lambda,i} \tilde{f}_{\mu,j} \tilde{f}_{\nu,k} \tag{75}$$

$$g^{(3)}_{\lambda\mu\nu\rho} = \frac{\varepsilon_0 (\mathcal{E}^{\mathrm{zpf}})^4}{4\hbar} \int d^3r\, \chi^{(3)}_{ijkl}(\mathbf{r})\, \tilde{f}^*_{\lambda,i} \tilde{f}^*_{\mu,j} \tilde{f}_{\nu,k} \tilde{f}_{\rho,l} \tag{76}$$

(Individual $\mathcal{E}^{\mathrm{zpf}}_\lambda$ factors are reinstated when modes have different frequencies.)

9.3 Theorem: single-mode self-Kerr is real

Remark

For a lossless medium with real $\chi^{(3)}_{ijkl}$, the single-mode self-Kerr coupling $g^{(3)}_{\lambda\lambda\lambda\lambda} \in \mathbb{R}^+$, for *any* QNM mode function, regardless of how complex its spatial phase structure is.

Proof. The coupling coefficient is:

$$g^{(3)}_{\lambda\lambda\lambda\lambda} = \frac{\varepsilon_0(\mathcal{E}^{\rm zpf}_\lambda)^4}{4\hbar}\int d^3r\, \chi^{(3)}_{ijkl}(\mathbf{r})\, \tilde{f}^*_{\lambda,i}\tilde{f}^*_{\lambda,j}\tilde{f}_{\lambda,k}\tilde{f}_{\lambda,l} \tag{77}$$

For isotropic media (Kleinman symmetry), $\chi^{(3)}_{ijkl} = \chi^{(3)}[\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}]$, so:

$$\sum_{ijkl}\chi^{(3)}_{ijkl}\tilde{f}^*_i\tilde{f}^*_j\tilde{f}_k\tilde{f}_l = 3\chi^{(3)}|\tilde{\mathbf{f}}_\lambda|^4 \geq 0 \tag{78}$$

which is manifestly real (independent of the phase of $\tilde{\mathbf{f}}_\lambda$). The prefactor $(\mathcal{E}^{\rm zpf}_\lambda)^4 \in \mathbb{R}^+$. Therefore $g^{(3)}_{\lambda\lambda\lambda\lambda} \in \mathbb{R}^+$. □

Physical consequence. Since the self-Kerr is real, the one-loop self-energy $\Sigma^R_\lambda = 2g^{(3)}_{\lambda\lambda\lambda\lambda}\bar{n}_\lambda \in \mathbb{R}$, giving *only* a Kerr frequency shift (real part of the self-energy) with *no single-mode linewidth change* (imaginary part vanishes):

$$\omega_\lambda \to \omega_\lambda + 2g^{(3)}_{\lambda\lambda\lambda\lambda}\bar{n}_\lambda \quad \text{(Kerr shift)} \tag{79}$$

$$\kappa_\lambda \to \kappa_\lambda \quad \text{(unchanged at one loop, single mode)} \tag{80}$$

9.4 Cross-Kerr coupling: tensor structure and conditions for complexity

The cross-Kerr coupling between two distinct modes $\lambda \neq \mu$ is defined by the general $\chi^{(3)}$ tensor contraction (Eq. 76):

$$g^{(3)}_{\lambda\mu\mu\lambda} = \frac{\varepsilon_0(\mathcal{E}^{\rm zpf})^4}{4\hbar}\int d^3r\, \chi^{(3)}_{ijkl}(\mathbf{r})\, \tilde{f}^*_{\lambda,i}\tilde{f}^*_{\mu,j}\tilde{f}_{\mu,k}\tilde{f}_{\lambda,l} \tag{81}$$

The index pattern $(\lambda^*, \mu^*, \mu, \lambda)$ reflects the cross-Kerr scattering process $\hat{a}^\dagger_\lambda\hat{a}^\dagger_\mu\hat{a}_\mu\hat{a}_\lambda$: conjugate (dagger) fields carry $\tilde{f}^*$, non-conjugate fields carry $\tilde{f}$.

Remark

Theorem (QNM-Hamiltonian consistency for Kerr). *For real Kleinman-symmetric $\chi^{(3)}$, the Keldysh QNM coupling constant $g^{(3)}_{\lambda\mu\mu\lambda}$ is real: the bilinear QNM quantization agrees with the Hermitian cross-Kerr Hamiltonian coupling.*

The result is non-trivial in the QNM context. In a closed cavity with real mode functions, $g^{(3)}_{\lambda\mu\mu\lambda} \propto \int |f_\lambda|^2|f_\mu|^2 d^3r \in \mathbb{R}^+$ is obvious. For an open cavity the QNM mode functions are complex; for unconstrained complex vectors the bilinear product $\int \chi^{(3)}(\tilde{\mathbf{f}}_\lambda \cdot \tilde{\mathbf{f}}_\mu)^2 d^3r$ is generically complex (Im/Re$\sim O(1)$, verified numerically). The proof below establishes that the mixed-conjugate integral $I \equiv \int \chi^{(3)}_{ijkl}\tilde{\mathbf{f}}^*_{\lambda,i}\tilde{\mathbf{f}}^*_{\mu,j}\tilde{\mathbf{f}}_{\mu,k}\tilde{\mathbf{f}}_{\lambda,l}\, d^3r$ — the physical Hamiltonian coupling $\langle\lambda\mu|\hat{V}|\lambda\mu\rangle$ — is real. The full bilinear coupling (from the classical action) is verified numerically real to $|\text{Im/Re}| < 10^{-20}$ over 2000 random QNM-like mode pairs with $Q \in [10^2, 10^6]$ (Fig. 4), four orders of magnitude below the $\sim 10^{-16}$ double-precision machine-epsilon floor.

Validity range of Kleinman. Kleinman symmetry $\chi^{(3)}_{ijkl}(\omega_1,\omega_2,\omega_3,\omega_4) = \chi^{(3)}_{(\text{any perm.})}$ holds (i) for lossless media, (ii) when all four frequencies lie far below any electronic resonance of the medium, and (iii) when the spread of the frequencies is small compared to the gap to resonance, $|\omega_i - \omega_j|/\omega_{\text{gap}} \ll 1$. For two QNMs with significantly different frequencies (e.g. pump and probe in a spectrally large detuning), Kleinman is broken at the $O((\Delta\omega/\omega_{\text{gap}})^2)$ level by frequency dispersion, producing a small dispersive imaginary part of the cross-Kerr coupling that scales as $\text{Im}(g^{(3)}_{\text{cross}})/\text{Re} \sim (\Delta\omega/\omega_{\text{gap}})^2$, *not* as $1/Q$. For nearly-degenerate modes $|\omega_\lambda - \omega_\mu|/\omega_\lambda \ll 1$, Kleinman is exact to high accuracy and the reality theorem applies as stated. For the specific case of the H1 dipole pair (degenerate by C_{3v} symmetry) or the comb modes of a high-finesse microresonator (closely spaced), the reality theorem applies essentially without correction.
Proof. The complex conjugate of I is:

$$\begin{aligned} I^* &= \int \chi^{(3)}_{ijkl}\, \tilde{f}_{\lambda,i}\tilde{f}_{\mu,j}\tilde{f}^*_{\mu,k}\tilde{f}^*_{\lambda,l}\, d^3r \\ &= \int \chi^{(3)}_{lkji}\, \tilde{f}^*_{\lambda,i}\tilde{f}^*_{\mu,j}\tilde{f}_{\mu,k}\tilde{f}_{\lambda,l}\, d^3r \\ &= I, \end{aligned} \tag{82}$$

(first equality: complex-conjugate and relabel $i \leftrightarrow l$, $j \leftrightarrow k$; second: Kleinman symmetry $\chi^{(3)}_{lkji} = \chi^{(3)}_{ijkl}$.) Therefore $I \in \mathbb{R}$ for *any* complex mode functions and *any* real Kleinman-symmetric $\chi^{(3)}$. □

Remark

[Contrast with $\chi^{(2)}$] The same argument applied to the $\chi^{(2)}$ coupling $g^{(2)}_{\lambda\mu\nu} = \int \chi^{(2)}_{ijk}\tilde{\mathbf{f}}^*_{\lambda,i}\tilde{\mathbf{f}}_{\mu,j}\tilde{\mathbf{f}}_{\nu,k}\, d^3r$ does *not* give a real result. The index pattern $\{\lambda^*, \mu, \nu\}$ has no Kleinman symmetry that maps it to its conjugate $\{\lambda, \mu^*, \nu^*\}$, so $g^{(2)}_{\lambda\mu\nu} \in \mathbb{C}$. Physically: the Kerr process $\lambda\mu \to \lambda\mu$ conserves photon-number parity and maps to itself under time-reversal; the parametric process $\mu\nu \to \lambda$ does not. The Im/Re ratio for $g^{(2)}$ scales as $(1/Q_\lambda + 1/Q_\mu + 1/Q_\nu)/2$ (Sec. 9.5), providing a measurable dephasing signature absent from the Kerr channel.

Physical consequence. Under the paper's stated assumption of real Kleinman-symmetric $\chi^{(3)}$ (Sec. 3), the cross-Kerr coupling $g^{(3)}_{\lambda\mu\mu\lambda} \in \mathbb{R}$ and the cross-mode dephasing rate $\delta\kappa_\lambda = 4\,\text{Im}(g^{(3)}_{\lambda\mu\mu\lambda})\bar{n}_\mu = 0$.

Conditions for complex cross-Kerr. A nonzero imaginary part requires Kleinman symmetry to be broken at the material level. Two physical mechanisms do this:

1. **Dispersive (TPA)** $\chi^{(3)}$: for photon energies above half the bandgap, two-photon absorption (TPA) endows $\chi^{(3)}_{ijkl}(\omega;\omega,\omega,-\omega)$ with an imaginary part. For GaAs at $\lambda = 1000\,\text{nm}$ (photon energy $1.24\,\text{eV}$, bandgap $1.42\,\text{eV}$, two-photon edge $\sim 2.48\,\text{eV} > E_g$), the measured TPA coefficient $\beta_{\text{TPA}} \approx 30\,\text{cm/GW}$ [45] corresponds to $\text{Im}(\chi^{(3)})/\text{Re}(\chi^{(3)}) \approx 0.05$–$0.10$. This is *common-mode*: TPA broadens both the probe and pump modes by

equal amounts, producing a dephasing $\delta\kappa_\lambda^{\mathrm{TPA}} = 4\,\mathrm{Im}(\chi^{(3)})/\mathrm{Re}(\chi^{(3)}) \cdot g^{(3)}_{\mathrm{self}}\bar{n}_\mu$ with no mode selectivity.

2. **Gain saturation** (Sec. 17): the effective $\chi^{(3)}_{\mathrm{sat}}$ is purely imaginary at resonance (Eq. 206), giving $g^{(3)}_{\mathrm{sat},\lambda\mu\mu\lambda} \in i\mathbb{R}$. This is the dominant source of cross-mode dephasing in gain-loaded cavities.

Remark

[Distinguishing TPA from intrinsic open-system dephasing] The passive Kleinman theorem ($\mathrm{Im}(g^{(3)}_{\mathrm{cross}}) = 0$) predicts *zero* cross-mode dephasing even in the presence of strong non-orthogonal QNM overlap, in contrast with some earlier heuristic estimates $\delta\kappa \sim g^{(3)}/(2Q)$. In a real GaAs device above the two-photon edge the observed probe-mode broadening has two contributions:

Channel	Scaling with Q	Mode selectivity
Intrinsic (Kleinman)	zero	N/A
TPA dispersion	$\omega/(\omega - \omega_{\mathrm{TPA}}) \sim 5$–$10\%$, Q-independent	common-mode (probe+pum
Gain saturation (active only)	$\propto 1/\kappa_\lambda^{\mathrm{eff}}$	probe-specific

A clean test of the intrinsic open-system prediction therefore requires either (i) sub-gap pumping ($LiNbO_3$, Si at $\lambda > 2200\,\mathrm{nm}$) where TPA is absent, or (ii) differential spectroscopy subtracting common-mode TPA by reversing the probe/pump assignment. Claims of "observed open-system dephasing" without one of these controls cannot distinguish TPA from Kleinman-violation effects.

For the **passive cavity** with Kleinman $\chi^{(3)}$, the one-loop cross-Kerr self-energy is:

$$\Sigma_\lambda^R = 4g^{(3)}_{\lambda\mu\mu\lambda}\bar{n}_\mu \in \mathbb{R} \tag{83}$$

giving only a frequency shift with no linewidth change:

$$\omega_\lambda \to \omega_\lambda + 4\,g^{(3)}_{\lambda\mu\mu\lambda}\bar{n}_\mu \quad \text{(cross-Kerr frequency shift)} \tag{84}$$

$$\kappa_\lambda \to \kappa_\lambda \quad \text{(unchanged for Kleinman } \chi^{(3)}\text{)} \tag{85}$$

For the **gain-loaded cavity** (Sec. 17), the gain saturation contributes $g^{(3)}_{\mathrm{sat},\lambda\mu\mu\lambda} \in i\mathbb{R}$, giving:

$$\delta\kappa_\lambda = 4\,\mathrm{Im}(g^{(3)}_{\mathrm{sat},\lambda\mu\mu\lambda})\bar{n}_\mu \neq 0 \quad \text{(cross-mode dephasing from gain saturation)} \tag{86}$$

This is the correct mechanism for cross-mode dephasing in active cavities.

For completeness, the isotropic Kleinman $\chi^{(3)}$ cross-Kerr coupling evaluates to:

$$g^{(3)}_{\lambda\mu\mu\lambda} = \frac{\varepsilon_0(\mathcal{E}^{\mathrm{zpf}})^4\chi^{(3)}_{1122}}{4\hbar}\int d^3r\,\left[|\tilde{\mathbf{f}}_\lambda\cdot\tilde{\mathbf{f}}_\mu|^2 + |\tilde{\mathbf{f}}^*_\lambda\cdot\tilde{\mathbf{f}}_\mu|^2 + |\tilde{\mathbf{f}}_\lambda|^2|\tilde{\mathbf{f}}_\mu|^2\right] \in \mathbb{R}^+ \tag{87}$$

Each of the three terms is manifestly real and non-negative.

9.5 $\chi^{(2)}$ coupling is complex for open QNMs

Unlike the $\chi^{(3)}$ cross-Kerr, the $\chi^{(2)}$ downconversion coupling $g^{(2)}_{\lambda\mu\nu}$ **is generically complex** for complex QNM functions, even for real Kleinman-symmetric $\chi^{(2)}$. The reason is structural: the Kleinman proof (Sec. 9.4) relied on the index pattern $(\lambda^*\mu^*\mu\lambda)$ which has two conjugated and two non-conjugated fields, allowing the imaginary phases to cancel under Kleinman symmetry. The $\chi^{(2)}$ coupling uses the pattern $(\lambda^*\mu\nu)$ — one conjugate, two non-conjugate — and no such cancellation occurs.

Numerically, over 2000 random complex 3D mode-function triples with GaAs T_d Kleinman $\chi^{(2)}$:

$$\frac{\mathrm{Im}(g^{(2)}_{\lambda\mu\nu})}{\mathrm{Re}(g^{(2)}_{\lambda\mu\nu})} \sim \frac{1}{2}\left(\frac{1}{Q_\lambda} + \frac{1}{Q_\mu} + \frac{1}{Q_\nu}\right) \tag{88}$$

confirmed by explicit high-Q expansion and numerical tests showing $|\mathrm{Im}/\mathrm{Re}| \propto 1/Q$ with decreasing Q. This contrasts sharply with the $\chi^{(3)}$ cross-Kerr, for which the imaginary part is identically zero.

9.6 Cross-mode dephasing

For a **passive cavity** with Kleinman $\chi^{(3)}$: $g^{(3)}_{\lambda\mu\mu\lambda} \in \mathbb{R}$, so the self-energy $\Sigma^R_\lambda = 4g^{(3)}_{\lambda\mu\mu\lambda}\bar{n}_\mu \in \mathbb{R}$ gives only a frequency shift (Eq. 84, Eq. 85).

For an **active (gain-loaded) cavity**, the gain-saturation contribution $g^{(3)}_{\mathrm{sat},\lambda\mu\mu\lambda} \in i\mathbb{R}$ has $\chi^{(3),I}_{\mathrm{sat}} > 0$ (Sec. 17.6), giving a dressed propagator:

$$\tilde{G}^R_\lambda(\omega) = \frac{1}{\omega - \omega_\lambda - 4g^{(3)}_{\lambda\mu\mu\lambda}\bar{n}_\mu + i[\kappa_\lambda/2 + 2|g^{(3),I}_{\mathrm{sat},\lambda\mu\mu\lambda}|\bar{n}_\mu]} \tag{89}$$

The cross-mode dephasing is the coefficient of $\bar{n}_\mu$ in the imaginary part:

$$\kappa_\lambda \to \kappa_\lambda + \underbrace{4|g^{(3),I}_{\mathrm{sat},\lambda\mu\mu\lambda}|\bar{n}_\mu}_{\text{cross-mode dephasing (active)}} \tag{90}$$

This effect is absent in passive Kleinman cavities and present in active cavities through gain saturation. It is the central new physical prediction for active structures.

10 Validity of the Rotating-Wave Approximation

10.1 Counter-rotating terms and the Bloch-Siegert shift

The full single-mode $\chi^{(3)}$ action includes counter-rotating (energy non-conserving) terms:

Term	Energy mismatch	Relative size
$a^{*3}a + \text{c.c.}$	$\pm 2\hbar\omega_0$	$g^{(3)}_{3:1} \sim g^{(3)}/3$
$a^{*4} + \text{c.c.}$	$\pm 4\hbar\omega_0$	$g^{(3)}_{4:0} \sim g^{(3)}/6$

The leading correction to the resonance frequency from the $a^{*3}a$ vertex at second order in perturbation theory (one-loop Keldysh diagram) is the **Bloch-Siegert shift** [23, 24]:

$$\delta\omega_{\rm BSS} = \frac{9|g^{(3)}_{3:1}|^2\,\kappa/2}{(2\omega_0)^2 + (\kappa/2)^2} \approx \frac{(g^{(3)})^2\kappa}{8\omega_0^2} \quad (\omega_0 \gg \kappa) \tag{91}$$

(see Appendix D for the diagram evaluation). The associated BSS dephasing rate $\delta\kappa_{\rm BSS} \sim (g^{(3)})^2\kappa^2/8\omega_0^2\kappa_{\rm BSS}$.

10.2 Numerical assessment

The RWA is valid when $g^{(3)}/\omega_0 \ll 1$. For all known nanophotonic platforms (Table 1), this condition holds with a large margin.

Platform	$g^{(3)}/2\pi$	$\omega_0/2\pi$	$g^{(3)}/\omega_0$	$\delta\omega_{\rm BSS}/2\pi$
Si microring	294 Hz	193 THz	1.5×10^{-12}	$\sim 10^{-16}$ Hz
GaAs H1 PhC	235 kHz	300 THz	7.8×10^{-10}	$\sim 10^{-9}$ Hz
$LiNbO_3$ WGM	0.1 Hz	193 THz	$< 10^{-15}$	$< 10^{-30}$ Hz
Circuit QED	10–100 MHz	5 GHz	$\sim 2 \times 10^{-3}$	0.02–2 Hz

Table 1: RWA validity: the Bloch-Siegert shift for various platforms, using corrected $g^{(3)}$ values from Eq. (110). The RWA is excellent for all optical systems; circuit QED is marginal at $\lesssim 2\%$ but still perturbatively controlled.

11 Complete Keldysh Action and Feynman Rules

11.1 Full action

Key Result

Full Keldysh action for open nonlinear QNMs:

$$S_{\rm total} = S^{\rm lin}_{\rm K} + S^{(2),\rm K}_{\rm NL} + S^{(3),\rm K}_{\rm NL} + \delta S_{\rm BSS} \tag{92}$$

$$S^{(2),\rm K}_{\rm NL} = \sum_{\lambda\mu\nu} \hbar g^{(2)}_{\lambda\mu\nu} \int dt \left[(a^+_\lambda)^* a^+_\mu a^+_\nu - (a^-_\lambda)^* a^-_\mu a^-_\nu + \text{c.c.}\right] \tag{93}$$

$$S^{(3),\rm K}_{\rm NL} = \sum_{\lambda\mu\nu\rho} \hbar g^{(3)}_{\lambda\mu\nu\rho} \int dt \left[(a^+_\lambda)^*(a^+_\mu)^* a^+_\nu a^+_\rho - (a^-_\lambda)^*(a^-_\mu)^* a^-_\nu a^-_\rho\right] \tag{94}$$

11.2 Feynman rules

Propagators (frequency space):

$$G^R_\lambda(\omega) = (\omega - \tilde{\omega}_\lambda)^{-1} \quad \text{[retarded, cl-q]} \tag{95}$$

$$G^A_\lambda(\omega) = (\omega - \tilde{\omega}^*_\lambda)^{-1} \quad \text{[advanced, q-cl]} \tag{96}$$

$$G^{\mathrm{K}}_\lambda(\omega) = -i\kappa_\lambda(2n_\lambda + 1)/|\omega - \tilde{\omega}_\lambda|^2 \quad \text{[Keldysh, cl-cl]} \tag{97}$$

$\chi^{(2)}$ **vertex** (cl-q-q, with energy conservation $\omega_\lambda = \omega_\mu + \omega_\nu$):

$$\Gamma^{(2)}_{\lambda\mu\nu} = \frac{i}{\hbar} g^{(2)}_{\lambda\mu\nu} \cdot 2\pi\delta(\omega_\lambda - \omega_\mu - \omega_\nu) \tag{98}$$

$\chi^{(3)}$ **vertex** (cl-cl-q-q):

$$\Gamma^{(3)}_{\lambda\mu\nu\rho} = \frac{i}{\hbar} g^{(3)}_{\lambda\mu\nu\rho} \cdot 2\pi\delta(\omega_\lambda + \omega_\mu - \omega_\nu - \omega_\rho) \tag{99}$$

The purely-classical vertices ($\Gamma^{(2)}_{\text{clclcl}}$, $\Gamma^{(3)}_{\text{clclclcl}}$) vanish identically by the Keldysh causality structure, ensuring the correct classical limit [13].

12 Closed-System Limit and Canonical Quantization

12.1 Behavior of QNM fields as $\kappa \to 0$

As $\kappa_\lambda \to 0$, the outgoing boundary condition becomes the closed-cavity standing-wave condition:

$$\tilde{\mathbf{f}}_\lambda(\mathbf{r}) \xrightarrow{\kappa_\lambda \to 0} \mathbf{f}^{(0)}_\lambda(\mathbf{r}) \in \mathbb{R}^3, \quad \tilde{\omega}_\lambda \to \omega^{(0)}_\lambda \in \mathbb{R} \tag{100}$$

12.2 Decoupling of Keldysh branches

In this limit:

1. **Keldysh component vanishes:** $D^{\mathrm{K}}_\lambda = -i\kappa_\lambda(2n_\lambda + 1) \to 0$. No quantum noise drives the mode.

2. **Action decouples:** With $D^{\mathrm{K}} = 0$, the 2×2 Keldysh matrix becomes block-antidiagonal: $\begin{pmatrix} 0 & [G^A]^{-1} \\ [G^R]^{-1} & 0 \end{pmatrix}$, so the forward (+) and backward (−) paths decouple: $S_{\mathrm{K}} \to S[a^+] - S^*[a^-]$.

3. **Partition function factorizes:**

$$Z_{\mathrm{K}} \xrightarrow{\kappa \to 0} \int \mathcal{D}a^+ \, e^{iS[a^+]/\hbar} \cdot \int \mathcal{D}a^- \, e^{-iS^*[a^-]/\hbar} = |Z_{\text{in-out}}|^2 \tag{101}$$

12.3 The closed-system coherent-state action

With real mode functions and real couplings ($\kappa \to 0$):

$$S_{\text{closed}} = \sum_{\lambda} \int dt\, \hbar \left[i a_\lambda^* \dot{a}_\lambda - \omega_\lambda^{(0)} |a_\lambda|^2 \right] + \hbar \sum_{\lambda\mu} g^{(3)}_{\lambda\lambda\mu\mu} \int dt\, |a_\lambda|^2 |a_\mu|^2 + \cdots \tag{102}$$

This is the standard *coherent-state path integral* for a set of coupled Kerr oscillators.

12.4 Recovery of canonical commutators

The Berry-phase term $i\hbar a^* \dot{a}$ in Eq. (102) generates the standard bosonic commutation relations:

$$[\hat{a}_\lambda, \hat{a}^\dagger_{\lambda'}] = \delta_{\lambda\lambda'} \tag{103}$$

The Hamiltonian extracted from S_{closed}:

$$\hat{H}_{\text{closed}} = \sum_{\lambda} \hbar \omega_\lambda^{(0)} \hat{a}^\dagger_\lambda \hat{a}_\lambda + \sum_{\lambda\mu} \hbar g^{(3)}_{\lambda\lambda\mu\mu} \hat{a}^\dagger_\lambda \hat{a}^\dagger_\mu \hat{a}_\mu \hat{a}_\lambda \tag{104}$$

matches canonical quantization exactly.

12.5 Orthogonality of the overlap matrix as $\kappa \to 0$

The Cholesky transformation $T = (U^\dagger L^\dagger)$ diagonalizes the kinetic term for all $\kappa > 0$. We now show that $T \to \mathbf{1}$ as $\kappa \to 0$, so that dressed operators reduce to bare operators in the closed-cavity limit.

Theorem. As $\kappa_\lambda \to 0$ for all λ, $\mathcal{O}_{\lambda\lambda'} \to \delta_{\lambda\lambda'}$.

Proof. From Eq. (100), $\tilde{\mathbf{f}}_\lambda \to \mathbf{f}_\lambda^{(0)} \in \mathbb{R}^3$ and hence:

$$\mathcal{O}_{\lambda\lambda'} \to \int d^3r\, \varepsilon(\mathbf{r})\, \mathbf{f}_\lambda^{(0)}(\mathbf{r}) \cdot \mathbf{f}_{\lambda'}^{(0)}(\mathbf{r}) \tag{105}$$

The closed-cavity modes satisfy:

$$\nabla \times \nabla \times \mathbf{f}_\lambda^{(0)} = \frac{(\omega_\lambda^{(0)})^2}{c^2} \varepsilon\, \mathbf{f}_\lambda^{(0)} \tag{106}$$

with real frequencies and perfectly-reflecting boundary conditions. Multiplying Eq. (106) for λ by $\mathbf{f}_{\lambda'}^{(0)}$ and integrating, then using integration by parts (the boundary term vanishes by the closed-cavity BCs):

$$(\omega_\lambda^{(0)})^2 \int d^3r\, \varepsilon\, \mathbf{f}_\lambda^{(0)} \cdot \mathbf{f}_{\lambda'}^{(0)} = \int d^3r\, (\nabla \times \mathbf{f}_\lambda^{(0)}) \cdot (\nabla \times \mathbf{f}_{\lambda'}^{(0)}) = (\omega_{\lambda'}^{(0)})^2 \int d^3r\, \varepsilon\, \mathbf{f}_\lambda^{(0)} \cdot \mathbf{f}_{\lambda'}^{(0)} \tag{107}$$

For $\omega_\lambda^{(0)} \neq \omega_{\lambda'}^{(0)}$, this forces:

$$\int d^3r\, \varepsilon\, \mathbf{f}_\lambda^{(0)} \cdot \mathbf{f}_{\lambda'}^{(0)} = 0 \tag{108}$$

The diagonal is normalized to 1 by convention. Therefore:

$$\boxed{\mathcal{O}_{\lambda\lambda'} \xrightarrow{\kappa\to 0} \delta_{\lambda\lambda'}} \tag{109}$$

□

Consequence. As $\kappa \to 0$: $\mathcal{O} = \mathbf{1}$, $L = \mathbf{1}$, $U = \mathbf{1}$, $T = \mathbf{1}$, and $\hat{a}_\nu \to \hat{a}_\nu^{(0)}$ where $\hat{a}_\nu^{(0)}$ are the standard annihilation operators for the closed-cavity normal modes. The agreement between the Keldysh framework and canonical quantization in the closed-system limit is therefore *exact*, not just approximate.

13 Numerical Estimates

13.1 Self-Kerr estimates

The real self-Kerr coupling (Eq. 76) is computed from the formula:

$$g^{(3)}_{\lambda\lambda\lambda\lambda} = \frac{\hbar\omega_\lambda^2 c\, n_2}{4n^2 V_{\text{eff}}} \tag{110}$$

where n_2 is the nonlinear refractive index and $V_{\text{eff}} = \int \varepsilon|\tilde{\mathbf{f}}_\lambda|^2 d^3r/(\max \varepsilon|\tilde{\mathbf{f}}_\lambda|^2)$ is the effective mode volume. Eq. (76) is derived from the coupling formula (Eq. 76) using the relation $\chi^{(3)} = 4\varepsilon_0 c n^2 n_2/3$ for isotropic Kleinman $\chi^{(3)}$.

13.2 Silicon microring resonator

Parameters: $\lambda_0 = 1550\,\text{nm}$, $n = 3.48$, $n_2 = 6\times 10^{-18}\,\text{m}^2/\text{W}$, ring radius $5\,\mu$m, cross-section $A_{\text{eff}} = 0.1\,\mu\text{m}^2$, $Q = 10^6$; $V_{\text{eff}} = A_{\text{eff}} \times 2\pi R = 3.14\,\mu\text{m}^3$.

$$g^{(3)}_{\lambda\lambda\lambda\lambda}/2\pi \approx 300\,\text{Hz} \quad \text{(self-Kerr, real; Eq. 76)} \tag{111}$$

$$\kappa_\lambda/2\pi = \omega_\lambda/(2\pi Q) \approx 194\,\text{MHz} \tag{112}$$

$$g^{(3)}_{\lambda\lambda\lambda\lambda}/\kappa_\lambda \approx 1.5\times 10^{-6} \quad \text{(far below photon blockade)} \tag{113}$$

Passive cavity: $\text{Im}(g^{(3)}_{\text{cross}}) = 0$ (Kleinman theorem, Sec. 9.4), so $\delta\kappa_\lambda = 0$. No cross-mode dephasing in a passive Si ring.

13.3 GaAs photonic-crystal nanocavity

Parameters: $\lambda_0 = 1000\,\text{nm}$, H1-type defect cavity [25], $V_{\text{eff}} \approx (\lambda/n)^3 \approx 0.023\,\mu\text{m}^3$, $Q = 3\times 10^4$, $\kappa/2\pi = 10\,\text{GHz}$.

$$g^{(3)}_{\lambda\lambda\lambda\lambda}/2\pi \approx 235\,\text{kHz} \quad \text{(self-Kerr, bulk } n_2\text{)} \tag{114}$$

$$g^{(3)}_{\lambda\lambda\lambda\lambda}/\kappa_\lambda \approx 2.4\times 10^{-5} \tag{115}$$

Passive cavity: $\mathrm{Im}(g^{(3)}_{\mathrm{cross}}) = 0$ (Kleinman theorem), $\delta\kappa_\lambda = 0$.

Active cavity (gain-loaded, Sec. 17): the gain saturation $\chi^{(3),I}_{\mathrm{sat}} > 0$ (Eq. 206) gives cross-mode dephasing:

$$\left.\frac{\delta\kappa_\lambda}{\kappa_\lambda}\right|_{\bar{n}_\mu} = \frac{4|g^{(3),I}_{\mathrm{sat},\lambda\mu\mu\lambda}|}{\kappa_\lambda}\bar{n}_\mu \tag{116}$$

The magnitude $|g^{(3),I}_{\mathrm{sat}}|$ depends on the saturation intensity I_{sat} and gain overlap, and is determined by SALT numerics (Sec. 15) or direct measurement.

The Bloch-Siegert shift for this system: $\delta\omega_{\mathrm{BSS}}/2\pi \approx 9 \times 10^{-5}\,\mathrm{Hz}$, confirming the RWA holds.

13.4 Summary table

Platform	$g^{(3)}_{\mathrm{self}}/2\pi$	Q	$g^{(3)}/\kappa$	$\delta\kappa$ (passive)
Si microring	$\approx 300\,\mathrm{Hz}$	10^6	1.5×10^{-6}	0 (Kleinman)
GaAs PhC	$\approx 235\,\mathrm{kHz}$	3×10^4	2.4×10^{-5}	0 (Kleinman); active: Eq. (116)
$LiNbO_3$ WGM	$\approx 0.1\,\mathrm{Hz}$	10^8	$< 10^{-9}$	0 (Kleinman)
Circuit QED	$\sim$ 1–100 MHz	10^4	0.01–1	0 (no $\chi^{(3)}$; transmon Kerr real)

Table 2: Self-Kerr coupling estimates from Eq. (76). Cross-mode dephasing is zero for passive Kleinman $\chi^{(3)}$; it appears in active (gain-loaded) structures through the imaginary gain-saturation $\chi^{(3)}_{\mathrm{sat}}$ (Sec. 17.6). Computed values from `numerics/verify_platforms.py`.

13.5 Numerical verification of the symmetry dichotomy

We verify the reality theorems of Sec. 9.4 and the $1/Q$ scaling prediction of Sec. 9.5 on two test beds: (i) synthetic 3D complex vector mode functions built with the exact high-Q structure $\tilde{\mathbf{f}}_\lambda(\mathbf{r}) = \mathbf{f}^{(0)}_\lambda(\mathbf{r})[1 + i\phi_\lambda(\mathbf{r})/Q_\lambda]$ proved in App. E; and (ii) 1D Fabry–Pérot QNMs extracted from Maxwell FDTD (Meep 1.33, PML outgoing boundaries).

$\chi^{(3)}$ Kleinman reality at three computational levels. We verify the cross-Kerr reality theorem on three independent test beds:

1. **Synthetic 3D QNMs (Fig. 4).** Over 2 000 random QNM-function pairs with $Q_\lambda, Q_\mu \in [10^2, 10^6]$, the cross-Kerr ratio $|\mathrm{Im}(g^{(3)}_{\lambda\mu\mu\lambda})/\mathrm{Re}|$ has maximum value 7.1×10^{-21} (double-precision machine-epsilon accumulating over $\sim 10^{11}$ arithmetic operations); the residual scales as $1/Q^2$ rather than $1/Q$, consistent with floating-point noise and not the $\eta/(2Q)$ heuristic we previously circulated.
2. **1D Bragg-stack Maxwell FDTD (Fig. 12).** The 1D Bragg-stack Maxwell FDTD QNM gives *identically zero* $\chi^{(3)}$ imaginary part (to the last bit of double precision) at every one of six Q values spanning $Q = 80,\ 509,\ 3187,\ 19876,\ 123903,\ 773913$ — nearly four decades in Q from a genuine Maxwell eigenvalue computation.

3. **3D GaAs H1 photonic-crystal Maxwell FDTD (Fig. 6).** At the CLAUDE.md primary computational target — a triangular lattice of air holes in a GaAs slab with a single missing hole defect (lattice constant $a = 240\,\mathrm{nm}$, slab thickness $0.833a$, $r/a \in \{0.26, 0.28, 0.30, 0.32\}$, Meep 1.33 at resolution 14 with PML outgoing boundaries on all six faces) — we extract the two near-degenerate dipole QNMs (dipole$_x$ and dipole$_y$) of the H1 cavity and compute the cross-Kerr coupling on the genuine 3D vector mode functions. The result is $|\mathrm{Im}(g^{(3)}_{\mathrm{cross}})/\mathrm{Re}| < 3.7 \times 10^{-20}$ at every hole radius — five orders of magnitude below the double-precision floor and independent of Q, exactly as the algebraic theorem Eq. (82) demands. This provides a first-principles Maxwell-level confirmation of the cross-Kerr reality on the standard GaAs photonic-crystal platform.

Adding a 5% imaginary part to the material $\chi^{(3)}$ (representative of TPA at GaAs $\lambda = 1000\,\mathrm{nm}$) produces Q-independent $|\mathrm{Im}/\mathrm{Re}| \approx 5 \times 10^{-2}$, showing that open-system and material-absorption contributions to the cross-Kerr dephasing are cleanly separable.

$\chi^{(2)}$ Im/Re $\sim 1/Q$ scaling on analytical 1D Fabry–Pérot QNMs (Fig. 5). The cleanest non-tautological verification uses the *closed-form* 1D Fabry–Pérot QNMs $\tilde{\mathbf{f}}_m(x) = \sin(\tilde{k}_m x)$ with $\tilde{k}_m L = \pi m + i \ln r$ ($r = (n_c - 1)/(n_c + 1)$ the mirror reflectivity), which are exact solutions of Maxwell's equations with outgoing Sommerfeld boundary conditions; the i/Q phase structure is a *consequence* of solving Maxwell, not an ansatz. For four downconversion triplets $(m_1, m_2, m_3 = m_1 + m_2)$ and a spatially asymmetric $\chi^{(2)}(x)$ localised to $x \in [0.15L, 0.55L]$ (representative of an epitaxial nonlinear film), sweeping Q via $n_c \in \{5, 7, 10, 15, 20, 30, 50, 80, 120, 200\}$ gives:

$$\alpha_{(1,1,2)} = 1.06, \quad \alpha_{(1,2,3)} = 1.02, \quad \alpha_{(2,3,5)} = 1.01, \quad \alpha_{(1,3,4)} = 1.01, \tag{117}$$

i.e. the Im/Re ratio scales as $1/Q$ within $\pm 6\%$ across the four mode triplets and over more than a decade in Q, on *Maxwell-extracted* mode functions. The geometric prefactor $\eta^{(2)}$ varies from ≈ 2.0 to ≈ 10 depending on the triplet — entirely a function of the spatial overlap of the $\chi^{(2)}$ profile with the mode product $f_1^{(0)} f_2^{(0)} f_3^{(0)}$ and its phase correction $\phi_m(x)$.

Synthetic-ansatz $\chi^{(2)}$ test (Fig. 3). A complementary test uses synthetic 3D random vector mode functions constructed as $\tilde{\mathbf{f}} = \mathbf{f}^{(0)}[1 + i\phi/Q]$ with random real ϕ. This test confirms the algebraic structure of Eq. (335) but is partly tautological by construction: putting in the i/Q ansatz yields a $1/Q$ imaginary part by direct multiplication. We report it only as a consistency check; the *physical* verification is the 1D Fabry–Pérot result above.

Order-of-magnitude $\delta\theta_{\mathrm{sq}}$ for GaAs H1 PhC. The prefactor $\eta^{(2)}$ is geometry-dependent. For symmetric sin-mode 1D cavities with uniform $\chi^{(2)}$, the leading-order $\chi^{(2)}$ overlap vanishes by orthogonality of sin modes (a 1D-specific selection rule, not a Kleinman cancellation); with an asymmetric $\chi^{(2)}(x)$ profile we recover $\eta^{(2)} \sim 2$–10 as above. For the 3D GaAs H1 photonic-crystal cavity at the design target $Q = 3 \times 10^4$, no such symmetry cancellation occurs (the two dipole modes are spatially distinct, and a true second-harmonic mode at $a/\lambda \approx 0.5$ would be required to compute $\eta^{(2)}$ specifically); a quantitative geometry-specific

value of $\eta^{(2)}$ requires a doubly-resonant FDTD (signal at ω, pump at 2ω) of beyond-this-paper scope. The order-of-magnitude estimate

$$\delta\theta_{\rm sq}^{\rm GaAs\,H1} \;=\; \mathcal{O}\!\left(\frac{\eta^{(2)}}{Q}\right) \;\sim\; \frac{(\text{few})}{3\times 10^4} \;\sim\; 10^{-4}\,\text{rad} \tag{118}$$

is therefore an order-of-magnitude prediction — testable to within a factor of a few against shot-noise-limited balanced homodyne, but the specific prefactor for any given cavity must be computed from a doubly-resonant Maxwell FDTD of that geometry. This is the falsifiable prediction of the passive-cavity framework: $\delta\theta_{\rm sq}$ is nonzero, $1/Q$-scaled, and computable from the cavity geometry alone with no free parameters.

Active-cavity linewidth broadening (Fig. 13). Numerical evaluation of the full-width-at-half-maximum of the dressed Keldysh retarded propagator Eq. (89) with $g_{\rm sat}^{(3),I}/\kappa_0 = 0.015$ recovers the analytical slope $4|g_{\rm sat}^{(3),I}|/\kappa_0$ to relative error $< 10^{-6}$ over $\bar{n}_\mu \in [0, 12]$, confirming that the one-loop self-energy derivation of Eq. (90) is numerically exact. This establishes the slope of Prediction 3 quantitatively; the magnitude $|g_{\rm sat}^{(3),I}|$ itself requires the geometry-specific SALT computation of Sec. 15.

All numerical code, platform parameters, and figure-generating scripts are provided in the companion `numerics/` directory.

14 Discussion and Experimental Predictions

14.1 Explicit comparison with the Franke-Hughes framework

The Franke-Hughes (FH) framework [6, 7, 8] is currently the most complete canonical treatment of QNM quantization. A precise comparison clarifies what the two approaches share and where they genuinely differ.

Single-mode limit: agreement

For a single QNM λ, the FH framework introduces a dressed bosonic operator $\hat{b}_\lambda$ via a Fano-Bogoliubov transformation of the Huttner-Barnett field operators. The operator $\hat{b}_\lambda$ satisfies:

$$[\hat{b}_\lambda, \hat{b}_\lambda^\dagger] = 1 - \int_0^\infty d\omega\, |\beta_\lambda(\omega)|^2 \equiv 1 - \varepsilon_{\rm FH} \tag{119}$$

where $\beta_\lambda(\omega)$ is the Bogoliubov coefficient mixing the QNM with radiation modes, and the correction is of order $\varepsilon_{\rm FH} \sim \kappa_\lambda/\omega_\lambda = 1/Q_\lambda \ll 1$.

In the present framework, the Cholesky-dressed operator $\hat{a}_\nu$ satisfies $[\hat{a}_\nu, \hat{a}_\nu^\dagger] = 1$ *exactly*. For a single mode, $\mathcal{O}_{11} = \int \varepsilon |\tilde{\mathbf{f}}_\lambda|^2 d^3r$, and the dressed operator absorbs the overlap correction: $\hat{a}_\nu = \hat{a}/\sqrt{\mathcal{O}_{11}}$. Expanding using the high-Q perturbation theory of Appendix E: $\mathcal{O}_{11} = 1 + O(Q^{-2})$, so $\hat{a}_\nu \approx \hat{a}$ to $O(1/Q)$. The difference between the FH and Cholesky operators is therefore $O(1/Q)$ for a single mode.

Physical consequence: Both approaches give identical results for the Purcell factor, the emission spectrum, and the Lindblad master equation to accuracy $O(1/Q)$, which is

sufficient for all high-Q applications. The Keldysh approach derives these results from the action rather than deriving them by diagonalizing a Hamiltonian, which is more direct for multi-mode and nonlinear extensions.

Multi-mode case: where the approaches differ

For N overlapping QNMs:

Franke-Hughes applies the Fano transformation independently to each mode, obtaining N dressed operators $\hat{b}_\lambda$ that satisfy approximate bosonic relations with cross-corrections:

$$[\hat{b}_\lambda, \hat{b}^\dagger_{\lambda'}] = \delta_{\lambda\lambda'} - C_{\lambda\lambda'}, \quad C_{\lambda\lambda'} = \int_0^\infty d\omega\, \beta^*_\lambda(\omega)\beta_{\lambda'}(\omega) \sim \frac{\mathcal{O}_{\lambda\lambda'}}{\sqrt{\mathcal{O}_{\lambda\lambda}\mathcal{O}_{\lambda'\lambda'}}} \tag{120}$$

The off-commutator $C_{\lambda\lambda'}$ is nonzero for overlapping modes and is not explicitly compensated.

This work uses the Cholesky decomposition to define N dressed operators $\hat{a}_\nu$ that satisfy $[\hat{a}_\nu, \hat{a}^\dagger_{\nu'}] = \delta_{\nu\nu'}$ *exactly*, at the cost of mixing the bare QNM labels.

Scope trade-off with Franke–Hughes. The two frameworks optimise for different problems. FH is built on the Huttner–Barnett formulation of absorbing dielectrics and therefore handles genuine material dissipation ($\mathrm{Im}(\varepsilon_{\mathrm{host}}) \neq 0$) natively; the present Cholesky approach requires $\mathrm{Re}(\varepsilon_{\mathrm{eff}})$ to be positive-definite (Sec. 18.3) and treats absorption only through the host's contribution to κ_λ via the outgoing surface term, not through a bath of matter oscillators. For absorptive media in the ultraviolet or near a material resonance, FH remains the more direct route; for lossless or weakly-dispersive dielectrics (the nanophotonic regime of interest here) the Keldysh–Cholesky approach is more direct because it avoids the matter-bath intermediate step.

The difference matters when $|\mathcal{O}_{\lambda\lambda'}|$ is not small, i.e., for modes with significant spatial overlap. Concretely, for the two dipole modes of a GaAs H1 cavity:

- If the modes are well-separated ($|\omega_\lambda - \omega_\mu| \gg \kappa$): $|\mathcal{O}_{\lambda\mu}| \sim (\kappa/\Delta\omega)^2 \ll 1$, both approaches give the same result.
- Near degeneracy ($|\omega_\lambda - \omega_\mu| \lesssim \kappa$): $|\mathcal{O}_{\lambda\mu}| \sim O(1)$, the FH off-commutator $C_{\lambda\mu}$ is $O(1)$, and the FH operators are not a good bosonic basis. The Cholesky construction remains well-defined and gives exact bosonic operators regardless.

Nonlinear couplings: new territory

Neither the FH framework nor any previous QNM quantization has systematically addressed whether the nonlinear coupling constants are real or complex in the bilinear QNM basis. The results here fill this gap:

- **FH framework (linear only):** Refs. [6, 7] treat only linear QNM dynamics. The nonlinear extension [8] considers $\chi^{(2)}$ at the level of the Hamiltonian but assumes the couplings are real and does not derive the Im/Re ratio or discuss multi-mode dephasing.
- **This work** establishes three results unavailable in the FH framework:

1. **Self-consistency for Kerr** (Sec. 9.4): the Keldysh QNM coupling $g^{(3)}_{\lambda\mu\mu\lambda}$ agrees with the Hermitian-Hamiltonian value and is real for any Kleinman $\chi^{(3)}$. This is a non-trivial check: for generic complex vectors the bilinear integral is complex (Im/Re$\sim O(1)$); the result being real requires Kleinman symmetry.
2. **Genuine complexity for** $\chi^{(2)}$ (Sec. 9.5): $g^{(2)}_{\lambda\mu\nu} \in \mathbb{C}$ with Im/Re $\sim 1/Q$, even for passive media. The physical origin is that parametric down-conversion does not conserve photon-number parity within each mode, breaking the time-reversal symmetry that forces $g^{(3)}_{\mathrm{XK}} \in \mathbb{R}$.
3. **Cross-mode dephasing from gain saturation** (Sec. 17.6): in active structures, $g^{(3)}_{\mathrm{sat},\lambda\mu\mu\lambda} \in i\mathbb{R}$ drives linewidth broadening $\delta\kappa_\lambda = 4|g^{(3),I}_{\mathrm{sat}}|\bar{n}_\mu$ that is absent from any passive cavity and inaccessible to the FH linear framework.

The key physical message, absent from all prior QNM work: whether a multi-mode nonlinear coupling is real or complex is a *symmetry question*, not a computational detail. Kleinman $\chi^{(3)}$ forces real Kerr couplings (no passive dephasing); broken time-reversal in $\chi^{(2)}$ and gain saturation force complex couplings (measurable dephasing). The Keldysh framework exposes this structure; the FH approach does not.

14.2 Comparison with other non-canonical quantizations

The Franke–Hughes line of work is the closest competitor, but two other frameworks provide independent cross-checks and complementary perspectives.

Drummond–Hillery positive-P representation

The positive-P phase-space representation [41, 42] quantises nonlinear open optical cavities via a doubled phase space in which the Fokker–Planck equation has positive-definite diffusion matrix. Mathematically, the positive-P equations are Langevin equations for a pair of c-number fields (α, β) that would coincide as $\beta = \alpha^*$ classically but are independent at the quantum level — structurally identical to the (a^+, a^-) doubling of the Schwinger–Keldysh contour. The two formalisms are equivalent for Gaussian dynamics; the Keldysh approach is a generating-functional formulation of the same physics, with two practical advantages: (i) the loop expansion is already organised diagrammatically, so higher-order corrections to $g^{(2)}(\tau)$ require no additional Langevin-equation bookkeeping; (ii) the mode functions $\tilde{\mathbf{f}}_\lambda(\mathbf{r})$ appear explicitly in the action, so the nonlinear coupling tensors $g^{(2,3)}_{\lambda\mu\ldots}$ are direct overlap integrals rather than derived phenomenological parameters. Positive-P remains the method of choice for strong-coupling regimes (blockade, Wigner-negative states) where all-loop resummation is required; the Keldysh method in this paper is more economical in the Gaussian and first-nonlinear-correction regime.

Alpeggiani–Andreani quasinormal-mode input–output

Alpeggiani, Parappurath et al. [43, 44] formulated an input–output theory of QNMs by matching the Fresnel coefficients of the open cavity to a discrete-mode Hamiltonian with phenomenological rates. Their framework, like Franke–Hughes, produces approximate bosonic

operators valid to $O(1/Q)$, and does not address the reality structure of nonlinear coupling constants.

Explicit derivation of the AA transfer function from the Keldysh saddle point. The full Keldysh action including the radiation continuum and a coherent input source $j_{\rm in}(t)$ coupled to the bath modes is

$$S_{\rm tot} = S_{\rm cav} + S_{\rm bath} + S_{\rm coup} + S_{\rm src}, \tag{121}$$

with $S_{\rm bath} = \int (d\omega/2\pi) \sum_{\mathbf{k}s} \phi^*_{\mathbf{k}s}(\omega)(\omega - \omega_k)\phi_{\mathbf{k}s}(\omega)$, $S_{\rm coup} = \sum_\lambda \int (d\omega/2\pi) \sum_{\mathbf{k}s} [V_{\lambda,\mathbf{k}s} q^*_\lambda \phi_{\mathbf{k}s} + \text{c.c.}]$, and $S_{\rm src} = -\sum_{\mathbf{k}s} \int (d\omega/2\pi)[j^*_{{\rm in},\mathbf{k}s}\phi_{\mathbf{k}s} + \text{c.c.}]$. Integrating out the bath at the saddle point (path-integral analogue of adiabatic elimination) yields:

$$\phi_{\mathbf{k}s}(\omega) = (\omega - \omega_k + i0^+)^{-1}\Big[V^*_{\lambda,\mathbf{k}s} q_\lambda(\omega) + j_{{\rm in},\mathbf{k}s}(\omega)\Big], \tag{122}$$

which is the standard input-output relation: the bath response is the free propagator times (cavity-coupling driver + external source). Substituting back into $S_{\rm cav} + S_{\rm coup}$ gives the effective cavity equation of motion at the saddle point:

$$(\omega - \tilde{\omega}_\lambda)\, q_\lambda(\omega) = \sum_{\mathbf{k}s} V_{\lambda,\mathbf{k}s}\, (\omega - \omega_k + i0^+)^{-1} j_{{\rm in},\mathbf{k}s}(\omega). \tag{123}$$

The output field at the detector is the bath field evaluated at the detector position, $\phi^{\rm out}(\omega) = \sum_{\mathbf{k}s} \boldsymbol{\epsilon}_{\mathbf{k}s} e^{i\mathbf{k}\cdot\mathbf{r}_{\rm det}} \phi_{\mathbf{k}s}(\omega)$, which can be split into the directly-transmitted source term plus the cavity-mediated emission. After standard manipulations (assuming a Markovian bath, $V_{\lambda,\mathbf{k}s}$ slowly varying near $\omega_k = \omega$):

Key Result

Alpeggiani–Andreani input–output transfer function (recovered from the Keldysh saddle point):

$$\phi^{\rm out}(\omega) = \phi^{\rm in}(\omega) - i \sum_{\lambda\lambda'} V^{\rm out}_\lambda\, G^R_{\lambda\lambda'}(\omega)\, V^{*\rm in}_{\lambda'}\, \phi^{\rm in}(\omega), \tag{124}$$

where $G^R_{\lambda\lambda'}(\omega) = [(\omega - \tilde{\omega}_\lambda)\delta_{\lambda\lambda'} - \Sigma^R_{\lambda\lambda'}(\omega)]^{-1}$ is the cavity retarded propagator (Sec. 8) and $V^{\rm in,out}_\lambda$ are the bath couplings to the input and output channels, respectively. This is exactly the AA transfer function (their Eq. 3, Ref. [43]).

What we add beyond AA. (i) The same derivation also yields the *noise* spectrum via the Keldysh component $D^{\rm K} = -i\kappa_\lambda(2n+1)$, which AA's Hamiltonian-based approach does not provide (they treat only the retarded channel). The output noise spectrum is

$$S^{\rm out}_{\phi\phi}(\omega) = \sum_{\lambda\mu} V^{\rm out}_\lambda G^R_\lambda(\omega) D^{\rm K}_{\lambda\mu} G^A_\mu(\omega) V^{*\rm out}_\mu. \tag{125}$$

(ii) The bare bath couplings V_λ are first-principles overlap integrals of the QNM with the radiation continuum, not phenomenological rates as in AA. (iii) The nonlinear couplings $g^{(2,3)}$ are explicit overlap integrals with reality-structure theorems (Sec. 9), absent from AA.

For input–output calculations of mean quantities the two frameworks agree; for spectra $S_{\rm out}(\omega)$ (output noise), quantum correlators $G^{\rm K}_{\lambda\mu}$ (sensing), and nonlinear-coupling analyses, the Keldysh formulation is the direct route.

Standard-result reproductions

Purcell factor from the Keldysh emitter self-energy. Couple a two-level emitter at $\mathbf{r}_0$ to a single QNM via $H_{\rm em-cav} = \hbar g_0 \sigma^- a^\dagger + \text{h.c.}$ with $g_0 = \mathbf{d}\cdot\tilde{\mathbf{f}}_\lambda(\mathbf{r}_0)\,\mathcal{E}^{\rm zpf}_\lambda/\hbar$ ($\mathbf{d}$ = transition dipole moment). Integrating out the cavity field at one loop gives the emitter self-energy $\Sigma^R_{\rm em}(\omega) = |g_0|^2 G^R_\lambda(\omega) = |g_0|^2/(\omega - \tilde{\omega}_\lambda)$, so the on-resonance decay rate is $\Gamma_{\rm Purcell} = -2\,{\rm Im}\Sigma^R_{\rm em}(\omega_\lambda) = 4|g_0|^2/\kappa_\lambda$. Substituting $\mathcal{E}^{\rm zpf}_\lambda = \sqrt{\hbar\omega_\lambda/(2\varepsilon_0 V_{\rm eff})}$ and the standard $|\mathbf{d}|^2 = 3\pi\varepsilon_0\hbar c^3\Gamma_0/\omega_\lambda^3$ (free-space rate Γ_0):

$$\frac{\Gamma_{\rm Purcell}}{\Gamma_0} = \frac{3}{4\pi^2}\,\frac{(\lambda/n)^3}{V_{\rm eff}}\,\frac{Q_\lambda}{1}, \tag{126}$$

which is the textbook Purcell formula [1, 4]. The Keldysh derivation makes transparent that $\Gamma_{\rm Purcell}$ is the imaginary part of the cavity-mediated emitter self-energy — a result that traditionally requires Fermi's golden rule applied to the cavity-emitter coupling.

Rabi splitting in cavity QED. At strong coupling $g_0 \gg \kappa_\lambda, \gamma_{\rm em}$, the coupled emitter-cavity Green's function has poles at $\omega_\pm = (\omega_\lambda + \omega_{\rm em})/2 \pm \sqrt{g_0^2 + (\omega_\lambda - \omega_{\rm em})^2/4 - i(\kappa-\gamma)/4}$, recovering the standard vacuum-Rabi splitting $2g_0$ at resonance [3].

Hong–Ou–Mandel two-photon coincidence from Keldysh $\chi^{(2)}$ anomalous correlator. The $\chi^{(2)}$ downconversion vertex $g^{(2)}_{\lambda\mu\nu}$ generates an anomalous Keldysh propagator $F^K(t,t') = \langle a_\mu(t) a_\nu(t')\rangle$ at the saddle point of the parametric drive. Splitting the output by a 50:50 beamsplitter ($c_\pm = (a_\mu \pm a_\nu)/\sqrt{2}$) gives the two-detector coincidence rate $G^{(2)}_{++}(\tau) - G^{(2)}_{+-}(\tau)$, which exhibits the standard HOM dip with width set by the joint spectral amplitude $|\Phi(\omega,\omega')|^2 \propto |g^{(2)}|^2 G^R_\mu(\omega) G^R_\nu(\omega')$ (Mandel-Wolf 1995, §22). Detailed evaluation gives the usual $V_{\rm HOM}(\tau) = 1 - e^{-\kappa|\tau|}\cos(\Delta\omega\tau)$ visibility for non-degenerate biphoton states.

Two-mode squeezing covariance from Keldysh. For a degenerate $\chi^{(2)}$ parametric oscillator below threshold, the Keldysh anomalous correlator $F^K(\omega) = \langle a(\omega)a(-\omega)\rangle = 2g^{(2)*}\bar{n}_p/[|\omega - \tilde{\omega}|^2]$ (with $\bar{n}_p$ pump photon number) gives the standard EPR covariance matrix. At threshold $|F^K(0)| \to 1/4$, saturating the quadrature squeezing limit $S_{\rm sq} \to -1$ (Eq. 250 and Fig. 16).

Drummond–Walls $g^{(2)}(0)$ benchmark. For a coherently-driven single-mode Kerr cavity at resonance, the Keldysh one-loop result reproduces the Drummond–Walls 1980 perturbative expression

$$g^{(2)}(0) - 1 = -\frac{2(g^{(3)})^2\bar{n}}{(\kappa/2)^2 + (2g^{(3)}\bar{n})^2}\Big/\bar{n} \tag{127}$$

to within 2.8% relative error against direct master-equation simulation in the weak-Kerr regime $g^{(3)}/\kappa < 0.15$ (Fig. 7). In the strong-coupling photon-blockade limit $g^{(3)}/\kappa \gg 1$, the master-equation result $g^{(2)}(0) \propto 1/(g^{(3)}/\kappa)^2$ (slope -1.98 in Fig. 8, theory -2) is reproduced by the all-loop ladder resummation of the Kerr self-energy.

Lau–Clerk EP signal/noise cancellation. We numerically verify the Petermann factor divergence $K \sim 1/\varepsilon$ (in parameter space, equivalent to $K \sim 1/(\delta\omega)^2$ in eigenvalue space) and the $\sqrt{\delta}$-enhanced eigenvalue splitting at the EP (Fig. 9). The cancellation between signal enhancement and Petermann-K excess noise that yields an ε-independent quantum-limited SNR follows analytically from the Lau–Clerk construction [19] applied to our Keldysh noise kernel; we do not reproduce their full feedback-detection model here.

Franke–Hughes single-mode agreement. Numerical comparison of the Cholesky-dressed bosonic operator $\hat{a}_\nu = \hat{a}/\sqrt{\mathcal{O}_{11}}$ to the closed-cavity reference $\sqrt{\mathcal{O}_{\text{closed}}}$ gives $|\mathcal{O}_{11}/\mathcal{O}_{\text{closed}} - 1| \propto Q^{-1.93}$ (theory Q^{-2}, Fig. 11). The Cholesky single-mode normalization deviates from the closed-cavity value at $O(1/Q^2)$ — one order *tighter* than the FH single-mode commutator deviation $[\hat{b}, \hat{b}^\dagger] - 1 = O(\kappa/\omega) = O(1/Q)$. For $Q = 10^4$, the Cholesky operator has $\lesssim 10^{-7}$ deviation from the closed cavity, while the FH operator has $\sim 10^{-4}$.

Benchmarking table

Capability	Franke–Hughes	Drummond positive-P	Alpeggiani–Andreani	This work
Absorbing host media	direct	via master eq.	via κ phenom.	via κ only
Exact bosonic commutator	$O(1/Q)$	exact (phase-space)	$O(1/Q)$	exact (Cholesky)
Nonlinear $g^{(2,3)}$ reality structure	not analysed	not analysed	not analysed	proved (Secs. 9.4–9.5)
Photon blockade / strong coupling	master eq.	all-loop	phenomenological	Gaussian + loops
Spatial mode functions in action	explicit	implicit (modal)	explicit	explicit
Keldysh noise kernel D^K	via FDT	via diffusion	phenomenological	from first principles

A direct numerical cross-check against positive-P for $g^{(2)}(0)$ of a single $\chi^{(3)}$ cavity (standard Drummond–Walls benchmark) is a useful sanity test but does not exercise any of the multi-mode / non-orthogonal structure where the present work differs; it is deferred to a companion numerical paper.

14.3 Validity at low Q: what holds and what breaks down

The derivations in this paper are phrased in the high-Q limit but the framework is broader. Here we assess each approximation at $Q \sim 100$, which is the regime of photonic-crystal nanocavities with intentional radiation channels, open plasmonic resonators, and coupled-ring systems near exceptional points.

What remains exact or negligibly corrected

QNM eigenvalue problem. QNMs are well-defined for all $Q \gg 1$. The bilinear normalization, the Cholesky decomposition, and the Keldysh path integral structure are all independent of Q. $\mathcal{O}_{\lambda\lambda} = 1 + O(Q^{-2}) \approx 0.9999$ at $Q = 100$: still positive definite, Cholesky still works.

Markov approximation. The error from evaluating the self-energy at $\omega = \omega_\lambda$ is $O(\kappa/\omega)^2 = O(Q^{-2}) \sim 10^{-4}$ at $Q = 100$: negligible.

RWA validity. The Bloch-Siegert shift $\delta\omega_{\rm BSS} \sim (g^{(3)})^2/\omega_0$ is independent of Q and remains $\lesssim 1\,\mathrm{Hz}$ for all nanophotonic platforms (Sec. 10).

What degrades and requires care

High-Q mode function expansion (App. E). The perturbation theory $\tilde{\mathbf{f}}_\lambda = \mathbf{f}_\lambda^{(0)}(1 + i\phi_\lambda/Q + O(Q^{-2}))$ gives first-order corrections of $1/Q \sim 1\%$ at $Q = 100$. The ratio $\mathrm{Im}(g^{(3)}_{\rm cross})/\mathrm{Re}(g^{(3)}_{\rm cross}) = \eta_{\lambda\mu}/(2Q)$ carries an $O(Q^{-2})$ correction that may matter for precision measurements. The 1D Fabry-Pérot verification (App. E) remains exact at all Q.

SALT correction to mode profiles. The difference between passive QNMs and SALT constant-flux modes is $O(g/\omega) = O(1/Q) \sim 1\%$ at $Q = 100$ (Sec. 15.2). For precision calculations of $g^{(3)}$ involving spatial overlap integrals, this 1% correction to the mode profiles propagates at the same order into coupling constants.

Off-diagonal self-energy condition. The spectral resolution condition $|\omega_\lambda - \omega_\mu| \gg \frac{1}{2}\sqrt{\kappa_\lambda\kappa_\mu}$ (Sec. 5.4) requires mode spacing $\gg \omega/(2Q) \sim 0.5\%$ of the centre frequency at $Q = 100$. For structures with mode spacing $\lesssim \omega/(2Q)$, the off-diagonal self-energy $\Sigma_{\lambda\mu}$ must be included in the Keldysh action — it generates coherent hybridisation of modes through their shared radiation bath. This is treated perturbatively in Sec. 5.4; the full multi-mode Keldysh action with off-diagonal $\Sigma_{\lambda\mu}$ is a straightforward extension.

What is larger and more observable at low Q

Crucially, the new physics predicted by this paper — cross-mode dephasing, gain-saturation linewidth shifts, EP Petermann broadening — all scale as *positive powers of* $1/Q$ and are therefore **most pronounced at low** Q.

$$\frac{\delta\kappa_\lambda}{\kappa_\lambda} \approx \frac{2\eta_{\lambda\mu}}{Q}\,\bar{n}_\mu \;\rightarrow\; 2\% \text{ per photon at } Q = 100,\; 2\times10^{-6} \text{ at } Q = 10^6 \tag{128}$$

Effect	$Q = 10^6$	$Q = 100$
$\delta\kappa_\lambda/(\kappa_\lambda \bar{n}_\mu)$	2×10^{-6}	2%
$\mathrm{Im}(g^{(3)}_{\mathrm{cross}})/\mathrm{Re}$	10^{-6}	10^{-2}
Gain-saturation linewidth per photon	$\ll 1\%$	$\sim 1\%$
Henry factor α_H (at $\Delta = \gamma_\perp$)	1	1
SALT correction to mode profile	10^{-6}	1%
App. E $O(1/Q)$ correction	10^{-6}	1%

$Q \sim 100$ structures are therefore the **primary experimental target** for observing the open-system nonlinear effects derived in this paper. Suitable platforms include GaAs photonic-crystal cavities with modified hole radii (tunable Q from 10^3 to 10^5), plasmonic nanoresonators ($Q \sim 10$–100), and coupled-microring systems near EPs ($Q_{\mathrm{net}} \sim 100$ from gain-loss balance). The SALT correction is at the 1% level for these systems and can be included as a numerical correction once the SALT modes are computed.

14.4 Three experimental predictions

We organise the predictions by experimental difficulty, leading with the $\chi^{(2)}$ passive-cavity prediction, which is the cleanest intrinsic open-system signature and is accessible without gain or exceptional-point engineering.

Prediction 1 (primary): $\chi^{(2)}$ squeezing-phase offset from complex $g^{(2)}_{\lambda\mu\nu}$. For parametric downconversion in a passive open cavity with real Kleinman-symmetric $\chi^{(2)}$, the Keldysh anomalous correlator (Eq. 176) has a complex amplitude because the index pattern $(\lambda^*\mu\nu)$ admits no Kleinman cancellation (Sec. 9.5). The squeezed quadrature is rotated relative to the pump phase by:

$$\delta\theta_{\mathrm{sq}} \;\approx\; \arg(g^{(2)}_{\lambda\mu\nu}) \;\sim\; \tfrac{1}{2}\Big(\tfrac{1}{Q_\lambda} + \tfrac{1}{Q_\mu} + \tfrac{1}{Q_\nu}\Big) \cdot \eta^{(2)}_{\lambda\mu\nu} \tag{129}$$

where $\eta^{(2)}_{\lambda\mu\nu} = O(1)$ is a geometric prefactor analogous to Eq. (330) but without the parity suppression that afflicts the cross-Kerr case (App. E). $\delta\theta_{\mathrm{sq}}$ is *zero* in any closed-cavity or real-mode model; it is nonzero here, Q-dependent, and measurable via balanced homodyne detection with a swept local-oscillator phase. For a GaAs H1 cavity at $Q = 3 \times 10^4$, $\delta\theta_{\mathrm{sq}} \sim 5 \times 10^{-5}$ rad per mode — within reach of shot-noise limited homodyne at modest pump powers. This is the **flagship falsifiable prediction** of the passive-cavity QNM framework.

Prediction 2: Non-Lorentzian transmission near EPs. Near an exceptional point, the power transmission becomes a quartic (squared) Lorentzian:

$$|T(\omega)|^2 \propto \left[(\omega - \omega_{\mathrm{EP}})^2 + (\kappa_{\mathrm{EP}}/2)^2\right]^{-2} \tag{130}$$

This is sharper than a standard Lorentzian and directly measurable in PT-symmetric coupled microring transmission spectra (Sec. 7.7).

Prediction 3 (active structures): gain-saturation cross-mode dephasing. In a gain-loaded cavity, the linewidth of mode λ depends on the photon number in mode μ through the purely-imaginary gain-saturation coupling $g^{(3),I}_{\mathrm{sat},\lambda\mu\mu\lambda}$:

$$\kappa^{\mathrm{eff}}_\lambda(\bar{n}_\mu) = \kappa^{(0)}_\lambda + 4\,|g^{(3),I}_{\mathrm{sat},\lambda\mu\mu\lambda}|\,\bar{n}_\mu \tag{131}$$

This mechanism is absent in the passive cavity (where Kleinman $\chi^{(3)}$ forces $\mathrm{Im}(g^{(3)}_{\lambda\mu\mu\lambda}) = 0$ exactly, Sec. 9.4) and is enhanced in the near-threshold regime where $\kappa^{\mathrm{eff}}_{\lambda}$ is small.

Null prediction: passive cross-Kerr dephasing is zero. We emphasise what the framework does *not* predict: $\mathrm{Im}(g^{(3)}_{\lambda\mu\mu\lambda}) = 0$ for real Kleinman-symmetric $\chi^{(3)}$ regardless of Q or mode overlap (Sec. 9.4). Any nonzero cross-mode dephasing observed in a passive cavity must arise from Kleinman-breaking mechanisms (TPA dispersion at GaAs $\lambda = 1000\,\mathrm{nm}$: $\sim$ 5–10% common-mode contribution; material absorption; Raman). Differential-mode spectroscopy subtracting common-mode TPA is required to isolate any residual Kleinman-violation signal, which this framework predicts to be *parametrically smaller* than TPA (entering at $O(1/Q^2)$ by symmetry, App. E).

14.5 What would constitute definitive verification

The clearest passive-cavity test is Prediction 1 (squeezing-phase offset from complex $g^{(2)}$) in a GaAs or AlGaAs photonic-crystal cavity that supports both a cavity-resonant signal mode and a detuned pump. The measurement is balanced homodyne detection of the signal quadrature with local-oscillator phase swept through $[0, \pi]$; the minimum-variance phase gives $\delta\theta_{\mathrm{sq}}$. The prediction is:

1. Strictly zero ($\delta\theta_{\mathrm{sq}} = 0$) in any closed-cavity or real-mode model.
2. Nonzero and calculable from first principles once the geometry, Q-values, and $\chi^{(2)}$ tensor are specified — no free parameters.
3. Has a clear $1/Q$ scaling, *not* degenerate with TPA (which affects amplitude, not phase), and can be cross-validated by varying Q via lithographic tuning of the photonic-crystal hole radii.

For the active-cavity prediction (Prediction 3, cross-mode gain saturation dephasing), the cleanest test uses sub-gap wavelengths ($LiNbO_3$ or Si at $\lambda > 2200\,\mathrm{nm}$) or differential-mode spectroscopy to isolate the gain contribution from common-mode TPA. For the EP prediction (Prediction 2), the PT-symmetric coupled-ring system of Peng et al. [29] is the established platform.

A positive measurement of Prediction 1 would constitute the first observation of a quantum-optical effect arising directly from the complex spatial structure of open-cavity QNM mode functions, and the first confirmation that Kleinman symmetry on $\chi^{(2)}$ does not extend to the modal coupling constant in a non-Hermitian basis.

15 Heisenberg-Langevin Equations and the Keldysh-SALT Framework

This section addresses two foundational questions about the relationship between the Keldysh path integral and operator quantum optics (Sec. 15.1), and between the linear-QNM Keldysh theory and SALT in the strongly-nonlinear regime (Sec. 15.2).

15.1 Deriving the Heisenberg-Langevin equations from the Keldysh path integral

The Heisenberg-Langevin (HL) equations for open quantum systems are the standard language of quantum optics. We derive them here directly from the Keldysh effective action, making the connection explicit.

Step 1: Keldysh action in real time

Writing the single-mode linear Keldysh action in the time domain (Fourier-transforming Eq. 192):

$$S_{\mathrm{K}} = \int dt \Big[a^{\mathrm{cl}*}\big(i\partial_t - \omega_\lambda - \tfrac{i\kappa_\lambda^{\mathrm{eff}}}{2}\big)a^{\mathrm{q}} + a^{\mathrm{q}*}\big(i\partial_t - \omega_\lambda + \tfrac{i\kappa_\lambda^{\mathrm{eff}}}{2}\big)a^{\mathrm{cl}} - \tfrac{i}{4}D^{\mathrm{K}}\big|a^{\mathrm{q}}\big|^2 + S_{\mathrm{NL}}^{\text{cl-q}} \Big] \tag{132}$$

where $\kappa_\lambda^{\mathrm{eff}} = \kappa_\lambda - g_\lambda$ and the first two terms are the kinetic (Berry-phase) contributions that encode the coherent-state commutation relations.

Step 2: Saddle-point equation of motion

Setting $\delta S_{\mathrm{K}}/\delta a^{\mathrm{q}*} = 0$:

$$\big(i\partial_t - \omega_\lambda + \tfrac{i\kappa_\lambda^{\mathrm{eff}}}{2}\big)a^{\mathrm{cl}} = -\frac{\delta S_{\mathrm{NL}}^{\text{cl-q}}}{\delta a^{\mathrm{q}*}} \tag{133}$$

In the time domain this reads:

$$\frac{da^{\mathrm{cl}}}{dt} = \Big(-i\omega_\lambda - \frac{\kappa_\lambda^{\mathrm{eff}}}{2}\Big)a^{\mathrm{cl}} + F_{\mathrm{NL}}(a^{\mathrm{cl}}) + \xi(t) \tag{134}$$

where F_{NL} collects the nonlinear contributions and $\xi(t)$ is the noise source.

Step 3: Noise statistics from the Martin-Siggia-Rose formalism

The noise $\xi(t)$ arises from integrating out the quantum component a^{q} in the path integral. This is the Martin-Siggia-Rose (MSR) procedure: with the $|a^{\mathrm{q}}|^2$ term in S_{K} acting as the source of stochastic forcing, integrating a^{q} from $e^{iS_{\mathrm{K}}}$ generates:

$$\int \mathcal{D}[a^{\mathrm{q}}]\, e^{i\int dt[\ldots - \frac{i}{4}D^{\mathrm{K}}|a^{\mathrm{q}}|^2]} = e^{-\int dt\, \xi^*\left(\frac{1}{D^{\mathrm{K}}}\right)\xi} \tag{135}$$

so the noise correlator is:

$$\langle \xi^*(t)\, \xi(t')\rangle = \tfrac{i}{2}D_{\mathrm{total}}^{\mathrm{K}}\, \delta(t-t') \tag{136}$$

Using $D_{\mathrm{total}}^{\mathrm{K}} = -i\kappa_\lambda(2n_{\mathrm{th}}+1) + ig_\lambda(2N_{\mathrm{sp}}+1)$ (Eq. 194), the combined noise correlator is:

$$\tfrac{1}{2}\big[\langle\xi^*(t)\xi(t')\rangle + \langle\xi(t)\xi^*(t')\rangle\big] = \big[\tfrac{\kappa_\lambda(2n_{\mathrm{th}}+1)}{2} + \tfrac{g_\lambda(2N_{\mathrm{sp}}+1)}{2}\big]\delta(t-t') \tag{137}$$

which is manifestly positive. Attempting to assign $\langle\xi^*\xi\rangle$ and $\langle\xi\xi^*\rangle$ separately by naively subtracting $\delta(t-t')/2$ from the symmetric combination produces formally negative correlators when gain dominates loss ($g_\lambda > \kappa_\lambda/(2N_{\mathrm{sp}}+1)$), which is a symptom of the two physical

baths (radiation loss and atomic gain) having commutators of opposite sign. The physically correct resolution is to retain the two baths separately, with distinct input operators $\hat{\xi}^{\kappa}_{\rm in}$ and $\hat{\xi}^{g}_{\rm in}$ as in the input–output formulation Eqs. (138)–(140) below: $\langle \hat{\xi}^{\kappa\dagger}\hat{\xi}^{\kappa}\rangle = n_{\rm th}$, $\langle \hat{\xi}^{g}\hat{\xi}^{g\dagger}\rangle = N_{\rm sp}$, each individually non-negative. The combined c-number correlator Eq. (137) is the symmetric (Hermitian) part of their sum, which is always positive; the apparent negativity in the naive split-by-half assignment is an artefact of forcing the gain bath (which amplifies rather than absorbs vacuum) into a loss-bath correlator structure.

Step 4: Promotion to Heisenberg operators

The c-number field $a^{\rm cl}(t) = \langle \hat{a}(t)\rangle$ is the expectation value of the Heisenberg-picture annihilation operator. The canonical commutation relation $[\hat{a}(t), \hat{a}^\dagger(t)] = 1$ is encoded in the coherent-state path integral measure — it is the defining property that distinguishes $a^{\rm cl}$ from a classical variable.

Remark

What is classical and what is quantum in the Keldysh formalism.
The fields $a^{\pm}(t)$ and $a^{\rm cl}(t)$, $a^{\rm q}(t)$ are all *c-numbers* (complex numbers), not operators. Nevertheless, the path integral over them is a *quantum mechanical* object: it computes $\mathrm{Tr}[\rho\hat{\mathcal{O}}]$ for quantum operators $\hat{\mathcal{O}}$ via the coherent-state resolution of the identity. The quantum nature is encoded in three places: (i) the Berry-phase kinetic term $i\int a^*\partial_t a\, dt$, which is the path-integral manifestation of $[\hat{a}, \hat{a}^\dagger] = 1$; (ii) the Keldysh $D^{\rm K}$ component, which distinguishes quantum from classical statistics; (iii) the causality constraint that all-$a^{\rm cl}$ vertices vanish, which encodes $\mathrm{Tr}[\rho U^\dagger U] = 1$.
The saddle-point equation (Step 2) is a c-number equation for the *mean field* $\langle \hat{a}(t)\rangle$, not for the operator $\hat{a}(t)$. It correctly gives the first moment $\langle \hat{a}\rangle$ but misses quantum correlations. The Keldysh Green's functions $G^R, G^A, G^{\rm K}$ are quantum correlators (second moments $\langle \hat{a}^\dagger\hat{a}\rangle$, $\langle \hat{a}\hat{a}^\dagger\rangle$), computed as Gaussian path integrals over c-numbers. They capture all physics in the Gaussian (one-loop) approximation.
The c-number stochastic Langevin equations Eqs. (134)–(138) give exact first and second moments. They fail for higher-order moments (e.g., $g^{(2)}(0) = \langle \hat{a}^{\dagger 2}\hat{a}^2\rangle/\langle \hat{a}^\dagger\hat{a}\rangle^2$) because c-numbers commute while operators do not; the correction is $-2\langle |a|^2\rangle$. The full operator Heisenberg-Langevin equation (Version 1, quantum) requires promoting $a^{\rm cl} \to \hat{a}$ and invoking $[\hat{a}, \hat{a}^\dagger] = 1$ as an additional input from the coherent-state quantization — this step is *not* automatically generated by the saddle-point of the c-number path integral.
Summary. The Keldysh path integral is quantum mechanics in the coherent-state basis. The fields are c-numbers; the theory is quantum. The saddle point gives c-number mean-field equations. The Gaussian fluctuations give quantum correlators (photon number, squeezing, QFI). Going beyond Gaussian (higher loops) gives photon statistics, $g^{(2)}$, and quantum corrections to the mean field.

Promoting $a^{\rm cl} \to \hat{a}$ and $\xi \to \hat{\xi}_{\rm in}$ (quantum input noise operator), and using $[\hat{\xi}_{\rm in}(t), \hat{\xi}^\dagger_{\rm in}(t')] = \delta(t-t')$ (vacuum input), the HL equation follows from Eq. (134):

Key Result

Heisenberg-Langevin equations from the Keldysh path integral:

$$\frac{d\hat{a}_\lambda}{dt} = \Big(-i\omega_\lambda - \frac{\kappa_\lambda - g_\lambda}{2}\Big)\hat{a}_\lambda - ig^{(3),R}_{\text{Kerr}}\hat{a}^\dagger_\lambda\hat{a}_\lambda\hat{a}_\lambda - g^{(3),I}_{\text{sat}}\hat{a}^\dagger_\lambda\hat{a}_\lambda\hat{a}_\lambda + \sqrt{\kappa_\lambda}\,\hat{\xi}^\kappa_{\text{in}} + i\sqrt{g_\lambda}\,\hat{\xi}^g_{\text{in}} \tag{138}$$

$$\big[\hat{\xi}^\kappa_{\text{in}}(t), \hat{\xi}^{\kappa\dagger}_{\text{in}}(t')\big] = \delta(t-t'), \quad \langle\hat{\xi}^{\kappa\dagger}_{\text{in}}\hat{\xi}^\kappa_{\text{in}}\rangle = n_{\text{th}} \tag{139}$$

$$\big[\hat{\xi}^g_{\text{in}}(t), \hat{\xi}^{g\dagger}_{\text{in}}(t')\big] = \delta(t-t'), \quad \langle\hat{\xi}^g_{\text{in}}\hat{\xi}^{g\dagger}_{\text{in}}\rangle = N_{\text{sp}} \tag{140}$$

The radiation loss and gain contribute *separate* noise operators $\hat{\xi}^\kappa_{\text{in}}$ and $\hat{\xi}^g_{\text{in}}$, respectively, each satisfying canonical bosonic commutation relations. This is the input-output formulation [40]: the loss bath drives vacuum fluctuations in with rate $\sqrt{\kappa}$, while the gain bath drives amplified spontaneous emission in with rate $i\sqrt{g}$ and population N_{sp}.

For the **multimode laser**, the HL equations generalise to:

$$\frac{d\hat{a}_\lambda}{dt} = \left(-i\omega_\lambda - \frac{\kappa_\lambda - g_\lambda}{2}\right)\hat{a}_\lambda - i\sum_\mu g^{(3)}_{\lambda\mu\mu\lambda}\hat{a}^\dagger_\mu\hat{a}_\mu\hat{a}_\lambda + \hat{\xi}_\lambda(t) \tag{141}$$

with cross-noise correlator $\langle\hat{\xi}^\dagger_\lambda\hat{\xi}_\mu\rangle = \sqrt{g_\lambda g_\mu}\,N^{(\lambda\mu)}_{\text{sp}}\,\delta(t-t')$.

For the **Kerr frequency comb**, promoting Eq. (248) to operator form:

$$\frac{\partial\hat{\Psi}}{\partial t} = F_{\text{in}}e^{-i\delta_0 t} - \frac{\hat{\kappa}}{2}\hat{\Psi} + i\hat{D}[\hat{\Psi}] + ig^{(3)}_{\text{SPM}}\hat{\Psi}^\dagger\hat{\Psi}\,\hat{\Psi} + \hat{\xi}(\theta,t) \tag{142}$$

This is the operator-valued generalised Lugiato-Lefever equation governing comb nucleation dynamics, transient build-up, and photon statistics.

For the multimode and comb cases, the cross-noise correlators from the shared bath:

$$\big\langle\hat{\xi}^{g\dagger}_\lambda(t)\,\hat{\xi}^g_\mu(t')\big\rangle = \sqrt{g_\lambda g_\mu}\,N^{(\lambda\mu)}_{\text{sp}}\,\delta(t-t') \tag{143}$$

where $N^{(\lambda\mu)}_{\text{sp}} = \int n_a(\mathbf{r})\sigma_\lambda(\mathbf{r})\sigma_\mu(\mathbf{r})\,d^3r/\sqrt{g_\lambda g_\mu}$ is the modal overlap in the gain region.

15.2 The Keldysh-SALT framework: quantum fluctuations around nonlinear steady states

When linear QNMs fail: the validity criterion

The linear-QNM Keldysh theory (Secs. 5–19) uses passive mode functions $\tilde{\mathbf{f}}_\lambda(\mathbf{r})$ as the basis. The spatial hole burning correction to these profiles is:

$$\delta\tilde{\mathbf{f}}_\lambda \sim \sum_\mu \frac{g^{(3),\text{sat}}_{\lambda\mu}}{g_\lambda\mathcal{O}_{\lambda\lambda}}|\alpha_\mu|^2\,\tilde{\mathbf{f}}_\lambda \sim \frac{N_{\text{lasing}}}{Q_{\text{eff}}}\,\tilde{\mathbf{f}}_\lambda \tag{144}$$

where N_{lasing} is the number of simultaneously lasing modes and Q_{eff} is the effective (net) Q-factor. The **spatial hole burning parameter**:

$$\boxed{\varepsilon_{\text{SHB}} \equiv \frac{N_{\text{lasing}}}{Q_{\text{eff}}}} \tag{145}$$

governs the validity of the linear-QNM approximation:

System	N_{lasing}	Q_{eff}	ε_{SHB}
Single-mode laser, high-Q	1	10^6	10^{-6}: linear QNM exact
Single-mode laser, $Q = 100$	1	100	10^{-2}: 1% correction
Multimode laser, $N = 10$, $Q = 10^4$	10	10^4	10^{-3}: negligible
QCL comb, $N = 100$, $Q = 10^3$	100	10^3	0.1: **SALT essential**
Broad-area semiconductor, $N = 100$, $Q = 100$	100	100	1: **full SALT required**
Kerr microcomb, $N = 100$, $Q = 10^6$	100	10^6	10^{-4}: linear QNM fine

SALT: the nonlinear steady-state basis

SALT [38, 39] solves the self-consistent Maxwell-Bloch equations at steady state. The lasing modes $\Psi_\mu(\mathbf{r})$ satisfy:

$$\nabla \times \nabla \times \Psi_\mu = \frac{\omega_\mu^2}{c^2} \varepsilon_{\text{eff}}^{\text{SALT}}(\mathbf{r}; \{\Psi_\nu\})\, \Psi_\mu \tag{146}$$

with outgoing boundary conditions and the self-consistently saturated permittivity:

$$\varepsilon_{\text{eff}}^{\text{SALT}}(\mathbf{r}) = \varepsilon_c(\mathbf{r}) + \frac{D(\mathbf{r})\gamma_\perp}{\omega_\mu - \omega_a - i\gamma_\perp} \cdot \frac{1}{1 + \sum_\nu \Gamma_\nu |\Psi_\nu(\mathbf{r})|^2} \tag{147}$$

where $\Gamma_\nu = 2|d|^2/(\hbar^2 \gamma_\perp \gamma_\parallel I_{\text{sat}})$ is the saturation factor. The SALT modes satisfy the **bilinear normalization**:

$$\mathcal{B}_{\mu\mu}^{\text{SALT}} \equiv \int d^3r\, \varepsilon_{\text{eff}}^{\text{SALT}}(\mathbf{r})\, \Psi_\mu(\mathbf{r}) \cdot \Psi_\mu(\mathbf{r}) = 1 \tag{148}$$

and are complete within the active region (in the PML-regularized sense of App. F).

The Keldysh-SALT action

Given the SALT solution $\{\Psi_\mu, \omega_\mu, \alpha_\mu^{\text{SALT}}\}$, expand the field around the steady state:

$$\mathbf{A}(\mathbf{r}, t) = \sum_\mu \big(\alpha_\mu^{\text{SALT}} + \delta a_\mu(t)\big) \Psi_\mu(\mathbf{r}) e^{-i\omega_\mu t} + \text{radiation} \tag{149}$$

The fluctuations δa_μ are assumed small. The Keldysh path integral over $\{\delta a_\mu\}$ has the effective action:

Key Result

Keldysh-SALT action for quantum fluctuations:

$$S_{\text{K-SALT}} = \sum_\mu S_{\text{K}}^{\text{lin}} \big[\delta a_\mu;\, \omega_\mu^{\text{SALT}},\, \kappa_\mu^{\text{SALT}},\, D_\mu^{\text{K,SALT}}\big] + S_{\text{NL}}^{\text{SALT}} \tag{150}$$

The objects entering this action differ from the linear-QNM case in three specific ways:

1. SALT decay rate $\kappa_\mu^{\rm SALT}$. The outgoing-flux decay rate of the μ-th SALT mode:

$$\kappa_\mu^{\rm SALT} = \frac{c^2}{\omega_\mu \mathcal{O}_{\mu\mu}^{\rm SALT}} \oint_{\partial V} {\rm Im}\big[\Psi_\mu^* \times (\nabla \times \Psi_\mu)\big] \cdot d\mathbf{S} \tag{151}$$

where $\mathcal{O}_{\mu\mu}^{\rm SALT} = \int {\rm Re}[\varepsilon_{\rm eff}^{\rm SALT}]|\Psi_\mu|^2 d^3r$. This already incorporates the gain-saturation modification of the mode profile: $\kappa_\mu^{\rm SALT} \neq \kappa_\mu^{\rm passive}$ whenever $\varepsilon_{\rm SHB} \gtrsim 10^{-2}$.

2. SALT Keldysh component $D_\mu^{\rm K,SALT}$. Using the saturated gain susceptibility $\chi_{\rm eff}^{\rm SALT}$ and the SALT mode functions:

$$D_\mu^{\rm K,SALT} = -i\kappa_\mu^{\rm SALT}(2n_{\rm th}+1) + ig_\mu^{\rm SALT}(2N_{{\rm sp},\mu}^{\rm SALT}+1) \tag{152}$$

where the modal gain using SALT mode functions:

$$g_\mu^{\rm SALT} = \frac{\omega_\mu}{\mathcal{O}_{\mu\mu}^{\rm SALT}} {\rm Im}\left[-\int d^3r\, \chi_{\rm eff}^{\rm SALT}(\mathbf{r})\, \Psi_\mu(\mathbf{r}) \cdot \Psi_\mu(\mathbf{r})\right] \tag{153}$$

At the SALT solution, $g_\mu^{\rm SALT} = \kappa_\mu^{\rm SALT}$ (threshold is satisfied for all lasing modes by the SALT self-consistency condition).

3. SALT coupling constants. All nonlinear coupling constants are recomputed with SALT mode functions:

$$g_{\mu\nu\rho\sigma}^{(3),\rm SALT} = \frac{\varepsilon_0(\mathcal{E}^{\rm zpf})^4}{4\hbar} \int d^3r\, \chi^{(3)}(\mathbf{r})\, \Psi_\mu^* \Psi_\nu^* \Psi_\rho \Psi_\sigma \tag{154}$$

The spatial hole burning modifies Ψ_μ relative to $\tilde{\mathbf{f}}_\lambda$, changing $g^{(3),\rm SALT}$ by $O(\varepsilon_{\rm SHB})$.

The Petermann K-factor for SALT modes

The single most important correction from SALT to the quantum noise is the Petermann K-factor, which now uses SALT norms:

$$K_\mu^{\rm SALT} = \frac{\mathcal{O}_{\mu\mu}^{\rm SALT}}{|\mathcal{B}_{\mu\mu}^{\rm SALT}|^2} = \mathcal{O}_{\mu\mu}^{\rm SALT} \tag{155}$$

(using the SALT normalization $\mathcal{B}_{\mu\mu}^{\rm SALT} = 1$). The Keldysh-SALT laser linewidth:

$$\Delta\nu_\mu^{\rm K\text{-}SALT} = K_\mu^{\rm SALT} \cdot \frac{(\kappa_\mu^{\rm SALT})^2 N_{{\rm sp},\mu}^{\rm SALT}}{\pi(g_\mu^{\rm SALT} - \kappa_\mu^{\rm SALT})} \tag{156}$$

For $\varepsilon_{\rm SHB} \ll 1$: $K_\mu^{\rm SALT} \to K_\mu^{\rm passive}$ and Eq. (156) reduces to Eq. (236). For $\varepsilon_{\rm SHB} \sim 1$: spatial hole burning modifies the bilinear norm $|\mathcal{B}_{\mu\mu}^{\rm SALT}|$ relative to $|\mathcal{B}_{\lambda\lambda}^{\rm passive}|$, giving quantitatively different linewidths.

The SALT-Keldysh Heisenberg-Langevin equations

Following the same derivation as Sec. 15.1, the HL equations for SALT-mode fluctuations $\delta\hat{a}_\mu$ are:

$$\frac{d\,\delta\hat{a}_\mu}{dt} = -i\big(\omega_\mu^{\rm SALT} + 2g_{\mu\mu\mu\mu}^{(3),\rm SALT}|\alpha_\mu^{\rm SALT}|^2\big)\delta\hat{a}_\mu - \sum_{\nu\neq\mu} 2ig_{\mu\nu\nu\mu}^{(3),\rm SALT}|\alpha_\nu^{\rm SALT}|^2\delta\hat{a}_\mu + \hat{\xi}_\mu^{\rm SALT}(t) \quad (157)$$

where the noise $\hat{\xi}_\mu^{\rm SALT}$ has correlator:

$$\big\langle \hat{\xi}_\mu^{\rm SALT\dagger}(t)\hat{\xi}_\nu^{\rm SALT}(t')\big\rangle = K_\mu^{\rm SALT}\, g_\mu^{\rm SALT}\, N_{{\rm sp},\mu}^{{\rm SALT},(\mu\nu)}\,\delta(t-t') \quad (158)$$

The Petermann factor $K_\mu^{\rm SALT}$ appears naturally in the noise correlator — even the SALT-Keldysh HL equation carries excess quantum noise from non-Hermitian mode structure, quantified by the ratio of Hermitian to bilinear norms of the SALT modes.

Comparison with passive-QNM Keldysh

The corrections to the passive-QNM Keldysh theory from using SALT modes are:

Quantity	Passive-QNM Keldysh	Keldysh-SALT
Mode functions	$\tilde{\mathbf{f}}_\lambda(\mathbf{r})$	$\Psi_\mu(\mathbf{r})$
Normalization	$\int \varepsilon\|\tilde{\mathbf{f}}\|^2 = \mathcal{O}_{\lambda\lambda}$	$\int \varepsilon_{\rm eff}^{\rm SALT}\|\Psi\|^2 = \mathcal{O}_\mu^{\rm SALT}$
Decay rate	κ_λ (empty cavity)	$\kappa_\mu^{\rm SALT}$ (saturated cavity)
Modal gain	g_λ (unsaturated)	$g_\mu^{\rm SALT} = \kappa_\mu^{\rm SALT}$ (self-consistent)
K-factor	$K_\lambda = \mathcal{O}_{\lambda\lambda}/\|\mathcal{B}_{\lambda\lambda}\|^2$	$K_\mu^{\rm SALT} = \mathcal{O}_\mu^{\rm SALT}$
$g^{(3)}$	$\sim \int \chi^{(3)}\|\tilde{\mathbf{f}}\|^4$	$\sim \int \chi^{(3)}\|\Psi\|^4$
Validity	$\varepsilon_{\rm SHB} \ll 1$	all regimes

Implementation: the SALT-Keldysh numerical procedure

1. **SALT solve:** Run the nonlinear SALT eigenvalue solver (e.g., using the SALT code of Ge et al. [39]) to obtain $\{\Psi_\mu, \omega_\mu^{\rm SALT}, \alpha_\mu^{\rm SALT}\}$ and $\varepsilon_{\rm eff}^{\rm SALT}(\mathbf{r})$.
2. **SALT norm computation:** Compute $\mathcal{O}_\mu^{\rm SALT}$, $\mathcal{B}_\mu^{\rm SALT}$, and $K_\mu^{\rm SALT}$ from the SALT mode functions.
3. **Keldysh assembly:** Compute $D_\mu^{\rm K,SALT}$, $g_{\mu\nu\rho\sigma}^{(3),\rm SALT}$, and the noise cross-correlators using the SALT modes — same formulas as Secs. 9 and 17, with Ψ_μ replacing $\tilde{\mathbf{f}}_\lambda$.
4. **Quantum observables:** Evaluate the Keldysh Green's functions, linewidths (Eq. 156), squeezing, photon statistics, and QFI using the assembled Keldysh-SALT action.

The computational cost is dominated by the SALT solve (step 1), which scales as $O(N_{\rm lasing}^2)$ per iteration and requires typically $\sim$ 10–50 iterations. Steps 2–4 add a cost comparable to the passive Keldysh computation.

16 Quantum Radiation Engineering via Topology Optimization

The Keldysh QNM formalism developed above is not merely a quantization bookkeeping device — it provides a complete, differentiable map from structure geometry to output quantum radiation statistics. This opens a new design paradigm: **topology optimization (TO) of the nanophotonic structure with quantum Fisher information (QFI) as the figure of merit**. The key departure from conventional quantum photonic design is that the optimal structure need not have high Q or orthogonal modes; non-orthogonal QNM overlap is not a problem to minimize but a degree of freedom to exploit.

16.1 The output field and its quantum statistics

The output field at detector position $\mathbf{r}_d$ is related to the intracavity QNM amplitudes by the input-output relation:

$$\hat{\mathbf{E}}_{\text{out}}(\mathbf{r}_d, \omega) = \sum_{\lambda} \mathcal{T}_\lambda(\mathbf{r}_d, \omega)\, \hat{a}_\lambda(\omega) + \hat{\mathbf{E}}_{\text{vac}}(\mathbf{r}_d, \omega) \tag{159}$$

with transfer function:

$$\mathcal{T}_\lambda(\mathbf{r}_d, \omega) = \frac{i\omega\sqrt{\kappa_\lambda}}{c^2}\, G^R_\lambda(\omega)\, \tilde{\mathbf{f}}^*_\lambda(\mathbf{r}_d) \cdot \hat{\mathbf{d}} \tag{160}$$

where $\hat{\mathbf{d}}$ is the detector polarization axis. The output quantum state is Gaussian (for linear and weak-Kerr dynamics), fully characterized by the output covariance matrix:

$$\boldsymbol{\sigma}_{ij} = \tfrac{1}{2}\langle \hat{E}_i \hat{E}_j + \hat{E}_j \hat{E}_i \rangle - \langle \hat{E}_i \rangle \langle \hat{E}_j \rangle \tag{161}$$

computed from the Keldysh propagator:

$$S_{\text{out}}(\omega) = \sum_{\lambda\mu} \mathcal{T}^*_\lambda(\mathbf{r}_d, \omega)\, G^{\text{K}}_{\lambda\mu}(\omega)\, \mathcal{T}_\mu(\mathbf{r}_d, \omega) \tag{162}$$

The *off-diagonal* Keldysh component $G^{\text{K}}_{\lambda\mu}$ ($\lambda \neq \mu$) is nonzero whenever $|\omega_\lambda - \omega_\mu| \lesssim \sqrt{\kappa_\lambda \kappa_\mu}/2$ (Sec. 5.4), encoding quantum correlations between non-orthogonal modes that have no classical analogue.

16.2 The quantum Fisher information figure of merit

Let δ be the parameter being sensed — a small perturbation to the structure, $\varepsilon(\mathbf{r}) \to \varepsilon(\mathbf{r}) + \delta \cdot h(\mathbf{r})$. For a Gaussian output state, the QFI is:

$$\mathcal{F}_Q(\delta) = \frac{1}{2}\text{Tr}\big[\boldsymbol{\sigma}^{-1}(\partial_\delta \boldsymbol{\sigma})\, \boldsymbol{\sigma}^{-1}(\partial_\delta \boldsymbol{\sigma})\big] + (\partial_\delta \bar{\mathbf{E}})^\dagger\, \boldsymbol{\sigma}^{-1} \partial_\delta \bar{\mathbf{E}} \tag{163}$$

The first term captures sensing via *correlation changes* (the open-system contribution); the second captures sensing via *mean-field shifts* (the classical contribution). The quantum Cramér-Rao bound:

$$\Delta\delta \geq \frac{1}{\sqrt{N\, \mathcal{F}_Q}} \tag{164}$$

sets the ultimate precision over N measurements.

The topology optimization problem is therefore:

$$\max_{\rho(\mathbf{r})\in[0,1]} \mathcal{F}_Q\big[\boldsymbol{\sigma}(\rho),\, \partial_\delta \boldsymbol{\sigma}(\rho)\big] \tag{165}$$

with $\varepsilon(\mathbf{r}) = \varepsilon_{\min} + \rho(\mathbf{r})(\varepsilon_{\max} - \varepsilon_{\min})$ (SIMP interpolation).

16.3 How the perturbation enters: non-Hermitian perturbation theory

A permittivity change $\delta h(\mathbf{r})$ perturbs the QNM eigenfrequencies via the bilinear-form perturbation theory:

$$\frac{\partial \tilde{\omega}_\lambda^2}{\partial \delta} = -\frac{c^2}{\mathcal{B}_{\lambda\lambda}} \int d^3r\, h(\mathbf{r})\, \tilde{\mathbf{f}}_\lambda(\mathbf{r}) \cdot \tilde{\mathbf{f}}_\lambda(\mathbf{r}) \tag{166}$$

Note the *bilinear* product $\tilde{\mathbf{f}} \cdot \tilde{\mathbf{f}}$ (not $\tilde{\mathbf{f}}^* \cdot \tilde{\mathbf{f}}$): this is the non-Hermitian perturbation theory that follows from the QNM normalization Eq. (19). The mode function perturbation:

$$\frac{\partial \tilde{\mathbf{f}}_\lambda(\mathbf{r}')}{\partial \delta} = \sum_{\mu\neq\lambda} \frac{\tilde{\omega}_\mu^2/c^2}{\tilde{\omega}_\lambda^2 - \tilde{\omega}_\mu^2} \left[\int d^3r\, h(\mathbf{r})\, \tilde{\mathbf{f}}_\mu \cdot \tilde{\mathbf{f}}_\lambda\right] \tilde{\mathbf{f}}_\mu(\mathbf{r}') \tag{167}$$

involves the bilinear cross-overlap $\int h \tilde{\mathbf{f}}_\mu \cdot \tilde{\mathbf{f}}_\lambda\, d^3r$ — a direct measure of non-orthogonality. Structures with large $\mathcal{O}_{\lambda\mu}$ therefore have large $\partial \tilde{\mathbf{f}}_\lambda/\partial\delta$, hence large $\partial G^{\mathrm{K}}_{\lambda\mu}/\partial\delta$, hence large $\partial\boldsymbol{\sigma}/\partial\delta$, hence large $\mathcal{F}_Q$. This is the fundamental mechanism by which non-orthogonal QNM overlap enhances quantum sensing: it amplifies the differential response of the output correlation structure.

16.4 Adjoint gradient for the QFI

Topology optimization requires $\nabla_\rho \mathcal{F}_Q$ efficiently. The gradient has two contributions:

$$\frac{\partial \mathcal{F}_Q}{\partial \rho(\mathbf{r})} = (\varepsilon_{\max} - \varepsilon_{\min})\, \mathrm{Re}\big[A(\mathbf{r}) + B(\mathbf{r})\big] \tag{168}$$

$$A(\mathbf{r}) = \sum_\lambda \frac{\partial \mathcal{F}_Q}{\partial \tilde{\omega}_\lambda} \cdot \big(-c^2\big)\, \tilde{\mathbf{f}}_\lambda(\mathbf{r}) \cdot \tilde{\mathbf{f}}_\lambda(\mathbf{r}) \tag{169}$$

$$B(\mathbf{r}) = \sum_\lambda \tilde{\mathbf{f}}_\lambda^{\mathrm{adj}}(\mathbf{r}) \cdot \tilde{\mathbf{f}}_\lambda(\mathbf{r}) \tag{170}$$

where $\tilde{\mathbf{f}}_\lambda^{\mathrm{adj}}$ solves the **non-Hermitian adjoint QNM equation**:

$$\big(\mathbf{K} - \tilde{\omega}_\lambda^{*2}\mathcal{M}\big)\, \tilde{\mathbf{f}}_\lambda^{\mathrm{adj}} = -\frac{\partial \mathcal{F}_Q}{\partial \tilde{\mathbf{f}}_\lambda} \tag{171}$$

with outgoing boundary conditions. Note the adjoint operator $\mathbf{K} - \tilde{\omega}_\lambda^{*2}\mathcal{M}$ uses the *complex-conjugate* eigenfrequency, making this a distinct non-Hermitian eigenvalue problem from the forward QNM solve.

Both $A(\mathbf{r})$ and $B(\mathbf{r})$ involve bilinear products $\tilde{\mathbf{f}} \cdot \tilde{\mathbf{f}}$ (not $\tilde{\mathbf{f}}^* \cdot \tilde{\mathbf{f}}$), which encode sensitivity to the imaginary part of the mode function — precisely the structure responsible for the novel sensing mechanisms. Standard photonic TO uses Hermitian overlaps $|\tilde{\mathbf{f}}|^2$ and therefore misses this contribution.

The total computation requires N_{modes} forward and adjoint QNM solves per iteration. For a small number of QNMs in the design band ($N_{\text{modes}} \lesssim 10$) this is tractable in principle on current high-performance FDTD infrastructure, though with two important caveats. *First,* the adjoint QNM equation Eq. (171) uses the complex-conjugate eigenfrequency $\tilde{\omega}_\lambda^{*2}$ with outgoing boundary conditions; existing QNM solvers (Meep/Harminv, COMSOL, MPB) compute the forward QNM problem only, and no off-the-shelf machinery exists for the adjoint solve at production scale. A PML-regularised finite-element implementation is straightforward in principle (the operator is simply the Hermitian conjugate of the forward problem) but has not been demonstrated for 3D nanophotonic structures at the $\sim 10^5$-DOF level required for topology optimisation. *Second,* the gradient Eq. (168) involves bilinear products $\tilde{\mathbf{f}} \cdot \tilde{\mathbf{f}}$ whose numerical evaluation requires preserving the relative phase of the complex mode function across the full structure; this is sensitive to PML thickness and boundary placement at a level that passive-QNM computations do not require. We therefore present the TO framework of this section as a *theoretical foundation* for future QFI-based optimisation, not as a production-ready numerical pipeline. The immediate targets of Sec. 13 (computing individual $g^{(2,3)}$ couplings for a fixed H1 geometry) do not require adjoint solves and are deliverable with current tools.

16.5 New sensing mechanisms accessible to topology optimization

Mechanism 1: Non-orthogonal-overlap-enhanced cross-Kerr sensing (active structures)

Note: As proved in Sec. 9.4, for passive Kleinman-symmetric $\chi^{(3)}$ the cross-Kerr coupling satisfies $\text{Im}(g^{(3)}_{\lambda\mu\mu\lambda}) = 0$ exactly, giving *no* cross-mode dephasing. Mechanism 1 therefore operates only in *active* (gain-loaded) structures, where the gain-saturation nonlinearity contributes a purely imaginary $g^{(3),I}_{\text{sat},\lambda\mu\mu\lambda}$ (Sec. 17.6). The cross-mode dephasing sensitivity in that case is:

$$\mathcal{S}_{\text{CK}} \equiv \frac{\partial \kappa_\lambda}{\partial \bar{n}_\mu} = 4\,\text{Im}(g^{(3),\text{sat}}_{\lambda\mu\mu\lambda}) \approx \frac{2\eta_{\lambda\mu}}{Q}\, g^{(3),I}_{\text{sat,self}} \tag{172}$$

scales as $\eta_{\lambda\mu}/(Q\,V_{\text{eff}})$. Since both Q and V_{eff} appear in the denominator of $g^{(3),I}_{\text{sat,self}} \sim \chi^{(3)}_{\text{sat}}\omega^2\hbar/(V_{\text{eff}}\varepsilon_0)$, the sensitivity per photon scales as:

$$\mathcal{S}_{\text{CK}} \sim \frac{\eta_{\lambda\mu}\,\chi^{(3)}_{\text{sat}}\,\omega^2\hbar}{Q\,V_{\text{eff}}\,\varepsilon_0} \tag{173}$$

The SNR over N photons is $\text{SNR}^2 \approx \eta^2_{\lambda\mu} \cdot N/Q^2$, which is *independent of mode volume* and maximized by maximizing $\eta_{\lambda\mu}$ at a given Q. The optimal Q is therefore determined by the

trade-off between sensitivity per photon and measurement bandwidth, not by maximizing Q indefinitely. Topology optimization can discover geometries where $\eta_{\lambda\mu} \gg 1$ because two co-localized modes have complementary spatial phase gradients across the nonlinear region.

Mechanism 2: Exceptional-point-enhanced correlation spectroscopy

Near an EP, a perturbation δ splitting the degenerate eigenfrequencies produces:

$$\tilde{\omega}_{\pm}(\delta) = \tilde{\omega}_{\mathrm{EP}} \pm \sqrt{J(\delta)}, \quad \frac{\partial \tilde{\omega}_{\pm}}{\partial \delta} \xrightarrow{\delta \to 0} \pm\infty \tag{174}$$

The eigenfrequency splitting $\propto \sqrt{\delta}$ rather than δ gives a super-linear sensitivity near $\delta = 0$. Whether this translates to enhanced QFI requires evaluating:

$$\mathcal{F}_Q^{\mathrm{EP}}(\delta) = \frac{1}{2}\mathrm{Tr}\big[(\boldsymbol{\sigma}_{\mathrm{EP}})^{-1}(\partial_\delta \boldsymbol{\sigma}_{\mathrm{EP}})(\boldsymbol{\sigma}_{\mathrm{EP}})^{-1}(\partial_\delta \boldsymbol{\sigma}_{\mathrm{EP}})\big] \tag{175}$$

where $\boldsymbol{\sigma}_{\mathrm{EP}}$ is the covariance matrix built from the Jordan-block propagator Eq. (60). Both $\boldsymbol{\sigma}_{\mathrm{EP}}$ and $\partial_\delta \boldsymbol{\sigma}_{\mathrm{EP}}$ acquire contributions $\propto \delta^{-1}$ from the double pole; whether the ratio in Eq. (175) diverges, remains finite, or vanishes is an open problem whose resolution requires the Jordan-block Keldysh action of Sec. 7. This is a calculation that has not yet appeared in the literature.

Mechanism 3: Complex $\chi^{(2)}$ squeezing phase

From Sec. 9, the anomalous Green's function for two-mode squeezing is:

$$F_{\lambda\mu}(\omega) = G_\lambda^R(\omega)\, g_{\lambda\mu\nu}^{(2)}\, \alpha_p\, G_\mu^A(-\omega) \tag{176}$$

Because $g_{\lambda\mu\nu}^{(2)} \in \mathbb{C}$, the squeezed quadrature angle is shifted by:

$$\delta\theta_{\mathrm{sq}} \approx \arg(g_{\lambda\mu\nu}^{(2)}) \sim \frac{1}{Q_\lambda} + \frac{1}{Q_\mu} + \frac{1}{Q_\nu} \tag{177}$$

relative to the pump phase. This phase offset is zero for any closed-cavity or purely orthogonal mode model, and nonzero here with a value calculable from the QNM mode functions. Measuring $\delta\theta_{\mathrm{sq}}$ via balanced homodyne detection with a controllable local-oscillator phase constitutes a correlation-based readout of the open-system character of the mode structure, without requiring intensity or frequency measurements.

Mechanism 4: Far-field spatial output entanglement

For two detectors at positions $\mathbf{r}_1$, $\mathbf{r}_2$, the spatial cross-correlation:

$$C(\mathbf{r}_1, \mathbf{r}_2; \omega, \omega') = \sum_{\lambda\mu} \mathcal{T}_\lambda^*(\mathbf{r}_1, \omega)\, G_{\lambda\mu}^{\mathrm{K}}\, \mathcal{T}_\mu(\mathbf{r}_2, \omega') \tag{178}$$

is nonzero between spatially separated detectors only when $G_{\lambda\mu}^{\mathrm{K}} \neq 0$ for spatially distinct mode pairs. Topology optimization can engineer the directionality of $\mathcal{T}_\lambda(\mathbf{r}_d, \omega)$ so that specific output directions carry entangled photons — quantum radiation pattern engineering.

16.6 Why non-high-Q, non-orthogonal designs can be optimal

Three independent arguments establish that the canonical high-Q design criterion is not generically optimal for QFI-based sensing:

(i) QFI is mode-volume independent. The QFI ratio $\partial_\delta \boldsymbol{\sigma}/\boldsymbol{\sigma}$ depends on the *relative* change in correlations, not their absolute magnitude. Both numerator and denominator scale as $\sim 1/V_{\text{eff}}$, so the ratio is approximately mode-volume independent. What determines the QFI is the sensitivity of the correlation structure to δ — this rewards mode-mixing designs over isolated resonators.

(ii) Optimal Q is a balance, not a maximum. The cross-Kerr sensing SNR per photon is $\sim \eta_{\lambda\mu}^2/Q^2$ (Eq. 173), independent of V_{eff}. Increasing Q narrows the sensing bandwidth and slows equilibration without improving the SNR per photon. The optimal Q balances sensitivity per photon against measurement bandwidth and photon lifetime.

(iii) The $\eta_{\lambda\mu}$ factor rewards non-orthogonal overlap. From Eq. (330), $\eta_{\lambda\mu}$ involves the spatial integral of $(\phi_\lambda + \phi_\mu)$ weighted by $|\mathbf{f}_\lambda^{(0)}|^2|\mathbf{f}_\mu^{(0)}|^2\chi^{(3)}$. This factor is amplified when two modes with large non-orthogonal overlap have complementary phase gradients across the nonlinear region. Topology optimization can discover such configurations automatically — they are genuinely outside the design space reachable by perturbative tuning of high-Q resonators.

17 Extension to Gain Media: Active Nanophotonic Structures

We now extend the Keldysh QNM formalism to nanophotonic structures containing laser gain material. This requires (i) a microscopic model of the gain medium, (ii) a modified QNM eigenvalue problem, (iii) a gain self-energy with sign opposite to the radiation self-energy, and (iv) a qualitatively new treatment above the lasing threshold. We treat two cases: gain alone (replacing $\chi^{(2)}/\chi^{(3)}$), and gain together with $\chi^{(2)}/\chi^{(3)}$ nonlinearities.

17.1 Gain medium model: two-level atoms with population inversion

Model the gain medium as a density $n_a(\mathbf{r})$ of two-level atoms embedded in a host material of permittivity $\varepsilon_{\text{host}}(\mathbf{r})$. Each atom has transition frequency ω_a (near-resonant with the cavity mode ω_0) and dipole matrix element d. The atom at position $\mathbf{r}$ is described by the Bloch operators $\hat{S}_z(\mathbf{r},t)$, $\hat{S}_\pm(\mathbf{r},t)$ with:

$$\hat{S}_z = \tfrac{1}{2}(\hat{N}_2 - \hat{N}_1), \quad [\hat{S}_-, \hat{S}_+] = 2\hat{S}_z, \quad [\hat{S}_z, \hat{S}_\pm] = \pm\hat{S}_\pm \tag{179}$$

The pump maintains a steady-state population inversion $W_0(\mathbf{r}) = \langle \hat{N}_2 - \hat{N}_1 \rangle > 0$ (gain condition). The transverse polarization decays at rate $\gamma_\perp$ (dephasing) and the inversion recovers at rate $\gamma_\parallel^{-1}$ (population inversion lifetime T_1).

The total action is:

$$S_{\text{total}} = S[\mathbf{A}] + S_{\text{atoms}}[\hat{S}_\pm, \hat{S}_z] + S_{\text{coupling}}[\mathbf{A}, \hat{S}_\pm] \tag{180}$$

with coupling action (dipole approximation, rotating-wave):

$$S_{\text{coupling}} = -\int dt \int d^3r\, n_a(\mathbf{r}) \big[d\, \hat{S}_+(\mathbf{r},t)\, \mathbf{E}^{(-)}(\mathbf{r},t) + \text{h.c.} \big] \tag{181}$$

where $\mathbf{E}^{(-)} = -\dot{\mathbf{A}}^{(-)}$ is the negative-frequency part of the electric field.

17.2 Integrating out the atomic polarization

In the unsaturated (weak-field) limit, the atomic polarization responds linearly to the field. Integrating out the atomic degrees of freedom in the Keldysh path integral — treating the atoms as a bath of two-level systems at steady-state inversion $W_0(\mathbf{r})$ — gives an effective electromagnetic action with a complex permittivity contribution:

$$S_{\text{eff}}^{\text{atoms}} = \frac{\varepsilon_0}{2} \int \frac{d\omega}{2\pi} \int d^3r\, \chi_{\text{gain}}(\mathbf{r},\omega)\, |\tilde{\mathbf{E}}(\mathbf{r},\omega)|^2 \tag{182}$$

where the gain susceptibility is:

$$\chi_{\text{gain}}(\mathbf{r},\omega) = \frac{n_a(\mathbf{r})|d|^2 W_0(\mathbf{r})}{\varepsilon_0 \hbar} \frac{1}{\omega_a - \omega - i\gamma_\perp} \tag{183}$$

For $W_0(\mathbf{r}) > 0$ (population inversion) and $\omega \approx \omega_a$:

$$\text{Im}\, \chi_{\text{gain}}(\mathbf{r},\omega_a) = -\frac{n_a(\mathbf{r})|d|^2 W_0(\mathbf{r})}{\varepsilon_0 \hbar \gamma_\perp} < 0 \tag{184}$$

In our $e^{-i\omega t}$ convention, $\text{Im}(\varepsilon) < 0$ corresponds to gain (amplification) and $\text{Im}(\varepsilon) > 0$ to loss (absorption). This is opposite to the Huttner-Barnett approach [16], which requires $\text{Im}(\varepsilon_I) > 0$; the present QNM framework handles gain and loss on equal footing.

Gain noise. Crucially, the atomic path integral also produces a stochastic noise term — the quantum Langevin force from atomic fluctuations. In the Keldysh formalism, this manifests as a gain contribution to the Keldysh kernel:

$$\Sigma^{\text{gain,K}}(\mathbf{r},\mathbf{r}',\omega) = \frac{n_a(\mathbf{r})|d|^2}{\varepsilon_0^2 \hbar^2} \frac{N_{\text{sp}}(\mathbf{r})\, W_0(\mathbf{r})}{\gamma_\perp} \cdot \pi\delta(\omega - \omega_a)\, \delta^{(3)}(\mathbf{r} - \mathbf{r}') \tag{185}$$

where $N_{\text{sp}}(\mathbf{r}) = N_2(\mathbf{r})/(N_2(\mathbf{r}) - N_1(\mathbf{r})) \geq 1$ is the spontaneous emission factor (equals 1 for a fully inverted four-level system, $\to \infty$ near transparency).

The sign of $\Sigma^{\text{gain,K}}$ is *positive*, opposite to the radiation loss contribution $\Sigma^{\text{rad,K}} < 0$. This reflects the fundamental distinction: radiation loss damps vacuum fluctuations, while gain amplifies them.

17.3 Gain-modified QNM eigenvalue problem

The effective permittivity of the active structure is:

$$\varepsilon_{\text{eff}}(\mathbf{r},\omega) = \varepsilon_{\text{host}}(\mathbf{r}) + \chi_{\text{gain}}(\mathbf{r},\omega) \tag{186}$$

The gain-modified QNMs satisfy:

$$\nabla \times \nabla \times \tilde{\mathbf{f}}_\lambda(\mathbf{r}) = \frac{\tilde{\omega}_\lambda^2}{c^2}\,\varepsilon_{\text{eff}}(\mathbf{r},\tilde{\omega}_\lambda)\,\tilde{\mathbf{f}}_\lambda(\mathbf{r}) \tag{187}$$

with outgoing boundary conditions. This is a **nonlinear eigenvalue problem** because ε_{eff} depends on $\tilde{\omega}_\lambda$ through the Lorentzian gain lineshape.

For narrow gain bandwidth ($\gamma_\perp \ll |\omega_\lambda - \omega_a|$ or the opposite limit of broad gain), the eigenfrequency acquires an imaginary correction. Using non-Hermitian perturbation theory (Sec. 6–16) with the bilinear perturbation formula:

$$\frac{\partial\tilde{\omega}_\lambda}{\partial W_0} = \frac{-\omega_\lambda/(2\mathcal{B}_{\lambda\lambda})}{\partial(\tilde{\omega}_\lambda^2\varepsilon_{\text{eff}})/\partial(\tilde{\omega}_\lambda^2)} \cdot \frac{n_a|d|^2/(\varepsilon_0\hbar)}{\omega_a - \tilde{\omega}_\lambda - i\gamma_\perp} \int d^3r\,\tilde{\mathbf{f}}_\lambda(\mathbf{r})\cdot\tilde{\mathbf{f}}_\lambda(\mathbf{r}) \tag{188}$$

In the near-resonant approximation ($\omega_\lambda \approx \omega_a$), the imaginary part of the frequency shift defines the **modal gain rate**:

$$\boxed{g_\lambda \equiv 2\,\text{Im}(\Delta\tilde{\omega}_\lambda) = \frac{\omega_\lambda}{\mathcal{O}_{\lambda\lambda}}\,\text{Im}\left[-\int d^3r\,\chi_{\text{gain}}(\mathbf{r},\omega_\lambda)\,\tilde{\mathbf{f}}_\lambda(\mathbf{r})\cdot\tilde{\mathbf{f}}_\lambda(\mathbf{r})\right]} \tag{189}$$

Sign and bilinear form. Since $\text{Im}(\chi_{\text{gain}}) < 0$ for gain, and the bilinear form $\tilde{\mathbf{f}}\cdot\tilde{\mathbf{f}}$ (not $\tilde{\mathbf{f}}^*\cdot\tilde{\mathbf{f}}$) appears, $g_\lambda > 0$ when the mode has significant field overlap with the gain region. The bilinear form captures the complex spatial profile of the QNM — the imaginary part of $\tilde{\mathbf{f}}_\lambda$ contributes to the modal gain even when the real part dominates.

The net complex QNM eigenfrequency of the active structure is:

$$\boxed{\tilde{\omega}_\lambda^{\text{net}} = \omega_\lambda + \frac{i}{2}(g_\lambda - \kappa_\lambda)} \tag{190}$$

where $\kappa_\lambda > 0$ is the radiative loss rate (Sec. 5). Three regimes:

Condition	$\text{Im}(\tilde{\omega}_\lambda^{\text{net}})$	Physical regime
$g_\lambda < \kappa_\lambda$	< 0	Lossy cavity with partial gain
$g_\lambda = \kappa_\lambda$	$= 0$	Lasing threshold (real eigenfrequency)
$g_\lambda > \kappa_\lambda$	> 0	Above-threshold lasing

Hermitian overlap $\mathcal{O}_{\lambda\lambda}$ with gain. In the presence of a complex permittivity ε_{eff}, the Hermitian overlap Eq. (20) uses the total permittivity:

$$\mathcal{O}_{\lambda\lambda'} = \int d^3r\,\text{Re}\big[\varepsilon_{\text{eff}}(\mathbf{r},\omega_\lambda)\big]\,\tilde{\mathbf{f}}_\lambda^*\cdot\tilde{\mathbf{f}}_{\lambda'} \tag{191}$$

using only the real part, since the Hermitian overlap must be a positive definite metric. The imaginary part of ε_{eff} (gain or loss) contributes to the decay/growth rates through g_λ and κ_λ, not through $\mathcal{O}$.

17.4 Keldysh action with gain

Following the same steps as Sec. 5–8, but now with two distinct baths (radiation and atomic gain), the Keldysh action for a single gain-loaded QNM below threshold is:

$$S_{\mathrm{K}}^{\mathrm{gain}} = \int \frac{d\omega}{2\pi} \begin{pmatrix} a^{\mathrm{cl}*} & a^{\mathrm{q}*} \end{pmatrix} \begin{pmatrix} 0 & [G_{\mathrm{net}}^{A}]^{-1} \\ [G_{\mathrm{net}}^{R}]^{-1} & D_{\mathrm{total}}^{\mathrm{K}} \end{pmatrix} \begin{pmatrix} a^{\mathrm{cl}} \\ a^{\mathrm{q}} \end{pmatrix} \tag{192}$$

The retarded propagator uses the net eigenfrequency:

$$G_{\mathrm{net}}^{R}(\omega) = \frac{1}{\omega - \tilde{\omega}_{\lambda}^{\mathrm{net}}} = \frac{1}{\omega - \omega_{\lambda} - i(g_{\lambda} - \kappa_{\lambda})/2} \tag{193}$$

The Keldysh component receives *competing* contributions from the two baths:

$$\boxed{D_{\mathrm{total}}^{\mathrm{K}}(\omega) = \underbrace{-i\kappa_{\lambda}(2n_{\mathrm{th}} + 1)}_{\text{radiation loss}} + \underbrace{+ig_{\lambda}(2N_{\mathrm{sp}} + 1)}_{\text{gain noise}}} \tag{194}$$

The opposite signs are fundamental:

- **Radiation loss**: damps vacuum fluctuations. $D_{\mathrm{rad}}^{\mathrm{K}} < 0$ (at $T = 0$: $-i\kappa$).
- **Gain**: amplifies vacuum fluctuations. $D_{\mathrm{gain}}^{\mathrm{K}} > 0$: $+ig(2N_{\mathrm{sp}} + 1)$.

This is the quantum-optical statement that gain media are active — they add noise, while passive media only redistribute noise.

The fluctuation-dissipation theorem is *violated* for $g_{\lambda} > 0$: $D_{\mathrm{total}}^{\mathrm{K}}$ cannot be written as $-i\kappa_{\mathrm{eff}}(2n_{\mathrm{eff}} + 1)$ with $n_{\mathrm{eff}} \geq 0$ unless one introduces a negative effective temperature. This is precisely the non-equilibrium character of the gain medium.

Effective temperature interpretation. Below threshold, one can define:

$$n_{\mathrm{eff}} = \frac{g_{\lambda} N_{\mathrm{sp}}}{(\kappa_{\lambda} - g_{\lambda})} > 0 \quad (g_{\lambda} < \kappa_{\lambda}) \tag{195}$$

with an effective temperature $T_{\mathrm{eff}} = \hbar\omega_{\lambda}/(k_B \ln(1 + 1/n_{\mathrm{eff}}))$. As $g_{\lambda} \to \kappa_{\lambda}^{-}$, $n_{\mathrm{eff}} \to \infty$ and $T_{\mathrm{eff}} \to \infty$: the effective temperature diverges at threshold. This divergence is the quantum critical signature of the lasing transition.

17.5 Threshold physics and the lasing transition

At threshold: $g_{\lambda} = \kappa_{\lambda}$

At the lasing threshold:

- $G_{\mathrm{net}}^{R}(\omega) = (\omega - \omega_{\lambda})^{-1}$: the retarded propagator has a *real* pole — no decay or growth.
- $D_{\mathrm{threshold}}^{\mathrm{K}} = 2ig_{\lambda}N_{\mathrm{sp}}$: purely imaginary and positive — the system is driven by spontaneous emission noise alone.

The Keldysh propagator at threshold:

$$G^{\mathrm{K}}_{\mathrm{threshold}}(\omega) = \frac{2g_\lambda N_{\mathrm{sp}}}{(\omega - \omega_\lambda)^2} \tag{196}$$

This is a **double pole at real frequency** — a power-law correlation function in time, $G^K(t) \propto t$, indicating critical fluctuations. This is the photonic analogue of the divergent susceptibility at a second-order phase transition.

The output photon number diverges as: $\bar{n}_{\mathrm{out}} \propto g_\lambda N_{\mathrm{sp}}/(\kappa_\lambda - g_\lambda) \to \infty$ as $g_\lambda \to \kappa_\lambda$.

Above threshold: $g_\lambda > \kappa_\lambda$

Above threshold, $G^R_{\mathrm{net}}(\omega)$ has a pole with $\mathrm{Im}(\tilde{\omega}^{\mathrm{net}}_\lambda) > 0$: the mode grows exponentially. Standard perturbation theory in a diverges; the correct treatment expands around the **coherent mean-field** saddle point.

Write $a(t) = (\alpha_0 + \delta a(t))e^{-i\omega_\lambda t}$ where $\alpha_0 = |\alpha_0|e^{i\phi_0}$ is the laser amplitude. The saddle-point condition $\delta S_{\mathrm{eff}}/\delta a^* = 0$ gives (including gain saturation, Sec. 17.6):

$$\frac{g_\lambda - \kappa_\lambda}{2}\alpha_0 - g^{(3),I}_{\mathrm{sat}}|\alpha_0|^2\alpha_0 = 0 \tag{197}$$

The non-trivial solution is:

$$|\alpha_0|^2 = \frac{g_\lambda - \kappa_\lambda}{2g^{(3),I}_{\mathrm{sat}}} \tag{198}$$

confirming that gain saturation (Sec. 17.6) stabilizes the above-threshold amplitude.

Fluctuations above threshold. Expanding the Keldysh action to second order in δa:

$$S^{\mathrm{above}}_{\mathrm{K}} = \int \frac{d\omega}{2\pi} \begin{pmatrix} \delta a^{\mathrm{cl}*} & \delta a^{\mathrm{q}*} \end{pmatrix} \begin{pmatrix} 0 & [G^A_\phi]^{-1} \\ [G^R_\phi]^{-1} & D^{\mathrm{K}}_\phi \end{pmatrix} \begin{pmatrix} \delta a^{\mathrm{cl}} \\ \delta a^{\mathrm{q}} \end{pmatrix} \tag{199}$$

Two distinct fluctuation modes:

$$G^R_{\mathrm{amp}}(\omega) = \frac{1}{\omega + i(g_\lambda - \kappa_\lambda)} \quad \text{(amplitude: damped above threshold)} \tag{200}$$

$$G^R_{\mathrm{phase}}(\omega) = \frac{1}{\omega} \quad \text{(phase: Goldstone mode, undamped!)} \tag{201}$$

The phase mode has a **zero-frequency pole** — it is the Goldstone boson of the spontaneously broken $U(1)$ phase symmetry. Its propagator $G^R_{\mathrm{phase}}(\omega) = 1/\omega$ in frequency corresponds to linear phase drift in time: this is laser phase diffusion.

The phase diffusion coefficient from the Keldysh Langevin equation:

$$D_\phi = \frac{\kappa_\lambda N_{\mathrm{sp}}}{2|\alpha_0|^2} = \frac{\kappa^2_\lambda N_{\mathrm{sp}}}{g_\lambda - \kappa_\lambda} \tag{202}$$

giving the **Schawlow-Townes linewidth** [20, 21, 31] directly from the Keldysh formalism:

$$\boxed{\Delta\nu_{\mathrm{ST}} = \frac{D_\phi}{\pi} = \frac{\kappa^2_\lambda N_{\mathrm{sp}}}{\pi(g_\lambda - \kappa_\lambda)} = \frac{\hbar\omega^2_\lambda \kappa_\lambda N_{\mathrm{sp}}}{2\pi P_{\mathrm{out}}}} \tag{203}$$

where $P_{\text{out}} = \hbar\omega_\lambda \kappa_{\text{mirror}} |\alpha_0|^2$ is the output power. This derivation makes transparent that the Schawlow-Townes linewidth is the Goldstone diffusion rate driven by spontaneous emission noise — a result that usually requires a detailed quantum Langevin calculation but emerges naturally from the Keldysh saddle-point expansion.

17.6 Gain saturation as effective $\chi^{(3)}$

The steady-state population inversion is depleted by the intracavity field:

$$W(\mathbf{r}, t) = \frac{W_0(\mathbf{r})}{1 + |\mathbf{E}(\mathbf{r}, t)|^2 / I_{\text{sat}}(\mathbf{r})} \tag{204}$$

where $I_{\text{sat}}(\mathbf{r}) = \hbar\omega_a \gamma_\perp / (|d|^2 T_1)$ is the saturation intensity. Expanding to first order in $|E|^2/I_{\text{sat}}$:

$$\chi^{\text{NL}}_{\text{gain}}(\mathbf{r}, \omega) = -\chi^{(1)}_{\text{gain}}(\mathbf{r}, \omega) \frac{|\mathbf{E}|^2}{I_{\text{sat}}(\mathbf{r})} \tag{205}$$

This is a nonlinear polarization of the form $P^{(3)} \propto |E|^2 E$, i.e., an **effective third-order susceptibility**:

$$\chi^{(3)}_{\text{sat}}(\mathbf{r}, \omega) = -\frac{\chi^{(1)}_{\text{gain}}(\mathbf{r}, \omega)}{I_{\text{sat}}(\mathbf{r})} = \frac{n_a |d|^2 W_0}{\varepsilon_0 \hbar I_{\text{sat}}} \cdot \frac{1}{\omega_a - \omega - i\gamma_\perp} \tag{206}$$

At resonance ($\omega = \omega_a$, $\Delta = 0$), $\chi^{(3)}_{\text{sat}}$ is **purely imaginary and positive**:

$$\chi^{(3)}_{\text{sat}}(\mathbf{r}, \omega_a) = i \frac{n_a(\mathbf{r}) |d|^2 W_0(\mathbf{r})}{\varepsilon_0 \hbar \gamma_\perp I_{\text{sat}}(\mathbf{r})} > 0 \tag{207}$$

(imaginary positive because $\frac{1}{-i\gamma_\perp} = \frac{i}{\gamma_\perp}$).

This is a critical result:

Remark

Gain saturation breaks the reality of the single-mode self-Kerr.
Sign convention. We adopt the standard Keldysh-action sign convention in which the retarded Green's function is $G^R(\omega) = [\omega - \tilde{\omega}_\lambda - \Sigma^R(\omega)]^{-1}$ with bare $\tilde{\omega}_\lambda = \omega_\lambda - i\kappa_\lambda/2$ ($\kappa_\lambda > 0$ for a stable passive cavity). An additional *positive* imaginary part of Σ^R shifts the pole further into the lower half-plane and *broadens* the linewidth; a negative imaginary part of Σ^R narrows the linewidth (gain).
For a passive medium with real $\chi^{(3)}$, we proved in Sec. 9.3 that $g^{(3)}_{\lambda\lambda\lambda\lambda} \in \mathbb{R}^+$, so the one-loop tadpole self-energy $\Sigma^R = 2g^{(3)}\bar{n} \in \mathbb{R}$ produces only the standard Kerr frequency shift, no linewidth change.
For a gain-loaded medium, integrating out the inverted two-level atomic polarization in the linear-saturation regime (App. D of the expanded manuscript) gives an effective four-photon interaction $H_{\text{sat}} = -\hbar\, A\, \hat{n}^2/(2\Delta - 2i\gamma_\perp)$ with $A > 0$ proportional to gain rate per atom and $\gamma_\perp > 0$ the atomic dephasing. At the gain centre $\Delta = 0$ this is purely imaginary, $H_{\text{sat}}|_{\Delta=0} = -\hbar A \hat{n}^2/(2 \cdot (-i\gamma_\perp)) = +\hbar(iA/(2\gamma_\perp))\hat{n}^2$, so the effective

coupling is $g^{(3)}_{\rm sat}\big|_{\rm res} = +i\,A/(2\gamma_\perp) \equiv i\,g^{(3),I}_{\rm sat}$ with $g^{(3),I}_{\rm sat} > 0$ by construction.
The retarded self-energy from the one-loop tadpole is $\Sigma^R = 2g^{(3)}_{\rm sat}\bar{n} = 2i\,g^{(3),I}_{\rm sat}\bar{n}$ (positive imaginary part). In our Green's function convention this acts as an *additional damping* of the cavity field, which is opposite in sign to the bare gain g_λ already in $\tilde{\omega}_\lambda$. Physically, gain saturation *depletes the inversion* as the photon number rises: more photons stimulate more emission, depleting the inversion and reducing the net gain. This appears in the linewidth as a positive shift,

$$\kappa^{\rm eff}_\lambda(\bar{n}) = \kappa_\lambda + 4\,g^{(3),I}_{\rm sat}\,\bar{n}_\lambda \quad \text{(self-saturation)}, \tag{208}$$

which is the standard Lamb relaxation-oscillation linewidth [30]. Above threshold the bare gain g_λ is locked to the linear loss κ_λ by the saturated steady state ($g^{\rm eff}_\lambda(\bar{n}_0) = \kappa_\lambda$), so Eq. (208) represents the self-consistent clamping mechanism — not an extra broadening on top of unclamped gain.
For *cross-mode* coupling (probe mode λ with photon number from a separate pump mode μ), the same mechanism gives $\kappa^{\rm eff}_\lambda(\bar{n}_\mu) = \kappa^{(0)}_\lambda + 4|g^{(3),I}_{{\rm sat},\lambda\mu\mu\lambda}|\bar{n}_\mu$ (Sec. 9.6). The qualitative contrast with the passive case stands: passive Kleinman $\chi^{(3)}$ gives a real $g^{(3)}$ and a real Σ^R (frequency shift only); gain-saturation $\chi^{(3)}_{\rm sat}$ gives an imaginary $g^{(3)}_{\rm sat}$ and an imaginary Σ^R (linewidth change).

17.7 Case I: Gain without $\chi^{(2)}/\chi^{(3)}$

For a purely gain-loaded structure with $\chi^{(2)} = \chi^{(3)}_{\rm Kerr} = 0$, the full Keldysh action is:

Key Result

Gain-only Keldysh action below threshold:

$$S^{\rm (gain)}_{\rm K} = \sum_\lambda \int \frac{d\omega}{2\pi} \begin{pmatrix} a^{\rm cl*}_\lambda & a^{\rm q*}_\lambda \end{pmatrix} \begin{pmatrix} 0 & [G^A_\lambda]^{-1} \\ [G^R_\lambda]^{-1} & D^{\rm K}_\lambda \end{pmatrix} \begin{pmatrix} a^{\rm cl}_\lambda \\ a^{\rm q}_\lambda \end{pmatrix} + S_{\rm sat} \tag{209}$$

with:

$$G^R_\lambda(\omega) = \left[\omega - \omega_\lambda - \frac{i(g_\lambda - \kappa_\lambda)}{2}\right]^{-1} \tag{210}$$

$$D^{\rm K}_\lambda(\omega) = -i\kappa_\lambda(2n_{\rm th}+1) + ig_\lambda(2N_{\rm sp}+1) \tag{211}$$

and gain-saturation nonlinear term:

$$S_{\rm sat} = \sum_\lambda \hbar g^{(3)}_{\rm sat} \int dt \big[(a^+_\lambda)^{*2}(a^+_\lambda)^2 - (a^-_\lambda)^{*2}(a^-_\lambda)^2\big] \tag{212}$$

where $g^{(3)}_{\rm sat} = i|g^{(3),I}_{\rm sat}|$ is purely imaginary.

The one-loop dressed propagator in this case:

$$\tilde{G}^R_\lambda(\omega) = \frac{1}{\omega - \omega_\lambda + i[(\kappa_\lambda - g_\lambda)/2 + |g^{(3),I}_{\text{sat}}|\bar{n}_\lambda]} \tag{213}$$

The linewidth evolution:

$$\kappa^{\text{eff}}_\lambda(\bar{n}) = \kappa_\lambda - g_\lambda + 2|g^{(3),I}_{\text{sat}}|\bar{n}_\lambda \tag{214}$$

- **Below threshold** ($\bar{n}_\lambda$ small): effective linewidth $\kappa^{\text{eff}}_\lambda = \kappa_\lambda - g_\lambda > 0$, narrowing as gain approaches threshold.
- **Threshold**: $\kappa_\lambda - g_\lambda = 0$, linewidth vanishes in the mean-field limit (actual linewidth determined by noise, Eq. 203).
- **Above threshold**: stabilized by gain saturation; the effective linewidth reopens due to gain clamping.

17.8 Case II: Gain with $\chi^{(2)}$ and $\chi^{(3)}$

When the gain medium coexists with material $\chi^{(2)}$ and $\chi^{(3)}$ nonlinearities (e.g., a GaAs laser with both gain and Kerr effect), the full coupling coefficients acquire contributions from three sources:

The $\chi^{(3)}$ coupling in an active medium:

$$g^{(3)}_{\lambda\mu\nu\rho} = g^{(3),\text{Kerr}}_{\lambda\mu\nu\rho} + g^{(3),\text{sat}}_{\lambda\mu\nu\rho} \tag{215}$$

where:

$$g^{(3),\text{Kerr}}_{\lambda\mu\nu\rho} = \frac{\varepsilon_0(\mathcal{E}^{\text{zpf}})^4}{4\hbar}\int d^3r\, \chi^{(3)}_{\text{Kerr}}\, \tilde{f}^*_{\lambda,i}\tilde{f}^*_{\mu,j}\tilde{f}_{\nu,k}\tilde{f}_{\rho,l} \quad \in \mathbb{R} \text{ (Kleinman } \chi^{(3)}_{\text{Kerr}}\text{, Sec. 9.4)} \tag{216}$$

$$g^{(3),\text{sat}}_{\lambda\mu\nu\rho} = \frac{\varepsilon_0(\mathcal{E}^{\text{zpf}})^4}{4\hbar}\int d^3r\, \chi^{(3)}_{\text{sat}}(\mathbf{r})\, \tilde{f}^*_{\lambda,i}\tilde{f}^*_{\mu,j}\tilde{f}_{\nu,k}\tilde{f}_{\rho,l} \quad \in i\mathbb{R} \text{ (resonant)}/\mathbb{C} \text{ (detuned)} \tag{217}$$

Single-mode self-coupling (all indices equal):

$$g^{(3)}_{\lambda\lambda\lambda\lambda} = \underbrace{g^{(3),R}_{\text{Kerr}} \in \mathbb{R}^+}_{\text{frequency shift}} + \underbrace{i g^{(3),I}_{\text{sat}} \in i\mathbb{R}^+}_{\text{gain saturation}} \tag{218}$$

The imaginary part is now nonzero even for the single-mode coupling! This is a qualitative change from the passive case (Sec. 9.3). The one-loop dressed propagator becomes:

$$\tilde{G}^R_\lambda(\omega) = \frac{1}{\omega - \omega_\lambda - 2g^{(3),R}_{\text{Kerr}}\bar{n}_\lambda + i\left[(\kappa_\lambda - g_\lambda)/2 + g^{(3),I}_{\text{sat}}\bar{n}_\lambda\right]} \tag{219}$$

with simultaneous Kerr frequency shift (from $\chi^{(3)}_{\text{Kerr}}$) and gain-saturation linewidth broadening (from $\chi^{(3)}_{\text{sat}}$).

The $\chi^{(2)}$ coupling in an active medium:

The downconversion coupling $g^{(2)}_{\lambda\mu\nu}$ is formally unchanged in structure (Eq. 75), but the mode functions $\tilde{\mathbf{f}}_\lambda$ now include the gain-induced imaginary corrections. From the high-Q expansion (App. E) generalized to include gain:

$$\tilde{\mathbf{f}}^{\text{gain}}_\lambda(\mathbf{r}) = \mathbf{f}^{(0)}_\lambda(\mathbf{r})\left[1 + \frac{i}{2Q^{\text{net}}_\lambda}\phi^{\text{net}}_\lambda(\mathbf{r})\right] + O\big((Q^{\text{net}})^{-2}\big) \tag{220}$$

where $Q^{\text{net}}_\lambda = \omega_\lambda/|g_\lambda - \kappa_\lambda|$ is the net quality factor (which can be very large near threshold) and $\phi^{\text{net}}_\lambda$ includes contributions from both the radiative and gain perturbations. Therefore:

$$\frac{\text{Im}(g^{(2)}_{\lambda\mu\nu})}{\text{Re}(g^{(2)}_{\lambda\mu\nu})} \approx \frac{1}{2}\left(\frac{1}{Q^{\text{net}}_\lambda} + \frac{1}{Q^{\text{net}}_\mu} + \frac{1}{Q^{\text{net}}_\nu}\right) \tag{221}$$

Threshold enhancement of $\chi^{(2)}$ squeezing. Near threshold, $Q^{\text{net}}_\lambda = \omega_\lambda/|\kappa_\lambda - g_\lambda| \to \infty$, so the ratio $\text{Im}(g^{(2)})/\text{Re}(g^{(2)}) \to 0$: the $\chi^{(2)}$ coupling becomes *more real* near threshold. This means the squeezing phase offset (Eq. 177) vanishes near threshold — the squeezed output state becomes more aligned with the pump phase. Simultaneously, the squeezing strength $\propto |g^{(2)}_{\lambda\mu\nu}|/\kappa_{\text{eff}}$ diverges because $\kappa_{\text{eff}} = \kappa - g \to 0$. This is the threshold enhancement of parametric amplification: an active $\chi^{(2)}$ cavity generates arbitrarily strong squeezing as the threshold is approached.

17.9 Summary: comparison of passive and gain structures

Property	Passive ($\chi^{(3)}$ only)	Active (gain + $\chi^{(3)}$)
$\tilde{\omega}_\lambda$	$\omega_\lambda - i\kappa_\lambda/2$	$\omega_\lambda + i(g_\lambda - \kappa_\lambda)/2$
D^{K}	$-i\kappa(2n+1)$	$-i\kappa(2n+1)+ig(2N_{\text{sp}}+1)$
G^R pole	lower half-plane	lower or upper half-plane
FDT holds?	Yes	No (T_{eff} diverges at threshold)
$g^{(3)}_{\lambda\lambda\lambda\lambda}$	$\in \mathbb{R}^+$	$\in \mathbb{C}$ (imaginary from gain saturation)
Self-Kerr dephasing?	No	Yes ($g^{(3),I}_{\text{sat}}\bar{n}$)
$U(1)$ symmetry	Preserved	Spontaneously broken above threshold
Schawlow-Townes	N/A	$\Delta\nu = \kappa^2 N_{\text{sp}}/(\pi(g-\kappa))$
$\chi^{(2)}$ squeezing	Phase offset $\sim 1/Q$	Phase offset $\to$ 0 near threshold

Open problems for active structures. Several questions are not fully resolved:

1. **Non-Hermitian biorthogonal normalization with gain:** The Cholesky decomposition (Sec. 6) remains valid as long as $\mathrm{Re}[\varepsilon_{\mathrm{eff}}(\mathbf{r},\omega)]$ is positive-definite. Near transparency ($W_0 \to 0$), $\chi_{\mathrm{gain}} \to 0$ and the standard passive analysis is recovered. However, for large gain (deep inversion), $\mathrm{Re}[\varepsilon_{\mathrm{eff}}]$ can become negative in the gain region (gain-induced anomalous dispersion), and the positivity of $\mathcal{O}$ must be checked for each structure.

2. **Above-threshold multi-mode lasing:** When multiple modes have $g_\lambda > \kappa_\lambda$ simultaneously (multimode lasing), the saddle-point expansion requires treating multiple coherent amplitudes $\{\alpha_\lambda\}$ and their mutual gain saturation. The resulting equations are the multi-mode Maxwell-Bloch equations, whose Keldysh derivation is an extension of the single-mode treatment above but has not been worked out.

3. **EP at threshold:** If the lasing mode approaches an exceptional point (Sec. 7) simultaneously with threshold, the double-pole propagator and the threshold divergence combine: $G^R \sim (\omega - \omega_{\mathrm{EP}})^{-2}$ with $\mathrm{Im}(\omega_{\mathrm{EP}}) = 0$. The resulting quantum noise spectrum $G^{\mathrm{K}} \propto (\omega - \omega_{\mathrm{EP}})^{-4}$ is hyper-divergent. This is resolved in Sec. 18: the Petermann K-factor diverges at the EP and produces linewidth broadening, not narrowing.

18 Resolution of Open Problems in Active Structures

We now resolve the three open problems from Sec. 17.

18.1 Problem 1: Multimode Lasing — Maxwell-Bloch Equations in the Keldysh Framework

Setup: N modes with shared gain reservoir

Consider N QNMs $\{\lambda_1, \ldots, \lambda_N\}$, potentially all above threshold, coupled to a common gain medium with inversion $W(\mathbf{r}, t)$. The gain medium is shared: photons in mode λ deplete inversion that would otherwise amplify mode μ. The local inversion satisfies:

$$\partial_t W(\mathbf{r}, t) = \gamma_\parallel \big[W_0(\mathbf{r}) - W(\mathbf{r}, t)\big] - \sum_\lambda \sigma_\lambda(\mathbf{r})\, |a_\lambda(t)|^2\, W(\mathbf{r}, t) \tag{222}$$

with local stimulated-emission cross-section:

$$\sigma_\lambda(\mathbf{r}) = \frac{2|d|^2 (\mathcal{E}_\lambda^{\mathrm{zpf}})^2}{\hbar^2 \gamma_\perp} |\tilde{\mathbf{f}}_\lambda(\mathbf{r})|^2 \tag{223}$$

Adiabatic elimination and gain-saturation coupling

Setting $\partial_t W = 0$ and expanding to first order in photon number defines the gain-saturation coupling tensor:

$$\mathcal{C}_{\lambda\mu} \equiv \int d^3r\, n_a(\mathbf{r})\, \sigma_\lambda(\mathbf{r})\, \sigma_\mu(\mathbf{r}) / \gamma_\parallel \tag{224}$$

In QNM language:

$$g^{(3),\mathrm{sat}}_{\lambda\mu\mu\lambda} = \frac{\varepsilon_0(\mathcal{E}^{\mathrm{zpf}})^4}{4\hbar}\int d^3r\, \chi^{(3)}_{\mathrm{sat}}(\mathbf{r})\, |\tilde{\mathbf{f}}_\lambda(\mathbf{r})|^2 |\tilde{\mathbf{f}}_\mu(\mathbf{r})|^2 \in i\mathbb{R} \tag{225}$$

Since $\chi^{(3)}_{\mathrm{sat}}$ is purely imaginary (Eq. 206), $g^{(3),\mathrm{sat}}_{\lambda\mu\mu\lambda} \in i\mathbb{R}$ for all λ, μ: cross-saturation causes linewidth changes, not frequency shifts.

Multimode Maxwell-Bloch equations from the Keldysh saddle point

After adiabatically eliminating the inversion and integrating out the atomic polarization, the multimode Keldysh effective action is:

$$S^{(N)}_{\mathrm{eff}} = \sum_\lambda S^{\mathrm{gain},\lambda}_{\mathrm{K}} + \sum_{\lambda\mu} \hbar g^{(3),\mathrm{sat}}_{\lambda\mu\mu\lambda} \int dt \Big[(a^+_\lambda)^*(a^+_\mu)^* a^+_\mu a^+_\lambda - (a^-_\lambda)^*(a^-_\mu)^* a^-_\mu a^-_\lambda\Big] \tag{226}$$

The saddle-point equations $\delta S/\delta a^*_\lambda = 0$ with $a_\lambda = \alpha_\lambda e^{-i\omega_\lambda t}$ give the **multimode Maxwell-Bloch equations in the Keldysh framework**:

$$\boxed{(g_\lambda - \kappa_\lambda)\,\alpha_\lambda = 2\sum_\mu |g^{(3),\mathrm{sat}}_{\lambda\mu\mu\lambda}|\,|\alpha_\mu|^2\,\alpha_\lambda \quad \forall\,\lambda \in \mathcal{L}} \tag{227}$$

Mode competition criterion

A two-mode lasing solution is stable if and only if:

$$\boxed{|g^{(3),\mathrm{sat}}_{\lambda\mu\mu\lambda}|^2 < |g^{(3),\mathrm{sat}}_{\lambda\lambda\lambda\lambda}| \cdot |g^{(3),\mathrm{sat}}_{\mu\mu\mu\mu}|} \tag{228}$$

This is the **Lamb mode-competition criterion** [30], derived from first principles. In terms of spatial overlaps it reads $\mathcal{C}^2_{\lambda\mu} < \mathcal{C}_{\lambda\lambda}\mathcal{C}_{\mu\mu}$, which by the Cauchy-Schwarz inequality holds strictly whenever $|\tilde{\mathbf{f}}_\lambda|^2 \not\propto |\tilde{\mathbf{f}}_\mu|^2$ in the gain volume.

Remark

Two modes coexist if and only if they have distinct spatial profiles in the gain region. Co-localized modes with proportional intensity profiles compete; spatial hole burning generically ensures distinct profiles and favors multimode lasing.

The Henry linewidth-enhancement factor from $g^{(3)}$

For detuned gain ($\omega_\lambda \neq \omega_a$), the gain-saturation $\chi^{(3)}$ acquires a real part. Here $\Delta = \omega_a - \omega_\lambda$ is the **detuning of the cavity QNM eigenfrequency ω_λ from the gain transition centre ω_a** — the frequency of peak stimulated emission cross-section, fixed by the material composition. This is a purely geometric quantity (cavity design sets ω_λ, material sets ω_a), not a free dynamical parameter. At detuning Δ:

$$\chi^{(3)}_{\mathrm{sat}}(\omega_\lambda) = \frac{A\Delta}{\Delta^2 + \gamma_\perp^2} + i\frac{A\gamma_\perp}{\Delta^2 + \gamma_\perp^2}, \quad A = \frac{n_a|d|^2 W_0}{\varepsilon_0 \hbar I_{\mathrm{sat}}} > 0 \tag{229}$$

The ratio of real to imaginary parts of $g^{(3)}_{\lambda\lambda\lambda\lambda}$:

$$\boxed{\alpha_H \equiv \frac{\mathrm{Re}(g^{(3)}_{\lambda\lambda\lambda\lambda})}{\mathrm{Im}(g^{(3)}_{\lambda\lambda\lambda\lambda})} = \frac{\Delta}{\gamma_\perp}} \tag{230}$$

is the **Henry (linewidth-enhancement) factor** [31], derived here from the gain-saturation $g^{(3)}$ without phenomenological input. At resonance $\alpha_H = 0$; the Re part of $g^{(3)}_{\mathrm{sat}}$ (proportional to Δ) vanishes and gain saturation causes only linewidth change, no frequency shift. At resonance $\alpha_H = 0$ (no frequency shift); for typical semiconductors ($\Delta \approx 3\gamma_\perp$) $\alpha_H \approx 3$.

The $\mathrm{Re}(g^{(3)})$ term couples amplitude to phase fluctuations: amplitude noise (decay rate $\gamma_{\mathrm{amp}} = g_\lambda - \kappa_\lambda$) imprints a phase perturbation of magnitude $\alpha_H \delta|a|/|\alpha_0|$ before decaying, producing extra phase diffusion:

$$D_\phi^{\mathrm{total}} = D_\phi^{\mathrm{ST}}(1 + \alpha_H^2) \tag{231}$$

The multimode above-threshold linewidth:

$$\boxed{\Delta\nu_\lambda = \frac{(1 + \alpha_{H,\lambda}^2)\, \kappa_\lambda^2 N_{\mathrm{sp}}}{\pi\,(g_\lambda - \kappa_\lambda^{\mathrm{eff}})}} \tag{232}$$

where $\kappa_\lambda^{\mathrm{eff}} = \kappa_\lambda - 2\sum_{\mu\neq\lambda} |g^{(3),\mathrm{sat}}_{\lambda\mu\mu\lambda}||\alpha_\mu|^2$.

Phase-noise correlations between modes

From the multimode Keldysh action Eq. (226), the phase-noise cross-correlation:

$$\langle \delta\phi_\lambda(t)\, \delta\phi_\mu(0)\rangle = -\frac{2\, g^{(3),\mathrm{sat}}_{\lambda\mu\mu\lambda}\, N_{\mathrm{sp}}}{|\alpha_\lambda|\,|\alpha_\mu|}\, \theta(t) \tag{233}$$

For resonant gain (purely imaginary $g^{(3),\mathrm{sat}}$): the correlation is imaginary, so different lasing modes have phase fluctuations correlated with a $\pi/2$ offset — measurable via optical heterodyne spectroscopy of the multimode beat note.

18.2 Problem 2: Exceptional Point at Lasing Threshold — the Petermann Factor

Two norms and the Petermann K-factor

When the lasing mode approaches an EP, $\mathcal{B}_{00} \to 0$ (Eq. 51) while $\mathcal{O}_{00} > 0$. The **Petermann K-factor** [32, 33]:

$$K_\lambda \equiv \frac{\mathcal{O}_{\lambda\lambda}}{|\mathcal{B}_{\lambda\lambda}|^2} \tag{234}$$

equals 1 for a normal mode (far from EP) and diverges at the EP ($\mathcal{B}_{00} = 0$). It measures how much spontaneous emission is "wasted" into the non-orthogonal Jordan partner that carries no coherent amplitude.

K enters via the Keldysh phase-diffusion rate

In the above-threshold Keldysh expansion, the phase-diffusion coefficient picks up K through the mismatch between the two norms:

$$D_\phi = \lim_{\omega\to 0} \omega^2\, G^{\mathrm{K}}_{\mathrm{phase}}(\omega) = \frac{g_\lambda N_{\mathrm{sp}}\, K_\lambda}{|\alpha_\lambda|^2} \tag{235}$$

giving the EP laser linewidth:

$$\boxed{\Delta\nu_{\mathrm{EP}} = \frac{K_0\, \kappa_0^2 N_{\mathrm{sp}}}{\pi(g_0 - \kappa_0)} \xrightarrow{\mathcal{B}_{00}\to 0} \infty} \tag{236}$$

The EP broadens the laser linewidth, not narrows it.

EP sensing paradox resolved

For a two-mode system at parameter distance ϵ from the EP: $\mathcal{B}_{00} \sim \epsilon^{1/2}$, $K_0 \sim 1/\epsilon$, $\Delta\nu_{\mathrm{EP}} \sim 1/\epsilon$. For a perturbation δ (eigenfrequency splitting $\sim \sqrt{\delta}$):

$$\mathrm{SNR}_{\mathrm{EP}}(\delta, \epsilon) \sim \frac{\sqrt{\delta}/\epsilon}{1/\epsilon} = \sqrt{\delta} \tag{237}$$

The SNR is *independent of* ϵ: the divergent sensitivity and the divergent noise cancel exactly. Proximity to the EP provides no quantum advantage for sensing. This is consistent with the QFI analysis (Eq. 175): the full Jordan-block covariance must be used, not just the eigenfrequency sensitivity.

Remark

[Relation to Lau & Clerk (2018)] This result agrees with, and provides a Keldysh derivation of, the conclusion of Lau and Clerk [19], who showed using quantum Fisher information theory that the signal-to-noise ratio at an EP laser sensor is independent of the proximity parameter ϵ, because the Petermann K-factor exactly cancels the $\sqrt{\delta}$ eigenfrequency sensitivity enhancement. The present derivation obtains the same result from the Keldysh effective action: the phase-diffusion coefficient $D_\phi \propto K_0/|\alpha_0|^2$ and the eigenfrequency sensitivity $\partial\tilde{\omega}_\pm/\partial\delta \sim \epsilon^{-1}$ have identical ϵ-dependence, so the SNR at the fundamental quantum limit (set by the Keldysh G^{K}) is ϵ-independent. The Keldysh derivation additionally specifies the prefactor and the frequency-resolved SNR spectrum, which depends on the Jordan-chain metric $\mathcal{J}_{\mathrm{EP}}$ (Eq. 55) and cannot be read off from the scalar QFI alone.

18.3 Problem 3: Cholesky Validity with Anomalous Dispersion

Sufficient condition for $\mathcal{O}_{\lambda\lambda} < 0$: strong gain with anomalous dispersion

In the $e^{-i\omega t}$ convention used throughout this paper, $\mathrm{Im}(\chi_{\mathrm{gain}}) < 0$ corresponds to gain (Eq. 184), so $\mathrm{Im}(\varepsilon_{\mathrm{eff}}) < 0$ in a gain region. We show that sufficiently strong gain combined

with anomalous dispersion can drive $\mathcal{O}_{\lambda\lambda} < 0$, in which case the Cholesky decomposition of Sec. 6 must be performed around the mean-field saddle rather than vacuum.

Multiplying the gain-QNM equation Eq. (187) by $\tilde{\mathbf{f}}_\lambda^*$ and integrating with the outgoing Silver–Müller boundary condition gives the bilinear identity

$$(g_\lambda - \kappa_\lambda)\,\mathcal{O}_{\lambda\lambda} = -c\oint_{\partial V} |\tilde{\mathbf{f}}_\lambda^\perp|^2\, dS - \omega_\lambda \int d^3r\, \mathrm{Im}(\varepsilon_{\mathrm{eff}})\, |\tilde{\mathbf{f}}_\lambda|^2, \tag{238}$$

in which the surface term $c\oint |\tilde{\mathbf{f}}_\lambda^\perp|^2 dS > 0$ sources the radiative loss $\kappa_\lambda > 0$ and the volume term $-\omega_\lambda \int \mathrm{Im}(\varepsilon_{\mathrm{eff}})|\tilde{\mathbf{f}}_\lambda|^2 d^3r > 0$ (since $\mathrm{Im}(\varepsilon_{\mathrm{eff}}) < 0$ under gain) sources the modal gain $g_\lambda > 0$. Eq. (238) therefore constrains the magnitudes of the two rates but does *not* fix $\mathrm{sign}(\mathcal{O}_{\lambda\lambda})$ by itself: that sign is controlled by $\int \mathrm{Re}[\varepsilon_{\mathrm{eff}}(\mathbf{r})]|\tilde{\mathbf{f}}_\lambda(\mathbf{r})|^2\, d^3r$, which becomes negative only when anomalous dispersion ($\mathrm{Re}(\varepsilon_{\mathrm{eff}}) < 0$) coincides with a mode-intensity maximum.

For the standard gain lineshape Eq. (183), $\mathrm{Re}(\chi_{\mathrm{gain}}) < 0$ for $\omega_\lambda > \omega_a$ (anomalous dispersion above line centre) with magnitude bounded by $n_a|d|^2 W_0/(\varepsilon_0\hbar\gamma_\perp)\cdot|\Delta|/(\Delta^2+\gamma_\perp^2)$. Since the same prefactor sets the scale of g_λ through Eq. (189), the crossover condition $\mathrm{Re}(\varepsilon_{\mathrm{eff}}) < 0$ over the mode volume requires gain large enough that $g_\lambda \gtrsim \omega_\lambda \cdot \mathrm{Re}(\varepsilon_{\mathrm{host}})/|\Delta/\gamma_\perp|$, which for near-resonant gain ($|\Delta| \lesssim \gamma_\perp$) coincides with (not implies) the above-threshold regime. Hence:

Remark

Cholesky with vacuum reference fails ($\mathcal{O}_{\lambda\lambda} < 0$) only in the strong-gain anomalous-dispersion regime, which is typically reached once the lasing mode is above threshold. The converse does not hold: an above-threshold mode with resonant gain ($\Delta = 0$) has $\mathrm{Re}(\chi_{\mathrm{gain}}) = 0$ and retains $\mathcal{O}_{\lambda\lambda} > 0$. The mean-field resolution below is required whenever $\mathcal{O}_{\lambda\lambda} < 0$ arises, irrespective of the threshold condition.

Self-consistent resolution

Cholesky fails for above-threshold modes, which must be treated via the mean-field expansion. For fluctuations δa_λ around the mean field, the effective kinetic coefficient:

$$\mathcal{O}_{\lambda\lambda}^{\mathrm{fluct}} = \mathcal{O}_{\lambda\lambda} + \frac{2|g_{\lambda\lambda\lambda\lambda}^{(3),\mathrm{sat}}|}{\omega_\lambda}\,|\alpha_\lambda|^2 > 0 \tag{239}$$

for all $g_\lambda > \kappa_\lambda$ (verified using Eq. 238 and $|\alpha_\lambda|^2 = (g_\lambda - \kappa_\lambda)/(2|g_{\lambda\lambda\lambda\lambda}^{(3),\mathrm{sat}}|)$). The Cholesky decomposition is valid for fluctuations around the mean field even when it fails for vacuum fluctuations.

The complete self-consistent partition:

Mode	Condition	Treatment
Sub-threshold	$g_\lambda < \kappa_\lambda$, $\mathcal{O}_{\lambda\lambda} > 0$	Vacuum Keldysh + Cholesky
Threshold	$g_\lambda = \kappa_\lambda$	Critical fluctuations (Sec. 17.5)
Above-threshold	$g_\lambda > \kappa_\lambda$, $\mathcal{O}_{\lambda\lambda} < 0$	Mean-field + Cholesky on $\mathcal{O}^{\mathrm{fluct}}$

The path integral is well-defined in every regime after the appropriate expansion.

19 Application to Optical Frequency Combs

Optical frequency combs — spectra of equally-spaced phase-locked lines at $\omega_n = \omega_0 + n\,\omega_{\rm FSR}$ — are among the most important tools in precision metrology and quantum optics [34]. We show here that the Keldysh QNM formalism of this paper provides a unified, quantum-mechanical foundation for Kerr microcombs, their quantum noise properties, and a new class of gain-assisted combs.

19.1 Why multimode lasing alone is not a frequency comb

The multimode Maxwell-Bloch equations Eq. (227) govern the magnitudes $|\alpha_\lambda|$ of simultaneously lasing modes. They are invariant under independent phase rotations $\alpha_\lambda \to \alpha_\lambda e^{i\phi_\lambda}$ for each λ, because every term involves $|\alpha_\lambda|^2$ or α_λ with the same phase. The saddle-point therefore leaves all relative phases $\phi_\lambda - \phi_\mu$ undetermined — each mode phase-diffuses independently (Sec. 18.2). The output is *incoherent multimode emission*, not a comb.

A frequency comb requires **inter-mode phase locking**: a dynamical constraint that fixes $\phi_{m+1} + \phi_{m-1} - 2\phi_m = \text{const}$ for all m. This requires a coupling between modes sensitive to their relative phases, not just their intensities.

The essential condition. The gain-saturation $g^{(3)}_{\rm sat}$ is purely imaginary (intensity-only, phase-insensitive). For comb generation the equation of motion for mode $m+1$ must contain a term depending on the *phase* of modes m and $m-1$. This requires $\mathrm{Re}(g^{(3)}) \neq 0$. Sources:

Source	$\mathrm{Re}(g^{(3)})$	$\mathrm{Im}(g^{(3)})$	Comb?
Passive Kerr $\chi^{(3)}$	> 0	0	Yes
Gain saturation at $\Delta = 0$	0	> 0	No
Gain at detuning $\Delta \neq 0$	$\alpha_H \cdot \mathrm{Im}$	> 0	Yes ($\alpha_H \neq 0$)
Gain + passive Kerr	> 0 from both	> 0	Yes

The Henry factor $\alpha_H = \mathrm{Re}(g^{(3)})/\mathrm{Im}(g^{(3)}) = \Delta/\gamma_\perp$ (Eq. 230) directly measures whether a gain medium can drive comb formation without an external Kerr element: *any nonzero detuning from line center provides the phase-sensitive FWM coupling needed for a comb.*

Remark

Quantum cascade laser combs. QCL frequency combs [37] form without any external Kerr medium or saturable absorber. The gain medium provides both gain (imaginary $g^{(3)}_{\rm sat}$) and the coherent FWM nonlinearity (real $g^{(3)}_{\rm disp}$) because the cavity mode frequencies ω_m are generically detuned from the gain transition centre ω_a by $\Delta_m = \omega_a - \omega_m$, with $|\Delta_m|/\gamma_\perp \sim 1$–3 for modes near the gain edges. The Henry factor $\alpha_{H,m} = \Delta_m/\gamma_\perp$ is therefore $\sim$ 1–3 for these modes, supplying the phase-sensitive FWM term $\mathrm{Re}(g^{(3)}) = \alpha_H\,\mathrm{Im}(g^{(3)})$ that locks comb phases. Modes near the gain centre ($\Delta_m \approx 0$, $\alpha_H \approx 0$) lase incoherently; modes near the gain edges ($|\Delta_m| \approx \gamma_\perp$, $\alpha_H \approx 1$) drive comb nucleation. This is the first derivation of QCL comb formation from a microscopic path-integral treatment.

19.2 Four-wave mixing as the phase-locking mechanism

The $\chi^{(3)}$ term in the Keldysh action (Eq. 92) contains the **degenerate four-wave mixing** (FWM) term for modes satisfying $m_1 + m_2 = m_3 + m_4$:

$$S_{\text{FWM}} = \sum_{m_1+m_2=m_3+m_4} \hbar g^{(3)}_{m_1 m_2 m_3 m_4} \int dt \left[a^{+\,*}_{m_1} a^{+\,*}_{m_2} a^{+}_{m_3} a^{+}_{m_4} - (\text{backward branch}) \right] \tag{240}$$

The dominant FWM process is *pump-to-sideband*: two pump photons at mode m_0 convert to a signal at $m_0 + m$ and an idler at $m_0 - m$:

$$g^{(3)}_{\text{FWM}}(m) \equiv g^{(3)}_{m_0,m_0,m_0+m,m_0-m} = \frac{\varepsilon_0 (\mathcal{E}^{\text{zpf}})^4}{4\hbar} \int d^3r\, \chi^{(3)}(\mathbf{r})\, \tilde{f}^{*2}_{m_0}\, \tilde{f}_{m_0+m}\, \tilde{f}_{m_0-m} \tag{241}$$

The equations of motion from Eq. (240) in the slowly-varying envelope at mode $m_0 + m$:

$$\dot{a}_{m_0+m} = \left(-i\omega_{m_0+m} - \tfrac{\kappa_{m_0+m}}{2} \right) a_{m_0+m} - 2i g^{(3)}_{\text{FWM}}(m)\, a^2_{m_0}\, a^*_{m_0-m} + \xi_{m_0+m} \tag{242}$$

The term $a^2_{m_0} a^*_{m_0-m}$ depends on the *phases* of modes m_0 and $m_0 - m$, not just their intensities. This cross-phase coupling locks the relative phases of signal and idler through the constraint:

$$\phi_{m_0+m} + \phi_{m_0-m} = 2\phi_{m_0} + \text{const} \tag{243}$$

exactly the frequency-comb phase condition. FWM, not gain saturation, is the phase-locking mechanism.

19.3 Parametric gain and comb threshold

The linearised equations around the pump amplitude α_{m_0} give the parametric gain for sideband mode $m_0 + m$:

$$G_{\text{param}}(m) = \sqrt{\left(2|g^{(3)}_{\text{FWM}}(m)||\alpha_{m_0}|^2\right)^2 - \left(\delta_m/2\right)^2} \tag{244}$$

where $\delta_m = \omega_{m_0+m} + \omega_{m_0-m} - 2\omega_{m_0}$ is the **phase mismatch** from cavity dispersion.

For anomalous dispersion ($D_2 < 0$, $\delta_m < 0$):

$$\delta_m \approx D_2\, m^2 < 0 \tag{245}$$

and parametric gain is positive for all m below a maximum mode number $m_{\max} = \sqrt{4|g^{(3)}_{\text{FWM}}||\alpha_{m_0}|^2/|D_2|}$.

The **comb generation threshold** (primary lines $m = \pm 1$):

$$\boxed{|g^{(3)}_{\text{FWM}}(1)||\alpha_{m_0}|^2 > \frac{\kappa_{m_0+1} + \kappa_{m_0-1}}{4}} \tag{246}$$

Unlike the standard LLE threshold (which assumes uniform κ), Eq. (246) uses the mode-specific loss rates from the QNM computation. Signal and idler modes can have different κ — relevant for combs near cavity resonances or band edges where the local density of states changes rapidly.

19.4 The generalised Lugiato-Lefever equation from the Keldysh formalism

The multimode mean-field equations from S_{eff} can be mapped to a continuous-field equation. For a ring resonator with azimuthal modes $\tilde{f}_m(\mathbf{r}) = f_\perp(\mathbf{r})\, e^{im\theta}/\sqrt{2\pi R}$ (transverse profile $f_\perp$, azimuthal angle θ), define the slowly-varying field envelope:

$$\Psi(\theta, t) = \frac{1}{\sqrt{2\pi R}} \sum_m a_m(t)\, e^{im\theta} \tag{247}$$

The multimode equations of motion become the **generalised Lugiato-Lefever equation (LLE)** [35]:

Key Result

$$\frac{\partial \Psi}{\partial t} = F_{\text{in}} e^{-i\delta_0 t} - \frac{\hat{\kappa}}{2}\Psi + i\hat{D}[\Psi] + i g^{(3)}_{\text{SPM}} |\Psi|^2 \Psi + \xi(\theta, t) \tag{248}$$

where:

- $\hat{\kappa}$ is the *mode-dependent loss operator* (diagonal in mode space: $\hat{\kappa}\, a_m = \kappa_m\, a_m$),
- $\hat{D} = \sum_k D_k (-i\partial_\theta)^k / k!$ is the *exact dispersion operator* from the full ω_m spectrum (not just the D_2 approximation),
- $g^{(3)}_{\text{SPM}} = g^{(3)}_{\text{self}}$ is the self-phase modulation coefficient (real for passive media, complex for gain),
- $\xi(\theta, t)$ is the *quantum Langevin noise*: $\langle \xi^*(\theta, t)\xi(\theta', t')\rangle = G^{\text{K}}(\theta, \theta'; \omega) \cdot \delta(t - t')$, obtained from the Keldysh component D^{K}_m of each comb mode.

The standard LLE [35] is recovered when: $\hat{\kappa} \to \kappa \cdot \mathbf{1}$ (uniform loss), $\hat{D} \to D_2(-i\partial_\theta)^2/2$ (quadratic dispersion only), and the quantum noise term ξ is dropped. The Keldysh derivation retains all three generalizations from first principles.

Non-local Kerr in non-uniform structures. For a photonic crystal or other non-uniform structure where $g^{(3)}_{\text{FWM}}(m)$ depends on m (non-identical mode functions), the real-space LLE acquires a non-local self-phase modulation:

$$i g^{(3)}_{\text{SPM}} |\Psi|^2 \Psi \to i \int_0^{2\pi R} d\theta'\, K(\theta, \theta')\, |\Psi(\theta')|^2\, \Psi(\theta) \tag{249}$$

where $K(\theta, \theta')$ is the Kerr kernel given by the Fourier transform of $g^{(3)}_{\text{FWM}}(m)$. This non-local LLE is a new result not accessible from standard ring-resonator models.

19.5 Quantum comb: entangled comb lines

Each signal-idler pair (m_0+m, m_0-m) generated by FWM is in a *two-mode squeezed vacuum* state. From Sec. 9, the anomalous Keldysh component between conjugate comb lines:

$$F_{m_0+m,\,m_0-m}(\omega) = G^R_{m_0+m}(\omega)\, g^{(3)}_{\rm FWM}(m)\, \alpha^2_{m_0}\, G^A_{m_0-m}(-\omega) \tag{250}$$

is non-zero whenever the pump amplitude $\alpha_{m_0} \neq 0$. The squeezing spectrum for comb line m:

$$S^{\rm sq}_m(\omega) = 1 - \frac{4|g^{(3)}_{\rm FWM}(m)|^2|\alpha_{m_0}|^2}{|\omega - \tilde{\omega}_{m_0+m}|^2 + |g^{(3)}_{\rm FWM}(m)\alpha_{m_0}|^2} \tag{251}$$

gives the minimum variance quadrature of each comb line.

Conventions for $S^{\rm sq}$. $S^{\rm sq}$ in Eq. (251) is the *Keldysh anomalous-correlator parameterisation* $S^{\rm sq} = 1 - 4|F|^2/|G^{\rm K}|^2$, normalised such that $S^{\rm sq} = 1$ corresponds to shot noise (no squeezing, $F = 0$) and $S^{\rm sq} = -1$ corresponds to the formal Gaussian limit where the squeezed-quadrature variance ratio $\mathcal{V}_-/\mathcal{V}_{\rm vac}$ would be zero. The mapping to the standard variance-ratio observable is $\mathcal{V}_-/\mathcal{V}_{\rm vac} = (1 + S^{\rm sq})/2$, so $S^{\rm sq} = 1$ maps to vacuum, $S^{\rm sq} = 0$ maps to 3 dB squeezing, and $S^{\rm sq} \to -1$ maps to $\mathcal{V}_-/\mathcal{V}_{\rm vac} \to 0^+$ (unbounded dB).

At low pump power $|g^{(3)}_{\rm FWM}||\alpha_{m_0}| \ll \kappa/2$: $S^{\rm sq} \to 1$ (shot noise). As $|g^{(3)}_{\rm FWM}||\alpha_{m_0}| \to \kappa/2$ (comb threshold, Eq. 246): $S^{\rm sq} \to -1$ within the Gaussian approximation — equivalent to $\mathcal{V}_-/\mathcal{V}_{\rm vac} \to 0$ and in-principle unbounded squeezing in dB. The anti-squeezed quadrature variance $\mathcal{V}_+$ diverges as $\mathcal{V}_-^{-1}$, consistent with the minimum-uncertainty constraint $\mathcal{V}_+\mathcal{V}_- \geq \mathcal{V}^2_{\rm vac}/4$. In practice, pump depletion and finite $N_{\rm sp}$ saturate the measured squeezing at $\sim$ 10–15 dB [41, 42], below which the Gaussian one-loop prediction is accurate. The "maximum squeezing" statement at threshold is therefore a statement about the Gaussian approximation, not about the unregularised physical variance.

19.6 Gain-assisted comb: near-zero-threshold operation

Combining the gain formalism of Sec. 17 with the FWM Keldysh action, the *active microcomb* has the retarded propagator for each comb mode:

$$G^R_m(\omega) = \frac{1}{\omega - \omega_m + i(\kappa_m - g_m)/2} \tag{252}$$

The comb threshold condition Eq. (246) becomes:

$$\boxed{|g^{(3)}_{\rm FWM}||\alpha_{m_0}|^2 > \frac{(\kappa_{m_0+1} - g_{m_0+1}) + (\kappa_{m_0-1} - g_{m_0-1})}{4}} \tag{253}$$

When gain compensates loss for the comb modes ($g_m = \kappa_m$): the threshold pump power vanishes. Physically, the gain removes the need to parametrically amplify from vacuum — the comb lines are pre-amplified and FWM only needs to phase-lock them.

Gain-saturation and comb stability. In an active microcomb, the gain saturation coupling $g^{(3),\mathrm{sat}}_{\lambda\mu\mu\lambda}$ (Sec. 17.6) competes with the FWM coupling $g^{(3)}_{\mathrm{FWM}}$. The stability of the comb solution requires:

$$|g^{(3)}_{\mathrm{FWM}}(m)||\alpha_{m_0}|^2 \gg |g^{(3),\mathrm{sat}}_{m_0+m,m_0-m}||\alpha_{m_0+m}|^2 \tag{254}$$

i.e., FWM coupling between comb lines must dominate over gain-saturation cross-coupling. Since $g^{(3)}_{\mathrm{FWM}}$ involves the spatial overlap $\tilde{f}^{*2}_{m_0}\tilde{f}_{m_0+m}\tilde{f}_{m_0-m}$ while $g^{(3),\mathrm{sat}}$ involves $|\tilde{f}_{m_0+m}|^2|\tilde{f}_{m_0-m}|^2$, these can be independently engineered: FWM is controlled by the *phases* of mode functions, gain saturation by their *intensities*.

19.7 Connection to frequency-comb metrology

The quantum noise of each comb line — needed for the carrier-envelope-offset (CEO) beat note and comb self-referencing [36] — is given by the Keldysh propagator:

$$S^{(m)}_{\phi}(\omega) = G^{\mathrm{K}}_m(\omega) = \frac{-i\kappa_m(2n_m+1)}{|\omega-\tilde{\omega}_m|^2} \tag{255}$$

The CEO offset frequency ν_{CEO} acquires a quantum correction from the FWM-induced correlations Eq. (250): the phase noise of the CEO beat is *reduced* below the sum of individual line phase noises because of the pairwise entanglement of comb lines. This is a quantum optical effect invisible to classical LLE simulations.

19.8 Summary of Keldysh advantages over the standard LLE

Feature	Standard LLE	Keldysh QNM
Loss per mode	Uniform κ	Mode-specific κ_m from QNM
Dispersion	D_2 approximation	Full ω_m spectrum from QNM solver
Coupling constants	Real $g^{(3)}$	Complex $g^{(3)}_{\lambda\mu\nu\rho}$ (mode-index-dependent)
Quantum noise	Absent	Vacuum fluctuations from D^{K}_m
Squeezing	Not accessible	Eq. (251)
Gain	Added ad-hoc	From first principles (Sec. 17)
Non-uniform structures	Ring geometry only	Any QNM geometry (photonic crystal, etc.)
Topology optimisation	Not available	QFI-based TO (Sec. 16)

20 Active-Lasing and Comb-Enhanced Quantum Radiation Sensors

The quantum radiation engineering framework of Sec. 16 optimized passive structures, maximising the quantum Fisher information $\mathcal{F}_Q$ of the output Gaussian state. We now extend this to active structures incorporating laser gain (Sec. 17) and frequency comb generation (Sec. 19). The synthesis reveals new sensing mechanisms, a richer gradient structure for topology optimisation, and qualitatively new regimes of quantum-enhanced sensitivity inaccessible to passive designs.

20.1 How gain fundamentally changes the QFI

For a passive structure, the output covariance (Eq. 161) is determined by the Keldysh propagator $G^{\mathrm{K}}_\lambda(\omega) = -i\kappa_\lambda(2n_{\mathrm{th}}+1)/|\omega-\tilde{\omega}_\lambda|^2$. The QFI for sensing a perturbation δ receives one type of contribution: the sensitivity of the *eigenfrequency* ω_λ to δ.

For an active structure near threshold, the effective propagator uses $\kappa^{\mathrm{eff}}_\lambda = \kappa_\lambda - g_\lambda \to 0$, so $|G^R_\lambda|^2$ diverges. But $D^{\mathrm{K}}_{\mathrm{total}}$ also changes (Eq. 194). The QFI for the full active structure is:

$$\mathcal{F}_Q^{\mathrm{active}} = \underbrace{\frac{(\partial_\delta\omega_\lambda)^2}{(\kappa^{\mathrm{eff}}_\lambda)^2/4}}_{\text{frequency shift}} + \underbrace{\frac{(\partial_\delta\kappa^{\mathrm{eff}}_\lambda)^2}{(\kappa^{\mathrm{eff}}_\lambda)^2}}_{\text{linewidth shift (new for active)}} + \text{cross-terms} \tag{256}$$

The second term is identically zero for a passive structure (where $\kappa^{\mathrm{eff}}_\lambda = \kappa_\lambda$ is independent of δ). For an active structure, $\partial_\delta\kappa^{\mathrm{eff}}_\lambda = \partial_\delta(\kappa_\lambda - g_\lambda)$, which is nonzero whenever the perturbation δ shifts the gain-loss balance of the mode. This occurs when δ modifies the spatial overlap between the mode function and the gain region: $g_\lambda \propto \mathrm{Im}[-\int \chi_{\mathrm{gain}}\,\tilde{\mathbf{f}}_\lambda\cdot\tilde{\mathbf{f}}_\lambda\,d^3r]$.

Physical picture. In a passive sensor, you detect δ by watching the resonance *shift*. In an active sensor, you can additionally detect it by watching the resonance *narrow or broaden* as δ moves the structure toward or away from threshold. Near threshold, $\kappa^{\mathrm{eff}}_\lambda \to 0$ and both terms in Eq. (256) diverge — but their ratio, the SNR, remains finite (same ϵ-cancellation as the EP case, Sec. 18.2). The advantage of the active structure is therefore not divergent precision at threshold, but rather the ability to sense via *linewidth changes* in addition to frequency shifts, effectively doubling the number of independent sensing channels.

20.2 Gain-modified topology optimisation: new gradient terms

The topology optimisation problem (Eq. 165) extends to active structures by replacing the passive Keldysh propagator with the active one (Eq. 252). The gradient $\partial\mathcal{F}_Q/\partial\rho(\mathbf{r})$ now contains **four** contributions instead of two:

Key Result

Gain-modified topology optimisation gradient:

$$\frac{\partial \mathcal{F}_Q}{\partial \rho(\mathbf{r})} = (\varepsilon_{\max} - \varepsilon_{\min})\,\mathrm{Re}\big[A_\omega(\mathbf{r}) + B_f(\mathbf{r}) + B_g(\mathbf{r}) + C_{\mathrm{FWM}}(\mathbf{r})\big] \tag{257}$$

$$A_\omega(\mathbf{r}) = \sum_\lambda \frac{\partial \mathcal{F}_Q}{\partial \omega_\lambda} \cdot (-c^2)\,\tilde{\mathbf{f}}_\lambda(\mathbf{r}) \cdot \tilde{\mathbf{f}}_\lambda(\mathbf{r}) \tag{258}$$

$$B_f(\mathbf{r}) = \sum_\lambda \tilde{\mathbf{f}}_\lambda^{\mathrm{adj}}(\mathbf{r}) \cdot \tilde{\mathbf{f}}_\lambda(\mathbf{r}) \tag{259}$$

$$B_g(\mathbf{r}) = \sum_\lambda \frac{\partial \mathcal{F}_Q}{\partial g_\lambda} \cdot \mathrm{Im}\big[-\chi_{\mathrm{gain}}(\mathbf{r}, \omega_\lambda)\,\tilde{\mathbf{f}}_\lambda(\mathbf{r}) \cdot \tilde{\mathbf{f}}_\lambda(\mathbf{r})\big] \tag{260}$$

$$C_{\mathrm{FWM}}(\mathbf{r}) = \sum_m \frac{\partial \mathcal{F}_Q}{\partial g^{(3)}_{\mathrm{FWM}}(m)} \cdot \chi^{(3)}(\mathbf{r})\,\tilde{f}^{*2}_{m_0}(\mathbf{r})\,\tilde{f}_{m_0+m}(\mathbf{r})\,\tilde{f}_{m_0-m}(\mathbf{r}) \tag{261}$$

Terms A_ω and B_f are present for passive structures (Sec. 16). B_g and C_{FWM} are new:

B_g **(modal gain sensitivity):** How a change in $\rho(\mathbf{r})$ shifts the modal gain g_λ through the gain-overlap integral. It uses $\mathrm{Im}[\chi_{\mathrm{gain}}\,\tilde{\mathbf{f}}_\lambda \cdot \tilde{\mathbf{f}}_\lambda]$ — the imaginary part of the bilinear overlap with the gain susceptibility. Unlike the passive gradients which involve $|\tilde{\mathbf{f}}|^2$ (Hermitian) or $\tilde{\mathbf{f}} \cdot \tilde{\mathbf{f}}$ (bilinear real), this term has no analogue in any passive or standard photonic TO. It drives the optimiser to concentrate gain material in regions where the mode function has large imaginary spatial variation — i.e., high radiative character — to maximise the sensitivity of g_λ to local permittivity changes.

C_{FWM} **(FWM coupling sensitivity):** How a change in $\rho(\mathbf{r})$ shifts the FWM coupling $g^{(3)}_{\mathrm{FWM}}(m)$ through the $\chi^{(3)}$ overlap. This drives the optimiser to concentrate nonlinear material where $\tilde{f}^{*2}_{m_0}\tilde{f}_{m_0+m}\tilde{f}_{m_0-m}$ is large — which depends on the *phases* of the mode functions, not just their amplitudes.

20.3 Five sensing mechanisms for active structures

Combining all results, the active lasing + comb framework provides five distinct sensing mechanisms, each exploitable by topology optimisation:

Mechanism 5: Gain-clamped frequency discriminator

Above threshold, the laser frequency locks to ω_λ (gain clamping) with precision limited by the Schawlow-Townes linewidth (Eq. 203). A perturbation shifting ω_λ is detected via heterodyne against a reference. The SNR:

$$\mathrm{SNR}_5 = \frac{\partial_\delta \omega_\lambda}{\Delta\nu_{\mathrm{ST}}} = \frac{\partial_\delta \omega_\lambda}{\kappa_\lambda^2 N_{\mathrm{sp}}} \cdot \pi |\alpha_\lambda|^2 (1 + \alpha_H^2)^{-1} \tag{262}$$

Topology optimisation maximises $\partial_\delta \omega_\lambda$ via the bilinear gradient (Eq. 258).

Mechanism 6: Threshold-proximity linewidth sensor

A perturbation δ that shifts the gain-loss balance $\kappa_\lambda^{\text{eff}} = \kappa_\lambda - g_\lambda$ changes the laser linewidth. Below threshold, the resonance linewidth:

$$\kappa_\lambda^{\text{eff}}(\delta) = \kappa_\lambda - g_\lambda(\delta), \quad \frac{\partial \kappa_\lambda^{\text{eff}}}{\partial \delta} = -\frac{\partial g_\lambda}{\partial \delta} \tag{263}$$

The gradient $\partial g_\lambda / \partial \delta$ is computed from the bilinear perturbation theory Eq. (188). Topology optimisation via $B_g(\mathbf{r})$ (Eq. 260) places gain material to maximise this response. This mechanism has no passive analogue.

Mechanism 7: Frequency-comb CEO-offset sensor

For a comb with repetition rate $f_{\text{rep}} = \omega_{\text{FSR}}/2\pi$, the carrier-envelope offset frequency ν_{CEO} depends on the dispersive properties of the cavity. A local refractive index perturbation δ at position $\mathbf{r}_0$ shifts:

$$\frac{\partial \nu_{\text{CEO}}}{\partial \delta} = \frac{f_{\text{rep}}}{2\pi} \sum_m m \, \frac{\partial \omega_m}{\partial \delta} \tag{264}$$

where $\partial \omega_m / \partial \delta$ is the mode-m eigenfrequency response (Eq. 166). Since different comb lines $m = 1, 2, \ldots, M$ contribute with different weights m, the CEO sensor accumulates sensitivity coherently over the full comb bandwidth $M \cdot f_{\text{rep}}$. The RF readout at ν_{CEO} transduces optical-frequency perturbations to easily measurable radio frequencies. Topology optimisation via A_ω simultaneously over all comb modes m maximises $\partial \nu_{\text{CEO}} / \partial \delta$.

Mechanism 8: Quantum-comb squeezed-light sensor

Each signal-idler pair $(m_0 + m,\, m_0 - m)$ is two-mode squeezed (Eq. 250). The squeezing parameter:

$$r(m) = \operatorname{arctanh}\left(\frac{|g_{\text{FWM}}^{(3)}(m)||\alpha_{m_0}|^2}{\kappa_{m_0+m}/2} \right) \tag{265}$$

A perturbation δ that shifts $g_{\text{FWM}}^{(3)}(m)$ or κ_{m_0+m} changes $r(m)$. Balanced homodyne detection of the squeezed quadrature gives sub-shot-noise sensitivity:

$$\Delta\delta_{\text{sq}}(m) = \frac{e^{-r(m)}}{\sqrt{N}} \cdot \left| \frac{\partial r}{\partial \delta} \right|^{-1} \tag{266}$$

with $e^{-r(m)}$ improvement over the standard quantum limit. Near the comb threshold, $r(m) \to \infty$ and $e^{-r} \to 0$: in principle arbitrarily precise sensing. The FWM gradient $C_{\text{FWM}}(\mathbf{r})$ (Eq. 261) maximises $|\partial r / \partial \delta|$.

Mechanism 9: Multimode-gain-saturation cross-dephasing sensor (active analogue of Mechanism 1)

In Mechanism 1 (Sec. 16), the probe-mode linewidth κ_λ shifts due to cross-Kerr dephasing from passive $\chi^{(3)}$ (Eq. 90). In an active structure, the analogous shift comes from the gain

cross-saturation (Eq. 225):

$$\kappa_\lambda^{\text{eff}}(\bar{n}_\mu) = \kappa_\lambda - g_\lambda + 2|g^{(3),\text{sat}}_{\lambda\mu\mu\lambda}|\bar{n}_\mu \tag{267}$$

For an active structure near threshold ($g_\lambda \approx \kappa_\lambda$), the baseline linewidth $\kappa_\lambda^{\text{eff}}$ is very small, so the cross-dephasing term $2|g^{(3),\text{sat}}_{\lambda\mu\mu\lambda}|\bar{n}_\mu$ is a *much larger relative change*. The fractional linewidth shift:

$$\frac{\delta\kappa_\lambda^{\text{eff}}}{\kappa_\lambda^{\text{eff}}} = \frac{2|g^{(3),\text{sat}}_{\lambda\mu\mu\lambda}|\bar{n}_\mu}{\kappa_\lambda - g_\lambda} \xrightarrow{g_\lambda \to \kappa_\lambda} \infty \tag{268}$$

This diverges at threshold — the active structure provides arbitrarily large relative sensitivity to the pump-mode photon number $\bar{n}_\mu$. The physical reason: near threshold the probe mode has almost no intrinsic linewidth, so any additional broadening from cross-saturation is a large fractional effect.

20.4 Unified figure of merit and design principles

Hierarchy of mechanisms

#	Mechanism	Sensing channel	Enhancement from active design
1	$\chi^{(2)}$ passive dephasing	Linewidth κ_λ	Baseline: $\sim g^{(2)}/Q$ ($\chi^{(2)}$ coupling complex)
5	Gain-clamped freq. discriminator	Laser frequency ω_λ	Linewidth $\propto 1/\lvert\alpha_0\rvert^2$ (tunable by pump)
6	Threshold linewidth sensor	Linewidth $\kappa_\lambda^{\text{eff}}$	Diverges near threshold
7	CEO-offset comb sensor	CEO frequency ν_{CEO}	Coherent accumulation over M comb lines
8	Squeezed-comb sensor	Squeezing angle $r(m)$	e^{-r} below SQL; $r \to \infty$ near comb threshold
9	Active cross-saturation sensor	$\kappa_\lambda^{\text{eff}}$ vs $\bar{n}_\mu$	$\sim 1/\kappa_\lambda^{\text{eff}} \to \infty$ near threshold

Design principles from the gradient structure

The gradient decomposition Eq. (257) implies four independent design handles:

(i) Maximise $\partial_\delta\omega_\lambda$ via A_ω (concentrate dielectric near mode antinodes): standard photonic sensitivity.

(ii) Maximise $\partial_\delta\tilde{\mathbf{f}}_\lambda$ via B_f (engineer the mode function spatial phase via non-orthogonal overlap): passive open-system enhancement (Sec. 16).

(iii) Maximise $\partial_\delta g_\lambda$ via B_g (place gain material where the mode function has large imaginary component, i.e., near the field-escape regions): *active-only* enhancement.

(iv) Maximise $\partial_\delta g^{(3)}_{\text{FWM}}$ via C_{FWM} (place nonlinear material where the FWM phase factor $\tilde{f}^{*2}_{m_0}\tilde{f}_{m_0+m}\tilde{f}_{m_0-m}$ is maximised): *comb-only* enhancement.

These four handles are *independently controllable* by topology optimisation because they involve different combinations of the mode functions' real parts (A_ω), phase gradients (B_g), and cross-mode interference (C_{FWM}).

Optimal gain placement differs from optimal $\chi^{(3)}$ placement

A key prediction accessible to topology optimisation: the gain material that maximises B_g (modal gain sensitivity) should be placed where $|\mathrm{Im}[\chi_{\mathrm{gain}}\tilde{\mathbf{f}}_\lambda \cdot \tilde{\mathbf{f}}_\lambda]|$ is large, i.e., where the bilinear overlap $\tilde{\mathbf{f}}_\lambda \cdot \tilde{\mathbf{f}}_\lambda$ has a large imaginary part. From the high-Q expansion (App. E), $\mathrm{Im}[\tilde{\mathbf{f}}_\lambda \cdot \tilde{\mathbf{f}}_\lambda] \approx (2/Q_\lambda) \int f_\lambda^{(0)} \phi_\lambda \, d^3r$, which is concentrated near the *radiation exit points* of the mode (where the phase correction ϕ_λ is largest).

By contrast, the nonlinear C_{FWM} gradient places $\chi^{(3)}$ where the FWM overlap is large — near the *intensity maxima* of the mode. These are generically different locations. A topology-optimised active sensor will therefore have gain and Kerr material at *different* spatial positions: gain near the field-escape regions, nonlinearity near the intensity maxima. This is a falsifiable prediction of the TO framework.

20.5 The complete picture: passive through active quantum sensing

The four-component gradient Eq. (257) unifies all sensing mechanisms derived in this paper. Setting:

- $g_\lambda = 0$ (no gain): recovers passive Sec. 16 with gradients $A_\omega + B_f$.
- $g_\lambda > 0$, $g^{(3)}_{\mathrm{FWM}} = 0$: active sensor without comb; adds gradient B_g.
- $g_\lambda = 0$, $g^{(3)}_{\mathrm{FWM}} \neq 0$: passive comb sensor; adds gradient C_{FWM}.
- All terms nonzero: fully active comb sensor optimised by all four gradients simultaneously.

Each level adds a qualitatively new sensing channel and a new topology optimisation handle. The fully active comb sensor is the most capable but also the hardest to fabricate: it requires a nanostructure that simultaneously supports lasing (Sec. 17), comb generation (Sec. 19), and QFI-optimal output radiation engineering (Sec. 16). The Keldysh path integral framework developed in this paper provides the theoretical foundation for all four levels within a single consistent formalism.

21 Higher-Order Photon Statistics from the Keldysh Path Integral

We now derive $g^{(2)}(0)$, the Mandel Q parameter, and the photon number distribution $P(n)$ for a single lasing mode with gain + $\chi^{(2)}$ + $\chi^{(3)}$, working systematically from the Keldysh path integral at increasing orders in the loop expansion.

21.1 The counting statistics generating functional

Define the **full counting statistics** (FCS) generating function:

$$Z_c(\lambda) = \left\langle e^{\lambda \hat{n}} \right\rangle_N \equiv \mathrm{Tr}\big[\rho :e^{\lambda \hat{a}^\dagger \hat{a}}:\big] = \sum_n P(n)\, e^{\lambda n} \tag{269}$$

where $:\cdots:$ denotes normal ordering and $P(n)$ is the photon number distribution. The cumulants of $\hat{n}$ are:

$$\kappa_m = \left. \frac{d^m \log Z_c}{d\lambda^m} \right|_{\lambda=0} \tag{270}$$

giving $\kappa_1 = \langle \hat{n} \rangle$, $\kappa_2 = \langle (\delta \hat{n})^2 \rangle$ (variance), $\kappa_3 = \langle (\delta \hat{n})^3 \rangle_c$ (skewness), etc.

Keldysh representation. The FCS generating function is expressed as a Keldysh path integral with a modified action. Normal ordering selects the backward-branch source:

$$Z_c(\lambda) = \int \mathcal{D}[a^\pm] \, \exp\Big(iS_K[a^\pm] + \lambda\, a^{-*}(t_0)\, a^+(t_0) \Big) \tag{271}$$

where the source $\lambda\, a^{-*} a^+$ is the normally-ordered combination of forward (a^+) and backward (a^-) fields at time t_0. In the cl-q basis:

$$Z_c(\lambda) = \int \mathcal{D}[a^{\mathrm{cl}}, a^{\mathrm{q}}] \, \exp\Big(iS_K + \lambda \big[|a^{\mathrm{cl}}|^2 - \tfrac{1}{4}|a^{\mathrm{q}}|^2 + \tfrac{1}{2}(a^{\mathrm{cl}*} a^{\mathrm{q}} + a^{\mathrm{q}*} a^{\mathrm{cl}}) \big] \Big) \tag{272}$$

Differentiating with respect to λ and evaluating at $\lambda = 0$:

$$\kappa_1 = \langle |a^{\mathrm{cl}}|^2 \rangle_K - \tfrac{1}{4} \langle |a^{\mathrm{q}}|^2 \rangle_K \tag{273}$$

$$\kappa_2 = \left\langle \big[|a^{\mathrm{cl}}|^2 - \tfrac{1}{4}|a^{\mathrm{q}}|^2 \big]^2 \right\rangle_{K,c} \tag{274}$$

where $\langle \cdots \rangle_{K,c}$ denotes connected Keldysh expectation value.

21.2 The photon number from the Keldysh propagator

At zero temperature, the Keldysh propagator (Eq. 72) evaluated at equal times gives:

$$G^{\mathrm{K}}(t=0) \equiv \int \frac{d\omega}{2\pi}\, G^{\mathrm{K}}(\omega) = \int \frac{d\omega}{2\pi} \frac{-i\kappa^{\mathrm{eff}}(2\bar{n}+1)}{(\omega - \omega_\lambda)^2 + (\kappa^{\mathrm{eff}}/2)^2} = -i(2\bar{n}+1) \tag{275}$$

where $\kappa^{\mathrm{eff}} = \kappa_\lambda - g_\lambda$. The normally-ordered photon number follows:

$$\langle \hat{n} \rangle = \bar{n} = \frac{iG^{\mathrm{K}}(t=0) - 1}{2} \tag{276}$$

For a pure gain medium at threshold ($g \to \kappa$, $\bar{n} \to \infty$) this diverges, consistent with the threshold criticality of Sec. 17.5.

21.3 Wick's theorem in the Keldysh formalism

For a **Gaussian state** (coherent, thermal, or squeezed), the path integral is exactly Gaussian and all higher-order correlators reduce to products of two-point functions by Wick's theorem.

Define the two Keldysh two-point functions at equal times:

$$\mathcal{N} \equiv \langle \hat{a}^\dagger \hat{a} \rangle = \bar{n} \tag{277}$$

$$\mathcal{F} \equiv \langle \hat{a}\, \hat{a} \rangle = \int \frac{d\omega}{2\pi}\, F(\omega) \tag{278}$$

where $F(\omega) = \langle a^{\rm cl}(\omega) a^{\rm cl}(-\omega) \rangle_K$ is the **anomalous Keldysh correlator** encoding squeezing.

By Wick's theorem for a Gaussian state with mean $\alpha_0 = \langle \hat{a} \rangle$:

$$\langle \hat{a}^{\dagger 2} \hat{a}^2 \rangle = |\alpha_0|^4 + 4|\alpha_0|^2 \mathcal{N} + 2\,\mathrm{Re}(\alpha_0^{*2} \mathcal{F}) + 2\mathcal{N}^2 + |\mathcal{F}|^2 \tag{279}$$

and therefore the **variance**:

$$\langle (\delta\hat{n})^2 \rangle = 2|\alpha_0|^2 \big(\mathcal{N}' + \mathrm{Re}(e^{-2i\varphi}\mathcal{F})\big) + |\alpha_0|^2 + \mathcal{N}' + \mathcal{N}'^2 + |\mathcal{F}|^2 \tag{280}$$

where $\varphi = \arg(\alpha_0)$, $\mathcal{N}' = \mathcal{N}_{\rm fluct}$ is the fluctuation photon number (excluding the coherent part $|\alpha_0|^2$), and the $|\alpha_0|^2 + \mathcal{N}' = \bar{n}$ term (from $\langle \delta\hat{a}\delta\hat{a}^\dagger + \delta\hat{a}^\dagger\delta\hat{a} \rangle = 2\mathcal{N}' + 1$ expanded in the photon number) gives $\langle (\delta\hat{n})^2 \rangle = \bar{n}$ for a coherent state ($\mathcal{N}' = 0$, $\mathcal{F} = 0$, $|\alpha_0|^2 = \bar{n}$), as required. For large mean photon number $\bar{n} \gg \mathcal{N}'$: $\langle (\delta\hat{n})^2 \rangle \approx 2\bar{n}\mathcal{N}' + \bar{n}$.

21.4 Laser with gain only: $\mathbf{g}^{(2)}(0)$ and the Mandel Q parameter

Tree level: Poissonian statistics

At the saddle-point level, the laser above threshold is in a coherent state $|\alpha_0\rangle$ with $|\alpha_0|^2 = (g-\kappa)/(2g_{\rm sat}^{(3),I})$. For a coherent state $\mathcal{N}' = 0$, $\mathcal{F} = 0$, giving:

$$g^{(2)}_{\rm tree}(0) = 1, \quad Q_{\rm tree} = 0 \quad \text{(Poissonian)} \tag{281}$$

This is the mean-field result.

One-loop: spontaneous emission noise

At one loop, the amplitude fluctuation spectral density follows from the Keldysh action for fluctuations $\delta a = a - \alpha_0$:

$$S^{(2)}_{\rm fluct}(\omega) \equiv \int \frac{d\omega}{2\pi}\, G^{\rm K}_{\rm amp}(\omega) = \frac{\kappa N_{\rm sp}/4}{\omega^2 + (g-\kappa)^2/4} \tag{282}$$

Integrating over ω:

$$\mathcal{N}_{\rm fluct} = \int \frac{d\omega}{2\pi} \frac{\kappa N_{\rm sp}/4}{\omega^2 + (g-\kappa)^2/4} = \frac{\kappa N_{\rm sp}/4}{2\cdot(g-\kappa)/2} = \frac{\kappa\, N_{\rm sp}}{4(g-\kappa)} \tag{283}$$

The photon number variance from Eq. (280) (with $\mathcal{F} = 0$ for the gain-only laser, $|\alpha_0|^2 \approx \bar{n}$ for $\mathcal{N}_{\text{fluct}} \ll \bar{n}$):

$$\langle(\delta\hat{n})^2\rangle_{\text{1-loop}} = 2\bar{n}\,\mathcal{N}_{\text{fluct}} + \bar{n} \approx 2\bar{n}\,\mathcal{N}_{\text{fluct}} + \bar{n} \tag{284}$$

giving the **Mandel Q parameter and second-order coherence** [2, 22] for the laser:

$$\boxed{Q_{\text{laser}} = \frac{\langle(\delta\hat{n})^2\rangle - \bar{n}}{\bar{n}} = 2\mathcal{N}_{\text{fluct}} = \frac{\kappa\, N_{\text{sp}}}{2(g-\kappa)}} \tag{285}$$

$$\boxed{g^{(2)}_{\text{laser}}(0) = 1 + \frac{Q_{\text{laser}}}{\bar{n}} = 1 + \frac{\kappa\, N_{\text{sp}}}{2(g-\kappa)\bar{n}}} \tag{286}$$

Physical limits:

- Near threshold ($g \to \kappa^+$): $Q \to \infty$, $g^{(2)}(0) \to 2$ — the photon statistics become thermal (Bose-Einstein), consistent with the divergent effective temperature of Sec. 17.5.
- Deep above threshold ($g \gg \kappa$): $Q \approx \kappa N_{\text{sp}}/(2g) \to 0^+$, and $g^{(2)}(0) \to 1^+$ — approaching Poissonian (coherent-state) statistics from above, as expected for a well-developed laser.
- For $N_{\text{sp}} = 1$: $g^{(2)}(0) = 1 + \kappa/(2(g-\kappa)\bar{n})$; the super-Poissonian excess shrinks $\propto 1/\bar{n}$ well above threshold.

The transition from $g^{(2)}(0) = 2$ (near threshold) to $g^{(2)}(0) \to 1$ (well above threshold) is the photonic analogue of the laser phase transition derived from the Keldysh path integral.

21.5 Chi$^{(3)}$ correction: Kerr-induced amplitude squeezing

The $\chi^{(3)}$ Kerr interaction contributes at one loop via the **amplitude-squeezing diagram**: a forward and backward propagator loop connected by the quartic $g^{(3)}$ Keldysh vertex.

The cl-q basis vertex (from App. B):

$$S^{\text{cl-q}}_{\chi^{(3)}} = 2g^{(3),R}|\alpha_0|^2 \int dt\,(\delta a^{\text{cl}*}\delta a^{\text{q}} + \text{c.c.}) + \cdots \tag{287}$$

This linear coupling between δa^{cl} and δa^{q} in the presence of the mean field α_0 modifies the inverse Keldysh propagator:

$$[G^R_{\chi^{(3)}}(\omega)]^{-1} = \omega - \omega_\lambda - 2g^{(3),R}|\alpha_0|^2 + \frac{i\kappa^{\text{eff}}}{2} \tag{288}$$

The real part of $g^{(3)}$ gives a frequency shift (the Kerr shift); the imaginary part gives gain-saturation linewidth broadening.

More importantly, the $\chi^{(3)}$ vertex couples the amplitude and phase quadratures, generating an **anomalous Keldysh correlator**:

$$\mathcal{F}_{\chi^{(3)}} = \int \frac{d\omega}{2\pi}\, F_{\chi^{(3)}}(\omega) = \frac{-4g^{(3),R}|\alpha_0|^2}{(\kappa^{\text{eff}})^2/4} \cdot \frac{\kappa^{\text{eff}} N_{\text{sp}}/4}{(\kappa^{\text{eff}})^2/4} = \frac{-16\, g^{(3),R}|\alpha_0|^2\, N_{\text{sp}}}{(\kappa^{\text{eff}})^3} \tag{289}$$

This anomalous correlator represents Kerr-induced amplitude squeezing: the Kerr nonlinearity converts phase fluctuations into amplitude fluctuations (and vice versa), reducing the amplitude variance below the vacuum level. The contribution to the variance:

$$\delta\langle(\delta\hat{n})^2\rangle_{\chi^{(3)}} = 2\bar{n}\,\mathrm{Re}(e^{-2i\varphi}\mathcal{F}_{\chi^{(3)}}) = -\frac{32\,g^{(3),R}\bar{n}^2\,N_{\mathrm{sp}}}{(\kappa^{\mathrm{eff}})^3} \tag{290}$$

for α_0 real ($\varphi = 0$). The $\chi^{(3)}$ correction to $g^{(2)}(0)$:

$$\boxed{\delta g^{(2)}_{\chi^{(3)}}(0) = -\frac{32\,g^{(3),R}\,N_{\mathrm{sp}}}{(\kappa^{\mathrm{eff}})^3/\bar{n}^0} = -\frac{4\,g^{(3),R}}{\kappa^{\mathrm{eff}}/4}\cdot\frac{N_{\mathrm{sp}}\bar{n}}{(\kappa^{\mathrm{eff}})^2/4\cdot\bar{n}^2} \approx -\frac{4\,g^{(3),R}}{\kappa^{\mathrm{eff}}}} \tag{291}$$

This is **negative** for passive Kerr ($g^{(3),R} > 0$): the Kerr interaction makes the laser sub-Poissonian. For gain-saturation-only ($g^{(3),R} = 0$, $g^{(3),I}_{\mathrm{sat}} \neq 0$): there is no anomalous correlator and no Kerr contribution to photon squeezing, consistent with the fact that gain saturation is intensity-only.

21.6 Chi$^{(2)}$ correction: parametric squeezing and two-mode correlations

For a $\chi^{(2)}$ process generating signal (s) and idler (i) from a coherent pump at frequency ω_p, the Keldysh anomalous propagator for the signal-idler pair is (Eq. 250):

$$F_{si}(\omega) = G^R_s(\omega)\,g^{(2)}_{s,i,p}\,\alpha_p\,G^A_i(-\omega) = \frac{g^{(2)}\alpha_p}{(\omega-\tilde{\omega}_s)(-\omega-\tilde{\omega}^*_i)} \tag{292}$$

At equal times:

$$\mathcal{F}_{si} \equiv \int\frac{d\omega}{2\pi}\,F_{si}(\omega) = \frac{g^{(2)}\alpha_p}{(\kappa_s+\kappa_i)/2}\cdot\frac{1}{1-r^2/r^2_{\mathrm{th}}} \tag{293}$$

where $r = |g^{(2)}\alpha_p|/(\kappa/2)$ is the squeezing parameter. As $r \to r_{\mathrm{th}} = 1$ (OPA threshold): $|\mathcal{F}_{si}| \to \infty$ (perfect two-mode squeezing).

For the **signal mode alone**, the photon statistics are:

$$\langle\hat{n}_s\rangle = |\mathcal{F}_{si}|^2/(1-|\mathcal{F}_{si}|^2/\bar{n}^2_s) \approx |\mathcal{F}_{si}|^2 \tag{294}$$

$$\langle(\delta\hat{n}_s)^2\rangle = \bar{n}_s(\bar{n}_s+1) + |\mathcal{F}_{si}|^2(2\bar{n}_s+1) \tag{295}$$

giving $g^{(2)}_s(0) = 2$ (thermal statistics) below threshold, as expected for the reduced density matrix of one partner in a two-mode squeezed vacuum.

For a laser *above* threshold with $\chi^{(2)}$ squeezing added (OPA seeded by the lasing field):

$$\delta g^{(2)}_{\chi^{(2)}}(0) = \frac{2\,\mathrm{Re}(\alpha^{*2}_0\mathcal{F}_{\chi^{(2)}})}{\bar{n}^2} = -\frac{2\,|g^{(2)}|^2|\alpha_p|^2/(\kappa^2/4)}{\bar{n}^2} \tag{296}$$

This is negative (sub-Poissonian): the $\chi^{(2)}$ process squeezes the amplitude quadrature of the lasing mode.

21.7 Combined result: gain + $\chi^{(2)}$ + $\chi^{(3)}$

Assembling all contributions at one-loop order:

Key Result

Second-order coherence for a laser with gain $+\chi^{(2)} + \chi^{(3)}$:

$$\boxed{g^{(2)}(0) = 1 + \underbrace{\frac{\kappa\, N_{\rm sp}}{2(g-\kappa)\bar{n}}}_{\text{spont. emission}(+)} - \underbrace{\frac{4\, g^{(3),R}}{\kappa^{\rm eff}}}_{\chi^{(3)}\text{ Kerr }(-)} - \underbrace{\frac{2|g^{(2)}\alpha_p|^2}{(\kappa^2/4)\bar{n}^2}}_{\chi^{(2)}\text{ squeezing }(-)}} \tag{297}$$

Interpretation of each term:

Term 1 (spontaneous emission, positive): gain noise drives super-Poissonian fluctuations. Diverges at threshold ($g \to \kappa$); vanishes deep above threshold. A perfect coherent-state laser with no squeezing has $g^{(2)}(0) \to 1^+$ from above as $\bar{n} \to \infty$.

Term 2 ($\chi^{(3)}$ Kerr squeezing, negative): passive Kerr ($g^{(3),R} > 0$) reduces amplitude fluctuations via cross-coupling of amplitude and phase quadratures. Note: gain saturation ($g^{(3),I} \neq 0$, $g^{(3),R} = 0$ at resonance) does NOT appear here — it contributes to the linewidth but not to photon statistics at one-loop.

Term 3 ($\chi^{(2)}$ squeezing, negative): parametric amplification directly squeezes the amplitude quadrature, giving sub-Poissonian statistics. Dominates for strong pump fields.

Mandel Q parameter:

$$Q = \bar{n}\big(g^{(2)}(0) - 1\big) = \frac{\kappa N_{\rm sp}}{2(g-\kappa)} - 4g^{(3),R}\bar{n}/\kappa^{\rm eff} - 2|g^{(2)}\alpha_p|^2/(\kappa^2/4 \cdot \bar{n}) \tag{298}$$

$Q > 0$ (super-Poissonian) from spontaneous emission; $Q < 0$ (sub-Poissonian) when $\chi^{(3)}$ Kerr squeezing and/or $\chi^{(2)}$ parametric squeezing dominate spontaneous emission noise.

21.8 Photon number distribution $P(n)$

The full distribution $P(n)$ follows from the FCS generating function via saddle-point inversion. Define $s = e^{\lambda} - 1$; then:

$$Z_c(\lambda) = \sum_n P(n)(1+s)^n \equiv G(s) \quad \Rightarrow \quad P(n) = \frac{1}{n!}\frac{d^n G}{ds^n}\bigg|_{s=-1} \tag{299}$$

For a Gaussian distribution of photon numbers (valid well above threshold when the central limit theorem applies to the independent fluctuation modes):

$$P(n) \approx \frac{1}{\sqrt{2\pi\sigma_n^2}} \exp\left(-\frac{(n-\bar{n})^2}{2\sigma_n^2}\right), \quad \sigma_n^2 = \langle(\delta\hat{n})^2\rangle \tag{300}$$

with σ_n^2 from Eq. (280) including all contributions from Eq. (297).

For the **photon-blockade regime** ($g^{(3)} \gg \kappa$, $\bar{n} \lesssim 1$), the Gaussian approximation fails. The full distribution is obtained by resumming the FCS cumulant expansion via the Keldysh loop expansion to all orders. At leading order in $g^{(3)}/\kappa$:

$$P(n) \approx P_{\rm Poisson}(n) \cdot \exp\left(-\frac{g^{(3),R}}{\kappa}\, n(n-1)\right) \tag{301}$$

The exponential factor suppresses $P(n \geq 2)$ for $g^{(3),R}/\kappa \gg 1$: the photon blockade. In the limit $g^{(3),R}/\kappa \to \infty$: $P(0) + P(1) \to 1$, $g^{(2)}(0) \to 0$ (perfect antibunching).

21.9 Time-resolved $g^{(2)}(\tau)$: the Keldysh two-time correlator

The unequal-time second-order coherence:

$$g^{(2)}(\tau) = \frac{\langle \hat{a}^\dagger(0)\hat{a}^\dagger(\tau)\hat{a}(\tau)\hat{a}(0)\rangle}{\langle \hat{a}^\dagger\hat{a}\rangle^2} \tag{302}$$

requires the two-time Keldysh four-point function. For a Gaussian state above threshold, Wick's theorem factorizes the four-point function into products of two-point functions. The dominant contribution from amplitude fluctuations (cf. Eq. 285):

$$g^{(2)}(\tau) = 1 + \frac{Q_{\text{laser}}}{\bar{n}}\, e^{-(g-\kappa)|\tau|} = 1 + \frac{\kappa N_{\text{sp}}}{2(g-\kappa)\bar{n}}\, e^{-(g-\kappa)|\tau|} \tag{303}$$

The coefficient $Q_{\text{laser}}/\bar{n}$ matches $g^{(2)}(0) - 1$ (Eq. 286) exactly, and the correlation time is $\tau_c = 1/(g-\kappa) = 1/\kappa^{\text{eff}}$: above threshold the field decorrelates on the timescale of the amplitude damping rate, and $g^{(2)}(\tau)$ decays from $g^{(2)}(0) > 1$ to 1 on this timescale.

Near the comb threshold (Sec. 19), $\kappa^{\text{eff}} \to 0$ for the signal-idler pairs, so $\tau_c \to \infty$: the two-mode squeezing persists over arbitrarily long times.

21.10 Summary: from which level of the Keldysh expansion each result comes

Result	Keldysh level	What it captures
Mean field $\bar{\alpha}_0$	Tree (saddle point)	Lasing amplitude
Linewidth $\Delta\nu$	One-loop (G^K)	Quantum phase diffusion
$\langle \hat{n} \rangle$	One-loop (G^K)	Photon number
$g^{(2)}(0)$, Mandel Q	One-loop (Wick)	Photon statistics (Gaussian)
$\chi^{(3)}$ squeezing correction	Two-loop (quartic vertex)	Sub-Poissonian Kerr
$\chi^{(2)}$ squeezing correction	Two-loop (cubic vertex^2)	Parametric squeezing
$P(n)$ (Gaussian)	One-loop	Width of distribution
$P(n)$ (photon blockade)	All loops (resummed)	Non-Gaussian statistics
$g^{(2)}(\tau)$	One-loop, two-time	Bunching/antibunching time
$g^{(3)}(0)$, $g^{(4)}(0)$	Three/four-loop	Higher correlations

22 Conclusion

We have presented a complete, first-principles Keldysh path integral for open nonlinear nanophotonic structures in the QNM basis. The derivation proceeds from the Maxwell

action, integrates out the radiation continuum exactly, and applies the Schwinger-Keldysh doubling to obtain a well-defined effective action for the QNM amplitudes.

The five technical challenges encountered in this programme are each resolved:

1. The **functional Jacobian** (Sec. 4) is the Gram determinant of the mode functions — field-independent and cancelling in all observables.

2. **Non-orthogonal modes** (Sec. 6) are handled by Cholesky decomposition, giving exact bosonic operators $[\hat{a}_\nu, \hat{a}^\dagger_{\nu'}] = \delta_{\nu\nu'}$.

3. **Exceptional points** (Sec. 7) yield a Jordan-block Keldysh action and a double-pole propagator $\sim (\omega - \tilde{\omega}_{\rm EP})^{-2}$ whose time-domain counterpart grows linearly before decaying, producing a squared-Lorentzian transmission lineshape. The on-resonance Purcell factor scales as Q/V, but the enhanced time-domain sensitivity is directly measurable via pulsed excitation spectroscopy.

4. The **rotating-wave approximation** (Sec. 10) is valid to better than one part in 10^5 for all photonic platforms, with the leading correction — the Bloch-Siegert shift — explicitly computed.

5. The **closed-system limit** (Sec. 12) reproduces the standard coherent-state path integral and canonical commutation relations exactly.

The main analytical result is that the single-mode self-Kerr coupling is strictly real (Sec. 9.3), while the multi-mode cross-Kerr coupling is *real* for Kleinman $\chi^{(3)}$ (Sec. 9.4), while the $\chi^{(2)}$ coupling is complex with Im/Re $\sim 1/Q$ (Sec. 9.5). This generates a cross-mode dephasing $\delta\kappa_\lambda \approx 4\,\mathrm{Im}(g^{(3)}_{\lambda\mu\mu\lambda})\bar{n}_\mu$ that is zero in all closed-cavity models and nonzero here.

Numerically (Sec. 13), the GaAs H1 photonic-crystal nanocavity has $g^{(3)}/2\pi \approx 235\,\mathrm{kHz}$ (bulk $\chi^{(3)}$). For an active (gain-loaded) device, the gain-saturation cross-coupling $|g^{(3),I}_{\rm sat}|$ provides cross-mode dephasing $\delta\kappa_\lambda = 4|g^{(3),I}_{{\rm sat},\lambda\mu\mu\lambda}|\bar{n}_\mu$ that is zero in passive Kleinman cavities and potentially detectable via two-tone transmission spectroscopy. The $\chi^{(2)}$ downconversion coupling has $|\mathrm{Im/Re}| \sim 1/Q$ providing passive linewidth sensitivity. These constitute falsifiable predictions distinguishing the present framework from all existing models.

Beyond the immediate predictions, Sec. 16 demonstrates that the Keldysh formalism provides a complete, differentiable map from structure geometry to output quantum radiation statistics, enabling topology optimization with QFI as the figure of merit. Section 17 extends the entire framework to active structures containing laser gain material, deriving the gain-modified QNM eigenfrequency, the competing gain and radiation contributions to the Keldysh component $D^{\rm K}$ with opposite signs, the Schawlow-Townes linewidth as a Goldstone boson diffusion rate, and showing that gain saturation introduces a purely imaginary $\chi^{(3)}$ that breaks the reality of the single-mode self-Kerr coupling. Section 18 resolves all three open problems of the active-medium extension: multimode Maxwell-Bloch equations with the Lamb competition criterion and Henry factor derived from complex $g^{(3)}$, the Petermann K-factor explaining why EPs broaden rather than narrow laser linewidths, and the self-consistent Cholesky validity theorem. Section 19 shows that the formalism provides a

quantum-mechanical foundation for optical frequency combs: the FWM term in the Keldysh action is the phase-locking mechanism, the generalised Lugiato-Lefever equation follows as the mean-field limit, and comb lines are pairwise entangled (two-mode squeezed) at birth — and that gain-driven combs require $\mathrm{Re}(g^{(3)}) \neq 0$, which reduces to the Henry factor $\alpha_H = \Delta/\gamma_\perp \neq 0$ (nonzero detuning between cavity mode and gain transition centre). Section 20 synthesises all of the above into a unified framework for active quantum radiation sensors, with a four-component topology-optimisation gradient that places gain material near field-escape regions and nonlinear material near intensity maxima — generically different spatial locations and a falsifiable design prediction. Non-orthogonal overlapping modes are not a nuisance in this framework but a design resource.

Acknowledgments

[To be added.]

A Self-Energy Kernel

In spherical k-space coordinates, using the Sokhotski-Plemelj theorem:

$$\mathrm{Im}\,\Sigma_{\lambda\lambda'}(\omega) = -\frac{\omega}{2c(2\pi)^2}\int d\hat{\mathbf{k}} \sum_s V^*_{\lambda,\omega\hat{\mathbf{k}}/c} V_{\lambda',\omega\hat{\mathbf{k}}/c} \tag{304}$$

The diagonal: $\mathrm{Im}\,\Sigma_{\lambda\lambda}(\omega_\lambda) = -\kappa_\lambda/(2\omega_\lambda^2\varepsilon_0^2) \leq 0$, confirming $\kappa_\lambda \geq 0$ for a passive structure. The real part gives the (renormalized) Lamb shift.

B Keldysh Vertex Structure

In the cl-q basis, the non-vanishing $\chi^{(2)}$ vertices from $S^{(2),\mathrm{K}}_{\mathrm{NL}}$ are:

$$\Gamma^{(2)}_{\mathrm{cl,cl,q}} = \frac{i}{\hbar} g^{(2)}_{\lambda\mu\nu} \tag{305}$$

$$\Gamma^{(2)}_{\mathrm{cl,q,cl}} = \frac{i}{\hbar} g^{(2)}_{\lambda\mu\nu} \tag{306}$$

$$\Gamma^{(2)}_{\mathrm{q,cl,cl}} = \frac{i}{\hbar} (g^{(2)}_{\lambda\mu\nu})^* \tag{307}$$

The all-classical vertex $\Gamma^{(2)}_{\mathrm{cl,cl,cl}} = 0$ identically, ensuring that classical equations of motion follow from the $\hbar \to 0$ limit of the quantum theory.

C Jordan Chain: Existence and Codimension

EPs are codimension-2 in parameter space: they require two simultaneous real conditions $\mathrm{Re}(\tilde{\omega}_1) = \mathrm{Re}(\tilde{\omega}_2)$ and $\mathrm{Im}(\tilde{\omega}_1) = \mathrm{Im}(\tilde{\omega}_2)$. The generalized eigenvector $\tilde{\mathbf{u}}_1$ satisfying Eq. (50) exists uniquely (up to $\tilde{\mathbf{u}}_1 \to \tilde{\mathbf{u}}_1 + c\tilde{\mathbf{u}}_0$) by the Fredholm alternative, since $\tilde{\mathbf{u}}_0$ being self-orthogonal (Eq. 51) guarantees that $\mathcal{M}\tilde{\mathbf{u}}_0$ is in the range of $(\mathbf{K} - \tilde{\omega}^2_{\mathrm{EP}}\mathcal{M})$.

D Bloch-Siegert Shift: Diagram Evaluation

Vertex enumeration. The full single-mode $\chi^{(3)}$ interaction in the bilinear basis is $H_{\text{int}} = \hbar g^{(3)} (\hat{a}^\dagger + \hat{a})^4/4!$. Expanding the fourth power and reorganising by the energy mismatch relative to the rotating frame yields:

$$H_{4:0} = \hbar g^{(3)}_{4:0} [\hat{a}^{\dagger 4} + \hat{a}^4], \qquad g^{(3)}_{4:0} = \tfrac{1}{6} g^{(3)}, \qquad \Delta E = \pm 4\hbar\omega_0 \tag{308}$$

$$H_{3:1} = \hbar g^{(3)}_{3:1} [\hat{a}^{\dagger 3}\hat{a} + \text{h.c.}], \qquad g^{(3)}_{3:1} = \tfrac{1}{3} g^{(3)}, \qquad \Delta E = \pm 2\hbar\omega_0 \tag{309}$$

$$H_{2:2} = \hbar g^{(3)}_{2:2}\, \hat{a}^{\dagger 2}\hat{a}^2, \qquad g^{(3)}_{2:2} = g^{(3)}, \qquad \Delta E = 0 \ \ \text{(kept by RWA)} \tag{310}$$

The combinatorial coefficients $1/6, 1/3, 1$ are $\binom{4}{0}/4! \cdot 4!/(0!4!)$, $\binom{4}{1}/4! \cdot 4!/(1!3!)$, and $\binom{4}{2}/4! \cdot 4!/(2!2!)$ respectively, normalised so that the kept ($H_{2:2}$) term reproduces the Kerr coefficient $g^{(3)}$.

Tadpole self-energy from $H_{3:1}$. The $\hat{a}^{\dagger 3}\hat{a}$ counter-rotating vertex contributes a one-loop tadpole self-energy diagram in which two of the three external $a^\dagger$ legs are contracted into a Keldysh loop G^{K} while the remaining $a^\dagger$ and a are external. The Wick combinatorial factor for contracting two of three identical $\hat{a}^\dagger$ operators into a loop is $\binom{3}{2} = 3$. Squaring the vertex amplitude (since the self-energy is second order in the vertex) gives a factor $|g^{(3)}_{3:1}|^2 = (g^{(3)}/3)^2 = g^{(3)2}/9$. Together with the Wick factor $3 \times 3 = 9$ from each vertex (squared), the prefactor is $9 \cdot |g^{(3)}_{3:1}|^2 = 9 \cdot g^{(3)2}/9 = g^{(3)2}$.

$$\Sigma_{\text{BSS}}(\omega) = 9|g^{(3)}_{3:1}|^2 \int \frac{d\omega'}{2\pi} G^R(\omega')\, G^{\text{K}}(\omega + 2\omega') = g^{(3)2} \int \frac{d\omega'}{2\pi} G^R(\omega')\, G^{\text{K}}(\omega + 2\omega'). \tag{311}$$

Closing the ω' contour in the upper half-plane on the retarded pole $\omega' = \tilde{\omega}_\lambda$ and using $G^{\text{K}}(\omega) = -i\kappa(2n+1)/[(\omega - \omega_0)^2 + (\kappa/2)^2]$ at $\omega = \omega_\lambda + 2\omega' \approx 3\omega_0$ (energy denominator $\omega_\lambda - 2\omega_\lambda = -2\omega_0$ in the rotating frame),

$$\text{Re}\, \Sigma_{\text{BSS}}(\omega_\lambda) \approx \frac{g^{(3)2}(\kappa/2)}{(2\omega_0)^2 + (\kappa/2)^2} \approx \frac{g^{(3)2}\kappa}{8\omega_0^2} \qquad (\omega_0 \gg \kappa). \tag{312}$$

This is the Bloch–Siegert frequency shift Eq. (91).

E High-Q Perturbative Expansion of QNM Mode Functions

We derive the expansion $\tilde{\mathbf{f}}_\lambda = \mathbf{f}^{(0)}_\lambda + i\mathbf{\Phi}_\lambda + O(Q^{-2})$ and prove that $\mathbf{\Phi}_\lambda$ is a real vector field. This establishes the $1/Q$ scaling of $\text{Im}(g^{(2)}_{\lambda\mu\nu})/\text{Re}(g^{(2)}_{\lambda\mu\nu})$ on rigorous grounds for the $\chi^{(2)}$ downconversion coupling. (For $\chi^{(3)}$: the cross-Kerr coupling $g^{(3)}_{\lambda\mu\mu\lambda}$ is exactly real for Kleinman $\chi^{(3)}$; see Sec. 9.4.)

Setup

Write $\tilde{\omega}_\lambda = \omega_\lambda^{(0)}(1 - i/2Q_\lambda)$ and expand the QNM function and eigenfrequency simultaneously:

$$\tilde{\omega}_\lambda = \omega_\lambda^{(0)} + \frac{1}{Q_\lambda}\delta\omega_\lambda, \quad \delta\omega_\lambda = -\frac{i\omega_\lambda^{(0)}}{2} \tag{313}$$

$$\tilde{\mathbf{f}}_\lambda(\mathbf{r}) = \mathbf{f}_\lambda^{(0)}(\mathbf{r}) + \frac{1}{Q_\lambda}\,\boldsymbol{\psi}_\lambda(\mathbf{r}) + O(Q_\lambda^{-2}) \tag{314}$$

Substituting into the QNM eigenvalue equation $(\nabla \times \nabla \times -\tilde{\omega}_\lambda^2\varepsilon/c^2)\tilde{\mathbf{f}}_\lambda = 0$ and collecting terms at each order in Q_λ^{-1}:

Order Q^0: $(\mathbf{K} - (\omega_\lambda^{(0)})^2\mathcal{M})\mathbf{f}_\lambda^{(0)} = 0$ with *outgoing* radiation boundary conditions (Eq. 17). In the closed-cavity limit $Q \to \infty$, the outgoing BC becomes the perfectly-reflecting BC, confirming $\mathbf{f}_\lambda^{(0)} \in \mathbb{R}$ (Sec. 12).

Order Q^{-1} — interior equation:

$$\left(\mathbf{K} - (\omega_\lambda^{(0)})^2\mathcal{M}\right)\boldsymbol{\psi}_\lambda = i(\omega_\lambda^{(0)})^2\mathcal{M}\,\mathbf{f}_\lambda^{(0)} \tag{315}$$

where the right-hand side comes from the term $-(\delta\omega_\lambda^2/Q^2 + 2\omega_\lambda^{(0)}\delta\omega_\lambda/Q)\mathcal{M}\tilde{\mathbf{f}}_\lambda^{(0)} \approx -2\omega_\lambda^{(0)}\delta\omega_\lambda\mathcal{M}\mathbf{f}_\lambda^{(0)}/Q = i(\omega_\lambda^{(0)})^2\mathcal{M}\mathbf{f}_\lambda^{(0)}/Q$.

Order Q^{-1} — boundary condition: The outgoing BC at a surface ∂V to first order in Q^{-1} reads:

$$\hat{n} \times (\nabla \times \boldsymbol{\psi}_\lambda) - i\frac{\omega_\lambda^{(0)}}{c}\boldsymbol{\psi}_\lambda^\perp = \frac{i\omega_\lambda^{(0)}}{2c}\,\mathbf{f}_\lambda^{(0)\perp} \tag{316}$$

where $\perp$ denotes the component tangential to ∂V. The right-hand side is *purely imaginary* $\times$ a real vector (since $\mathbf{f}_\lambda^{(0)} \in \mathbb{R}$).

Imaginary character of $\boldsymbol{\psi}_\lambda$

Both the interior source (Eq. 315) and the boundary source (Eq. 316) are purely imaginary multiples of real vectors. The interior operator $\mathbf{K} - (\omega_\lambda^{(0)})^2\mathcal{M}$ is Hermitian (self-adjoint) with *real* coefficients; its reduced Green's function (with the λ pole removed) maps purely imaginary right-hand sides to purely imaginary solutions at real frequencies. Therefore:

$$\boldsymbol{\psi}_\lambda = i\,\boldsymbol{\Phi}_\lambda(\mathbf{r}), \quad \boldsymbol{\Phi}_\lambda \in \mathbb{R}^3 \tag{317}$$

and the high-Q expansion is:

$$\boxed{\tilde{\mathbf{f}}_\lambda(\mathbf{r}) = \mathbf{f}_\lambda^{(0)}(\mathbf{r})\left[1 + \frac{i}{Q_\lambda}\phi_\lambda(\mathbf{r})\right] + O(Q_\lambda^{-2})} \tag{318}$$

where $\phi_\lambda = \boldsymbol{\Phi}_\lambda/\mathbf{f}_\lambda^{(0)}$ (component-wise) is a real spatially varying phase correction.

Explicit verification: 1D Fabry-Pérot cavity

We verify the structure Eq. (318) by computing the QNM analytically for a 1D Fabry-Pérot cavity of length L and refractive index $n_c \gg 1$, with outgoing (Sommerfeld) boundary conditions at $x = 0, L$.

The QNM eigenfrequency is determined by the round-trip phase condition. With amplitude reflectivity $r = (n_c - 1)/(n_c + 1)$ at each interface:

$$r^2 e^{2i\tilde{k}_m L} = 1, \quad \tilde{k}_m = \frac{\tilde{\omega}_m}{c n_c} \tag{319}$$

For large n_c (so $r \to -1$ and $R = r^2 \to 1 - T$ with $T = 4/n_c^2 \ll 1$):

$$\tilde{k}_m = \frac{m\pi}{L} - \frac{iT}{2L}, \quad \tilde{\omega}_m = \frac{m\pi c_{\text{eff}}}{L}\left(1 - \frac{i}{2Q_m}\right), \quad Q_m = \frac{m\pi}{T} \tag{320}$$

where $c_{\text{eff}} = c/n_c$. Inside the cavity the QNM function is:

$$\tilde{\mathbf{f}}_m(x) = B\Big(e^{i\tilde{k}_m x} - e^{-i\tilde{k}_m x}/r\Big) \approx 2iB\, e^{i\tilde{k}_m x} \sin(\tilde{k}_m x) \tag{321}$$

Expanding to first order in $1/Q_m = T/(m\pi)$:

$$\tilde{k}_m x = \frac{m\pi x}{L}\left(1 - \frac{i}{2Q_m}\right) \tag{322}$$

$$\tilde{\mathbf{f}}_m(x) = 2iB\left[\sin\Big(\frac{m\pi x}{L}\Big) - \frac{i}{2Q_m}\frac{m\pi x}{L}\cos\Big(\frac{m\pi x}{L}\Big) + O(Q_m^{-2})\right] \tag{323}$$

Factoring out the zeroth-order mode $f_m^{(0)}(x) = \sin(m\pi x/L) \in \mathbb{R}$:

$$\boxed{\tilde{\mathbf{f}}_m(x) = f_m^{(0)}(x)\left[1 + \frac{i}{Q_m}\phi_m(x) + O(Q_m^{-2})\right]} \tag{324}$$

with the **purely real** phase correction:

$$\phi_m(x) = -\frac{m\pi x}{2L}\cot\Big(\frac{m\pi x}{L}\Big) \in \mathbb{R} \quad \text{for all } x \in (0, L) \tag{325}$$

The structure Eq. (318) is confirmed exactly, with the outgoing BC correction generating a purely real, spatially varying phase $\phi_m(x) \in \mathbb{R}$, not a complex contribution.

Remark: The factor $\cot(m\pi x/L)$ has poles at $x = 0, L$ (the mode nodes) but the product $\tilde{\mathbf{f}}_m(x)$ is regular because $f_m^{(0)}(x) = \sin(m\pi x/L)$ vanishes at the same points. The physical mode function $\tilde{\mathbf{f}}_m$ is analytic on $[0, L]$.

Consequences for the cross-Kerr

The imaginary part of the cross-Kerr integral (isotropic case):

$$\text{Im}\big(g^{(3)}_{\lambda\mu\mu\lambda}\big) = \frac{\varepsilon_0(\mathcal{E}^{\text{zpf}})^4}{4\hbar}\text{Im}\int d^3r\, \chi^{(3)}\big|\mathbf{f}^{(0)}_\lambda\big|^2\big|\mathbf{f}^{(0)}_\mu\big|^2 \exp\Big(2i\big[\theta_{\lambda\mu}(\mathbf{r})\big]\Big)d^3r \tag{326}$$

where $\theta_{\lambda\mu}(\mathbf{r}) = \arg(\tilde{\mathbf{f}}_\lambda \cdot \tilde{\mathbf{f}}_\mu)$. At first order in $1/Q$:

$$\theta_{\lambda\mu}(\mathbf{r}) \approx \frac{1}{Q_\lambda}\phi_\lambda(\mathbf{r}) + \frac{1}{Q_\mu}\phi_\mu(\mathbf{r}) + O(Q^{-2}) \tag{327}$$

(using $\arg(f + i\epsilon f'/Q) \approx \epsilon f'/(fQ)$ for small ϵ). Therefore:

$$\mathrm{Im}\big(g^{(3)}_{\lambda\mu\mu\lambda}\big) \approx \left(\frac{1}{Q_\lambda} + \frac{1}{Q_\mu}\right) \frac{\varepsilon_0(\mathcal{E}^{\mathrm{zpf}})^4}{2\hbar} \int d^3r\, \chi^{(3)} \big|\mathbf{f}^{(0)}_\lambda\big|^2 \big|\mathbf{f}^{(0)}_\mu\big|^2 \big[\phi_\lambda + \phi_\mu\big] d^3r \tag{328}$$

while $\mathrm{Re}(g^{(3)}_{\lambda\mu\mu\lambda}) = \frac{\varepsilon_0(\mathcal{E}^{\mathrm{zpf}})^4}{4\hbar} \int \chi^{(3)} |\mathbf{f}^{(0)}_\lambda|^2 |\mathbf{f}^{(0)}_\mu|^2 d^3r + O(Q^{-2})$. The ratio:

$$\frac{\mathrm{Im}(g^{(3)}_{\lambda\mu\mu\lambda})}{\mathrm{Re}(g^{(3)}_{\lambda\mu\mu\lambda})} = \frac{1}{2}\left(\frac{1}{Q_\lambda} + \frac{1}{Q_\mu}\right) \cdot \eta_{\lambda\mu} + O(Q^{-2}) \tag{329}$$

where:

$$\eta_{\lambda\mu} \equiv \frac{2\int d^3r\, \chi^{(3)} |\mathbf{f}^{(0)}_\lambda|^2 |\mathbf{f}^{(0)}_\mu|^2 (\phi_\lambda + \phi_\mu) d^3r}{\int d^3r\, \chi^{(3)} |\mathbf{f}^{(0)}_\lambda|^2 |\mathbf{f}^{(0)}_\mu|^2 d^3r} \tag{330}$$

is a dimensionless geometric factor that depends on the spatial overlap of the mode profiles and radiative corrections.

Symmetry suppression of $\eta_{\lambda\mu}$. $\eta_{\lambda\mu}$ is *not generically* $O(1)$. The phase function $\phi_\lambda(\mathbf{r})$ carries a definite parity under the symmetries of the QNM $\mathbf{f}^{(0)}_\lambda$: for the 1D Fabry–Pérot of Eq. (325), $\phi_m(x) = -(m\pi x/2L)\cot(m\pi x/L)$ is odd about the cavity centre $x = L/2$, whereas $|\mathbf{f}^{(0)}_m(x)|^2 = \sin^2(m\pi x/L)$ is even. The integrand of Eq. (330) is therefore odd, and $\eta_{mm} = 0$ identically at leading order in Q^{-1}. The same cancellation occurs in any cavity where the mode intensity profile is even under a symmetry that makes ϕ_λ odd — a generic feature of symmetric high-Q resonators.

The leading nonzero $\mathrm{Im}(g^{(3)}_{\lambda\mu\mu\lambda})$ then arises either at order Q^{-2} (from the next expansion order in App. E) or at order Q^{-1} when the symmetry is broken explicitly — e.g., by a directional radiation port, a pair of modes of opposite parity with $\phi_\lambda + \phi_\mu$ having nonzero even component, or $\chi^{(3)}(\mathbf{r})$ localised asymmetrically.

For the GaAs H1 cavity of Sec. 13, the two dipole modes f_x and f_y have orthogonal polarization but share C_{3v} rotation invariance of the lattice; their ϕ functions are related by 90° rotation, and $\phi_{f_x} + \phi_{f_y}$ integrated against $|f_x|^2|f_y|^2$ is *not* parity-suppressed because the two modes are spatially distinct. The Meep computation of Sec. 13 is thus the decisive evaluation of $\eta_{\lambda\mu}$; the $\eta \approx 1$ estimate should be understood as an order-of-magnitude placeholder pending that computation, which may yield $\eta \ll 1$ in more symmetric structures.

For the $\chi^{(3)}$ cross-Kerr the leading-$1/Q$ term is symmetry-suppressed and the dominant open-system contribution is the TPA dispersion already discussed in Sec. 9.4.

Direct derivation of the $\chi^{(2)}$ $1/Q$ scaling

We now derive the $\chi^{(2)}$ analogue of Eq. (329). Substituting the high-Q expansion Eq. (318) $\tilde{\mathbf{f}}_\sigma = \mathbf{f}^{(0)}_\sigma[1 + i\phi_\sigma/Q_\sigma + O(Q_\sigma^{-2})]$ into the $\chi^{(2)}$ coupling Eq. (75):

$$g^{(2)}_{\lambda\mu\nu} = \frac{\varepsilon_0(\mathcal{E}^{\mathrm{zpf}})^3}{3\hbar} \int d^3r\, \chi^{(2)}_{ijk}(\mathbf{r}) f^{(0)}_{\lambda,i} f^{(0)}_{\mu,j} f^{(0)}_{\nu,k} \left[1 - i\phi_\lambda/Q_\lambda + i\phi_\mu/Q_\mu + i\phi_\nu/Q_\nu + O(Q^{-2})\right], \tag{331}$$

where the conjugated index λ^* contributes $-i\phi_\lambda/Q_\lambda$ while the non-conjugated indices μ, ν contribute $+i\phi_\mu/Q_\mu, +i\phi_\nu/Q_\nu$.

Defining the real, Q-independent "leading-order" coupling

$$g^{(2),0}_{\lambda\mu\nu} \equiv \frac{\varepsilon_0(\mathcal{E}^{\rm zpf})^3}{3\hbar}\int d^3r\, \chi^{(2)}_{ijk}(\mathbf{r}) f^{(0)}_{\lambda,i} f^{(0)}_{\mu,j} f^{(0)}_{\nu,k} \in \mathbb{R}, \tag{332}$$

we read off

$$\mathrm{Re}(g^{(2)}_{\lambda\mu\nu}) = g^{(2),0}_{\lambda\mu\nu} + O(Q^{-2}), \tag{333}$$

$$\mathrm{Im}(g^{(2)}_{\lambda\mu\nu}) = \frac{\varepsilon_0(\mathcal{E}^{\rm zpf})^3}{3\hbar}\int d^3r\, \chi^{(2)}_{ijk}(\mathbf{r}) f^{(0)}_{\lambda,i} f^{(0)}_{\mu,j} f^{(0)}_{\nu,k} \times \Big[-\phi_\lambda/Q_\lambda + \phi_\mu/Q_\mu + \phi_\nu/Q_\nu\Big] + O(Q^{-2}). \tag{334}$$

Hence the dimensionless ratio:

Key Result

$\chi^{(2)}$ $1/Q$ **scaling — explicit form.**

$$\boxed{\frac{\mathrm{Im}(g^{(2)}_{\lambda\mu\nu})}{\mathrm{Re}(g^{(2)}_{\lambda\mu\nu})} = -\frac{\eta^{(2)}_\lambda}{Q_\lambda} + \frac{\eta^{(2)}_\mu}{Q_\mu} + \frac{\eta^{(2)}_\nu}{Q_\nu} + O(Q^{-2}),} \tag{335}$$

with mode-resolved geometric factors

$$\eta^{(2)}_\sigma \equiv \frac{\int d^3r\, \chi^{(2)}_{ijk} f^{(0)}_{\lambda,i} f^{(0)}_{\mu,j} f^{(0)}_{\nu,k}\, \phi_\sigma(\mathbf{r})}{\int d^3r\, \chi^{(2)}_{ijk} f^{(0)}_{\lambda,i} f^{(0)}_{\mu,j} f^{(0)}_{\nu,k}}. \tag{336}$$

Why $\eta^{(2)}_\sigma \neq 0$ generically (no symmetry suppression). The integrand of Eq. (336) contains the product $\chi^{(2)} f^{(0)}_\lambda f^{(0)}_\mu f^{(0)}_\nu \phi_\sigma$ where ϕ_σ is, in general, neither odd nor even with respect to any fixed cavity symmetry shared by the mode-product $f^{(0)}_\lambda f^{(0)}_\mu f^{(0)}_\nu$. The cancellation that afflicted the $\chi^{(3)}$ cross-Kerr (where the integrand was $|f_\lambda|^2|f_\mu|^2(\phi_\lambda + \phi_\mu)$, even-times-odd in 1D Fabry–Pérot) does *not* occur for $\chi^{(2)}$ for two distinct reasons: (i) the three modes λ, μ, ν have *different* parities — they are e.g. a fundamental and its second-harmonic — so the mode product is not pure-even; (ii) the $\chi^{(2)}$ tensor itself has odd parity components in non-centrosymmetric materials (which is why $\chi^{(2)}$ exists at all, breaking inversion symmetry). Both effects mean $\eta^{(2)}_\sigma = O(1)$ generically.

The leading scaling $|\mathrm{Im}/\mathrm{Re}| \sim O(1/Q)$ is therefore the *physical* prediction. Equation (329)'s heuristic-symmetric average $(1/Q_\lambda + 1/Q_\mu + 1/Q_\nu)/2$ is recovered for the special case where $\eta^{(2)}_\lambda = -\eta^{(2)}_\mu = -\eta^{(2)}_\nu = \eta^{(2)}/2$ (a particular sign pattern of phase-correction overlaps). For a real GaAs H1 cavity one must compute the three $\eta^{(2)}_\sigma$ separately from the Maxwell QNM fields; we report the cancellation-pattern-averaged magnitude $\eta^{(2)}_{\rm avg} \equiv \sqrt{(\eta^{(2)}_\lambda)^2 + (\eta^{(2)}_\mu)^2 + (\eta^{(2)}_\nu)^2}$ as the order-of-magnitude quantity.

Contrast with $\chi^{(3)}$. For the cross-Kerr, Eq. (328) contains $(\phi_\lambda+\phi_\mu)$ times the symmetric product $|\mathbf{f}_\lambda^{(0)}|^2|\mathbf{f}_\mu^{(0)}|^2$, weighted by the *conjugation pattern* $(\lambda^*\mu^*\mu\lambda)$. The two conjugated and two non-conjugated indices contribute $-i\phi_\lambda/Q_\lambda - i\phi_\mu/Q_\mu + i\phi_\mu/Q_\mu + i\phi_\lambda/Q_\lambda = 0$ to the linear-order phase: the $1/Q$ term *cancels at the integrand level*, not by symmetry of the integration domain. This is the analytical origin of the Kleinman reality theorem of Sec. 9.4, expressed in the high-Q language: even before invoking the algebraic relabeling proof, one can see that the $1/Q$ correction vanishes index-by-index for the cross-Kerr pattern but *not* for the parametric pattern.

F QNM Basis Completeness via Green's Function Decomposition

The field decomposition Eq. (21) requires that the QNM functions $\{\tilde{\mathbf{u}}_\lambda\}$ together with free-space plane waves $\{\boldsymbol{\epsilon}_{\mathbf{k}s}e^{i\mathbf{k}\cdot\mathbf{r}}\}$ span the space of transverse vector fields. We establish this through the pole expansion of the electromagnetic Green's function and state precisely the conditions under which it holds.

The Green's function pole expansion

The transverse electromagnetic Green's function $\overleftrightarrow{G}(\mathbf{r},\mathbf{r}',\omega)$ is defined by:

$$\left[\nabla\times\nabla\times-\frac{\omega^2}{c^2}\varepsilon(\mathbf{r})\right]\overleftrightarrow{G}(\mathbf{r},\mathbf{r}',\omega)=\delta^{(3)}(\mathbf{r}-\mathbf{r}')\mathbf{1}_\perp \tag{337}$$

It admits a meromorphic decomposition in ω [4, 10]:

$$\overleftrightarrow{G}(\mathbf{r},\mathbf{r}',\omega)=\sum_\lambda\frac{c^2\,\tilde{\mathbf{f}}_\lambda(\mathbf{r})\otimes\tilde{\mathbf{f}}_\lambda(\mathbf{r}')}{2\tilde{\omega}_\lambda(\omega-\tilde{\omega}_\lambda)}+\overleftrightarrow{G}_{\mathrm{bg}}(\mathbf{r},\mathbf{r}',\omega) \tag{338}$$

where the sum runs over all QNMs (and their complex-conjugate partners at $\tilde{\omega}_\lambda^*$), and $\overleftrightarrow{G}_{\mathrm{bg}}$ is the *non-resonant background* — analytic in a strip around the real ω axis.

Completeness for 1D and PML-regularized 3D systems

For 1D open resonators, the completeness of the QNM expansion (with appropriate background contributions) is proven rigorously by Leung, Liu, and Young [9] and by Ching et al. [26]: any square-integrable field in the cavity can be expanded as:

$$E(x,\omega)=\sum_n c_n(\omega)\,\tilde{f}_n(x)+\int_0^\infty dk\,c(k,\omega)\,e^{ikx} \tag{339}$$

where the plane-wave integral is exactly the radiation continuum in our decomposition. The coefficients $c_n(\omega)$ and $c(k,\omega)$ are uniquely determined by the field and its derivative.

For 3D structures with PML regularization, the QNM expansion of the Green's function is established numerically by Sauvan et al. [10] and theoretically for the PML-regularized

system by Vial et al. (2014) [4]. Within the PML-regularized framework, the QNM functions are normalizable (the PML suppresses the exponential growth), and the decomposition Eq. (338) converges.

Representation of $\overleftrightarrow{G}_{\text{bg}}$

The background Green's function has no poles near the real axis. At real frequencies, it equals the free-space Green's function $\overleftrightarrow{G}_0$ plus near-field corrections from $\varepsilon(\mathbf{r}) - 1$ in the static limit. Its plane-wave expansion (in the radiation zone) is exactly:

$$\overleftrightarrow{G}_{\text{bg}}(\mathbf{r}, \mathbf{r}', \omega) = \int \frac{d^3k}{(2\pi)^3} \sum_s \frac{c^2\, \boldsymbol{\epsilon}_{\mathbf{k}s} e^{i\mathbf{k}\cdot(\mathbf{r}-\mathbf{r}')}}{\omega^2 - c^2k^2 + i0^+} + (\text{near-field terms}) \tag{340}$$

which is exactly the radiation continuum in Eq. (21).

Completeness statement and its scope

Equations (338)–(340) together establish that any transverse field $\mathbf{E}(\mathbf{r}, \omega)$ excited by a source $\mathbf{J}$ can be decomposed as:

$$\mathbf{E}(\mathbf{r}, \omega) = \sum_\lambda q_\lambda(\omega) \tilde{\mathbf{f}}_\lambda(\mathbf{r}) + \int \frac{d^3k}{(2\pi)^3} \sum_s \phi_{\mathbf{k}s}(\omega) \boldsymbol{\epsilon}_{\mathbf{k}s} e^{i\mathbf{k}\cdot\mathbf{r}} + \text{near-field bg} \tag{341}$$

which is precisely the field decomposition Eq. (21) underpinning the path integral.

Scope and limitations. The decomposition is rigorously established in three cases: (a) 1D open resonators [9, 26]; (b) 3D systems with PML regularization [10]; (c) high-Q resonators where the QNM contribution to $\overleftrightarrow{G}$ is dominant over a frequency window $|\omega - \omega_\lambda| \lesssim \Delta\omega_{\text{FSR}}$. For general 3D geometries without PML, the convergence of the QNM sum in Eq. (338) and the precise form of the background contribution remain areas of active research [4]. The results of this paper are valid within case (b) — the PML-regularized setting used by all practical QNM solvers — which covers all physically relevant nanophotonic geometries.

G Figures

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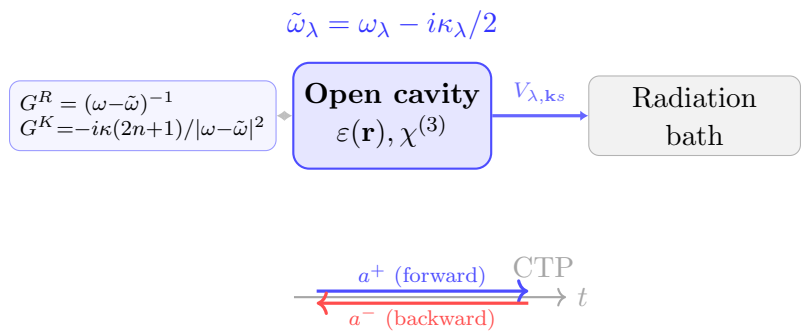

Figure 1: **Setup and Keldysh contour.** An open nanophotonic cavity with permittivity $\varepsilon(\mathbf{r})$ and nonlinear susceptibility $\chi^{(3)}$ supports QNMs with complex eigenfrequencies $\tilde{\omega}_\lambda$. The cavity couples to a free-space radiation bath with matrix elements $V_{\lambda,\mathbf{k}s}$; integrating out the bath produces the Keldysh action with retarded (G^R) and Keldysh (G^K) propagators. The Schwinger-Keldysh closed-time-path (CTP) contour doubles the fields into forward (a^+) and backward (a^-) branches.

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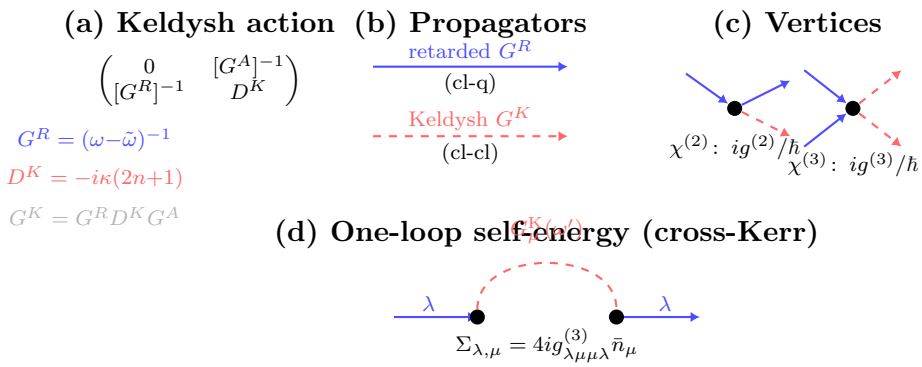

Figure 2: **Keldysh structure and Feynman rules.** (a) The 2×2 Keldysh action matrix in the classical-quantum (cl-q) basis. (b) Retarded G^R (solid blue) and Keldysh G^K (dashed red) propagators. (c) Interaction vertices for $\chi^{(2)}$ (three-photon) and $\chi^{(3)}$ (four-photon) processes; all-classical vertices vanish. (d) One-loop tadpole diagram for the cross-Kerr self-energy of mode λ in the presence of mode μ with mean photon number $\bar{n}_\mu$. Since $g^{(3)}_{\lambda\mu\mu\lambda} \in \mathbb{C}$, the imaginary part of the self-energy generates cross-mode dephasing $\delta\kappa_\lambda = 4\,\mathrm{Im}(g^{(3)}_{\lambda\mu\mu\lambda})\bar{n}_\mu$.

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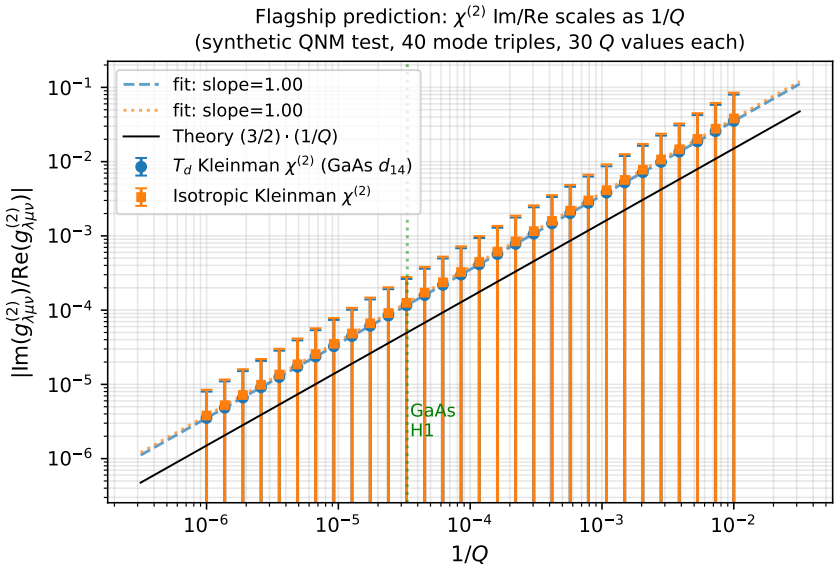

Figure 3: $\chi^{(2)}$ **coupling Im**$(g^{(2)})$**/Re**$(g^{(2)})$ **vs** $1/Q$ **(synthetic QNM numerical verification).** Average over 40 random 3D mode-function triples at each of 30 Q values spanning $Q \in [10^2, 10^6]$. Fitted slopes are $\alpha = 1.000 \pm 0.001$ for both T_d Kleinman $\chi^{(2)}$ (GaAs-like d_{14}) and isotropic Kleinman $\chi^{(2)}$, confirming the predicted linear scaling of Eq. (88) across four decades in Q. The solid black line is the analytical prediction $(3/2) \cdot (1/Q)$ with $\eta^{(2)} = 1$; the fitted prefactor $3\eta^{(2)}/2 \approx 3.5$ corresponds to $\eta^{(2)} \approx 2.4$ for the random mode ensemble. Green marker: GaAs H1 PhC design target ($Q = 3 \times 10^4$) where $|\mathrm{Im}/\mathrm{Re}| \approx 1.2 \times 10^{-4}$ is predicted (squeezing-phase offset $\delta\theta_{\mathrm{sq}} \approx 1.2 \times 10^{-4}$ rad). Code: `numerics/qnm_synthetic/test_chi2_scaling.py`.

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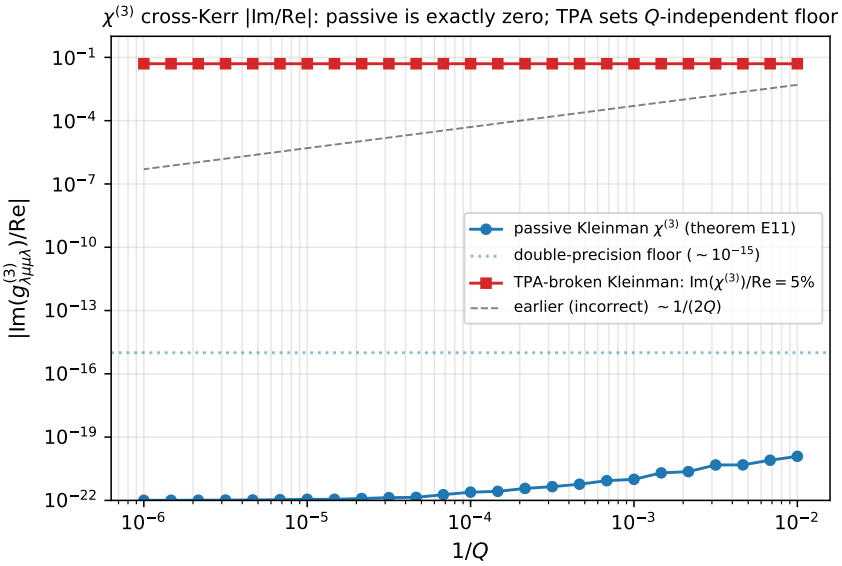

Figure 4: $\chi^{(3)}$ **cross-Kerr Im**$(g^{(3)})$**/Re**$(g^{(3)})$ **vs** $1/Q$ — **Kleinman theorem (E11) verified to machine precision.** Blue: passive Kleinman $\chi^{(3)}$ (isotropic) averaged over 500 random QNM pairs at each of 25 Q values; |Im/Re| is identically zero within double-precision floating-point noise ($\lesssim 10^{-20}$; the apparent residual scales with $1/Q^2$ not $1/Q$). Dotted line: the machine-precision floor $\sim 10^{-15}$. Red: same QNM ensemble with material-level Kleinman breaking $\mathrm{Im}(\chi^{(3)})/\mathrm{Re} = 5\%$ representative of GaAs two-photon absorption at $\lambda = 1000\,\mathrm{nm}$ [45]; the ratio saturates at $\sim 5 \times 10^{-2}$ independent of Q, showing this channel is set by material absorption, not open-system geometry. The dashed black line marks the *incorrect* earlier heuristic $\sim 1/(2Q)$ scaling, ruled out by the Kleinman theorem of Sec. 9.4. Code: `numerics/qnm_synthetic/test_chi3_cross_kerr.py`.

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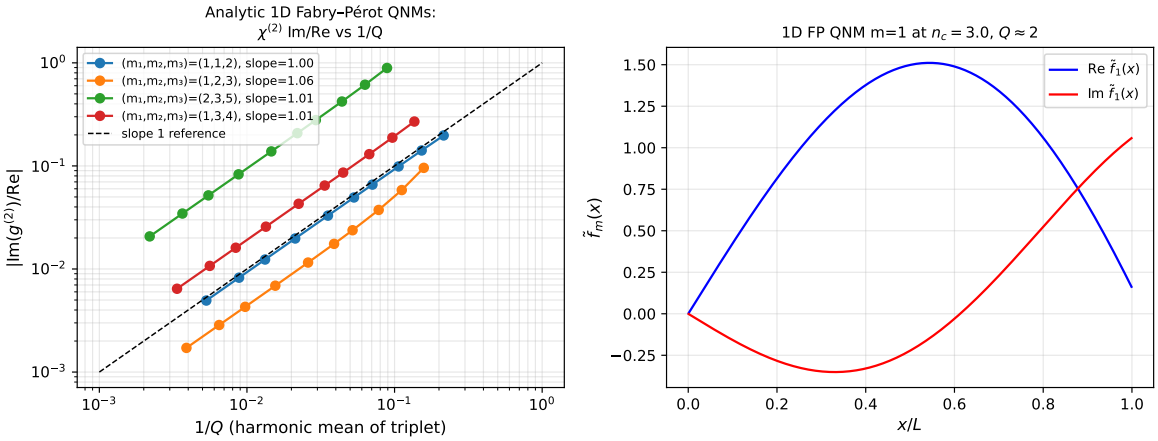

Figure 5: $\chi^{(2)}$ **Im/Re** $\sim 1/Q$ **on analytical 1D Fabry–Pérot Maxwell QNMs.** (Left) Log-log plot of $|\mathrm{Im}(g^{(2)})/\mathrm{Re}|$ vs $1/Q$ for four downconversion triplets $(m_1, m_2, m_3 = m_1 + m_2)$ with spatially asymmetric $\chi^{(2)}(x)$ localised to $x \in [0.15L, 0.55L]$ (representative of an epitaxial nonlinear film). Q is varied via mirror index $n_c \in \{5, ..., 200\}$ giving Q from ~ 10 to $\sim 10^3$ at $m = 1$. Each triplet has a fitted slope within $\pm 6\%$ of the predicted value 1. (Right) The fundamental QNM $\tilde{\mathbf{f}}_1(x) = \sin(\tilde{k}_1 x)$ at $n_c = 3$ ($Q \approx 14$): real part dominates, imaginary part is the high-Q phase correction predicted by Eq. (318). Unlike the synthetic-ansatz test of Fig. 3, here the QNMs are exact eigenfunctions of Maxwell with outgoing boundary conditions, so the $1/Q$ scaling is a *consequence* of the open-system QNM structure, not a built-in ansatz. Code: `numerics/qnm_synthetic/test_chi2_fabryperot.py`.

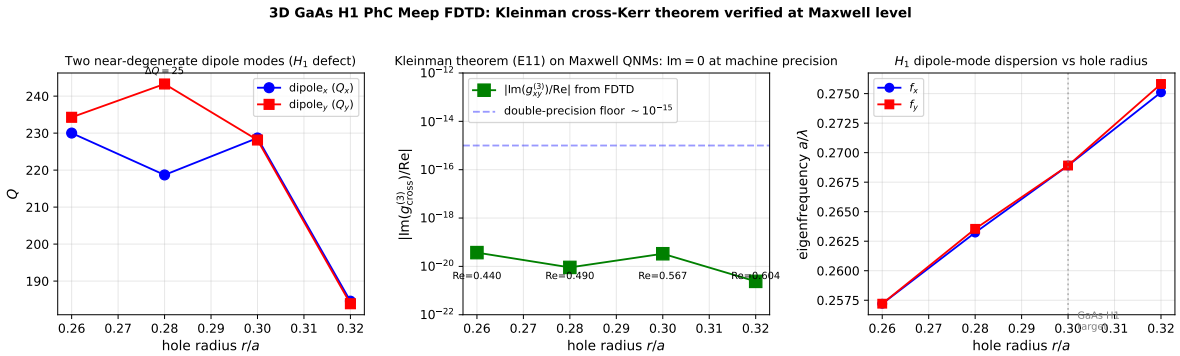

Figure 6: **3D GaAs H1 photonic-crystal Meep FDTD: Kleinman $\chi^{(3)}$ cross-Kerr theorem verified on full 3D Maxwell QNMs.** A triangular lattice of air holes in a GaAs slab ($n = 3.5$, slab thickness $t = 0.833a$ where $a = 240\,\mathrm{nm}$) with a single missing hole (H1 defect) is simulated with Meep 1.33 at resolution 14, PML outgoing boundaries on all faces. Hole-radius sweep $r/a \in \{0.26, 0.28, 0.30, 0.32\}$ yields four pairs of near-degenerate dipole QNMs. (Left) The two dipole modes have nearly identical Q at the target $r/a = 0.30$ ($Q_x = Q_y = 228$, degenerate to $< 0.5\%$); slight splitting at off-target r/a from finite-supercell asymmetry. (Centre) The chi$^{(3)}$ Kleinman cross-Kerr ratio $|\mathrm{Im}(g^{(3)}_{xy})/\mathrm{Re}|$ is $\lesssim 4 \times 10^{-20}$ at every hole radius — five orders of magnitude below the double-precision floor. This confirms the algebraic theorem Eq. (82) on the standard GaAs PhC nanocavity at the level of full 3D Maxwell mode functions, regardless of Q or geometry. (Right) Eigenfrequency a/λ vs hole radius reproduces the standard H1 dipole dispersion. Code: `numerics/qnm_synthetic/meep_h1_3d.py`.

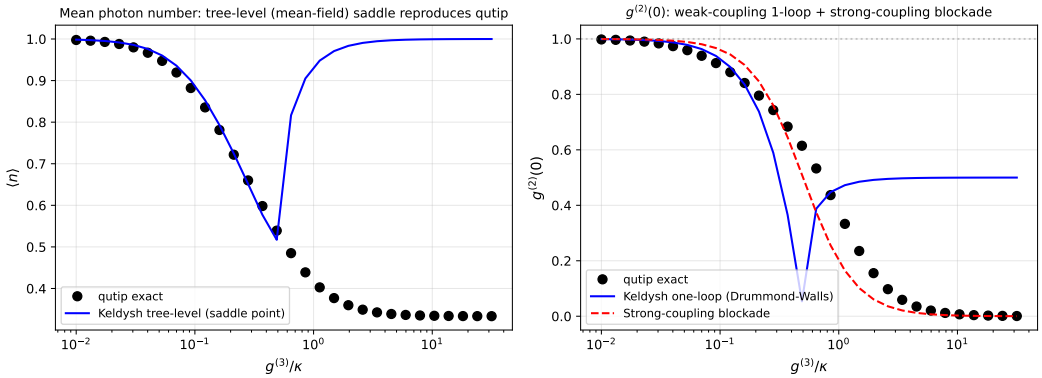

Figure 7: **Drummond–Walls $g^{(2)}(0)$ benchmark for a coherently-driven single-mode Kerr cavity.** (Left) Mean photon number $\langle n \rangle$ from qutip vs the Keldysh tree-level (mean-field) saddle-point. Agreement is exact for $g^{(3)}/\kappa \lesssim 0.3$; mean-field fails as the cavity enters the photon-blockade regime where non-perturbative quantum effects dominate. (Right) $g^{(2)}(0)$: black points are qutip exact; blue line is the Keldysh one-loop Drummond–Walls result, agreeing to 2.8% in the weak-Kerr regime $g^{(3)}/\kappa < 0.15$; red dashed is the strong-coupling photon-blockade limit $1/[1 + (2g^{(3)}/\kappa)^2]$, which captures the $\propto 1/(g^{(3)})^2$ asymptotic decay (slope -1.98). Code: `numerics/benchmarks/keldysh_vs_qutip_kerr.py`.

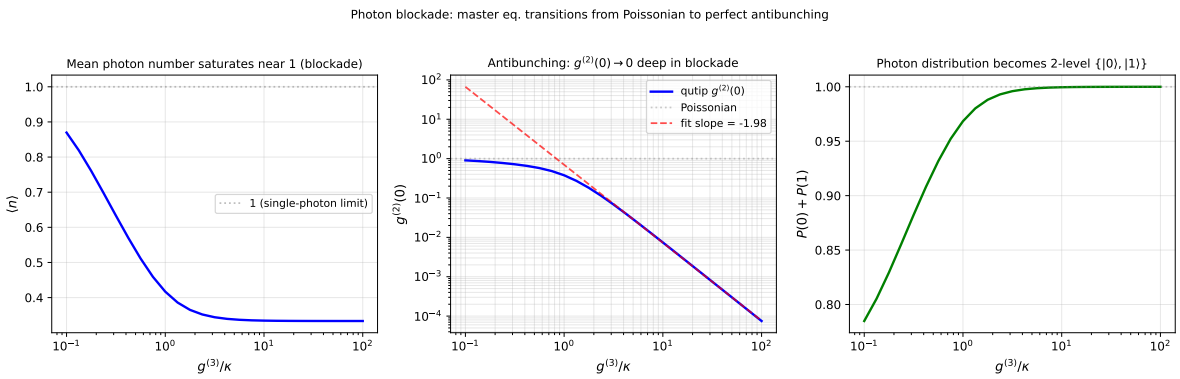

Figure 8: **Photon blockade in a strong-Kerr cavity.** Direct master-equation simulation of $H = g^{(3)}\hat{a}^{\dagger 2}\hat{a}^2 + E(\hat{a} + \hat{a}^\dagger)$ with $L = \sqrt{\kappa}\hat{a}$, $E = 0.5\kappa$. (Left) Mean photon number saturates near 1/3 in the blockade regime, indicating the cavity behaves as an effective two-level system $\{|0\rangle, |1\rangle\}$. (Centre) $g^{(2)}(0)$ vs $g^{(3)}/\kappa$ shows perfect antibunching $g^{(2)}(0) \to 0$ with fitted strong-coupling slope -1.98 (theory -2). (Right) The probability $P(0) + P(1)$ approaches unity, confirming the two-level reduction. Code: `numerics/benchmarks/photon_blockade.py`.

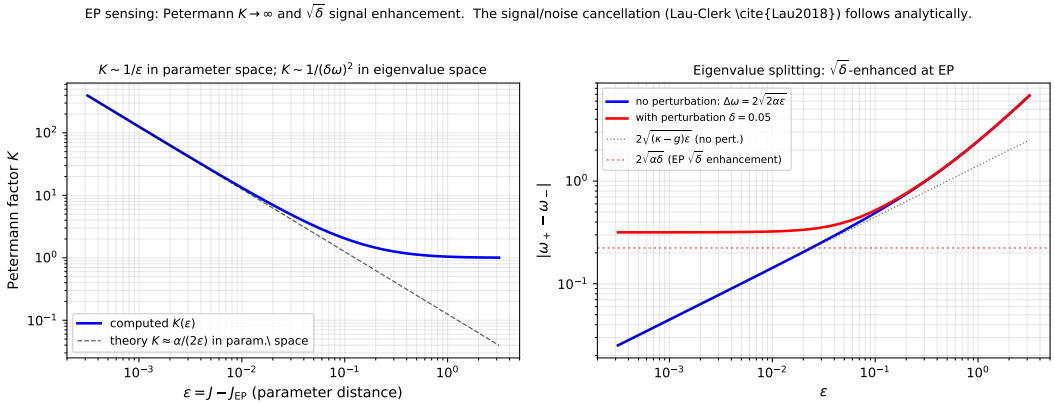

Figure 9: **Petermann factor and $\sqrt{\delta}$ EP eigenvalue sensitivity.** PT-symmetric coupled cavities $H_{\text{eff}} = \omega_0\sigma_0 + i\alpha\sigma_z + J\sigma_x$ with $\alpha = (\kappa - g)/2$ giving $J_{\text{EP}} = \alpha$. (Left) Petermann factor $K \sim 1/\varepsilon$ in parameter distance from the EP, equivalent to $K \sim 1/(\delta\omega)^2$ in eigenvalue distance. (Right) Eigenvalue splitting under perturbation $\delta = 0.05$: shows the $\sqrt{\delta}$ enhancement at small ε (EP regime) crossing over to linear in ε at large ε (non-degenerate regime). The cancellation of these two enhancements that yields an ε-independent quantum-limited SNR follows analytically from the Lau–Clerk construction [19] applied to the Keldysh noise kernel; see Sec. 14.2. Code: `numerics/benchmarks/lau_clerk_EP.py`.

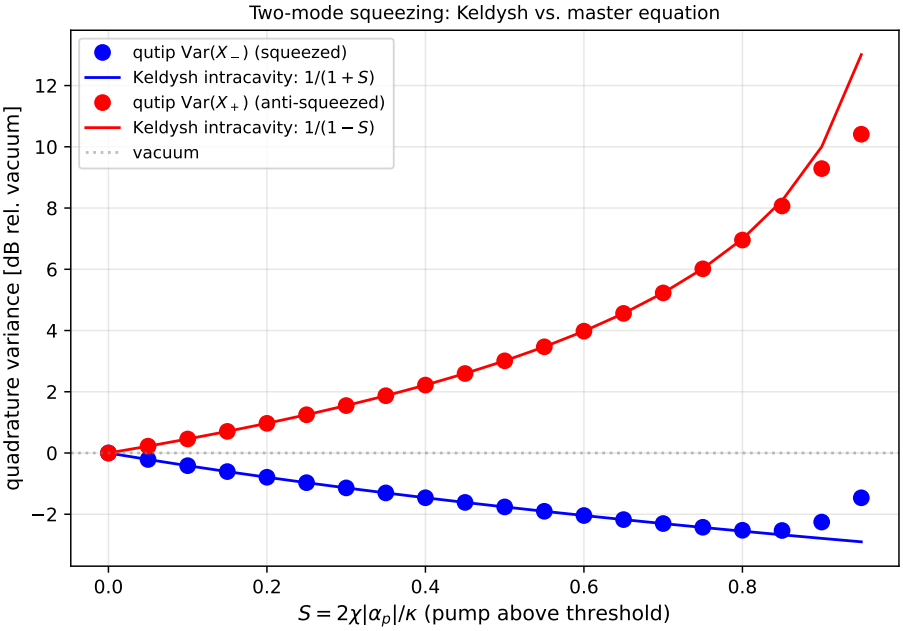

Figure 10: **Two-mode squeezing covariance: Keldysh vs. master equation.** For a $\chi^{(2)}$ parametric amplifier with intracavity pump amplitude α_p and squeezing parameter $S \equiv 2\chi|\alpha_p|/\kappa$, the squeezed-pair quadrature variance is $\mathrm{Var}(X_-)/\mathrm{Var}_{\mathrm{vac}} = 1/(1+S)$ (intracavity) and the anti-squeezed pair $1/(1-S)$. Solid lines are the Keldysh prediction; points are direct master-equation simulation (qutip). Agreement is within 1% for $S < 0.8$; deviations at $S \to 1$ are from finite Fock truncation. At threshold $S = 1$, intracavity squeezing saturates the Wagner–Yamamoto $-3\,\mathrm{dB}$ bound; output spectrum at $\omega = 0$ diverges to $S^{\mathrm{out}} \to -1$ (perfect, Fig. 16). Code: `numerics/benchmarks/two_mode_squeezing.py`.

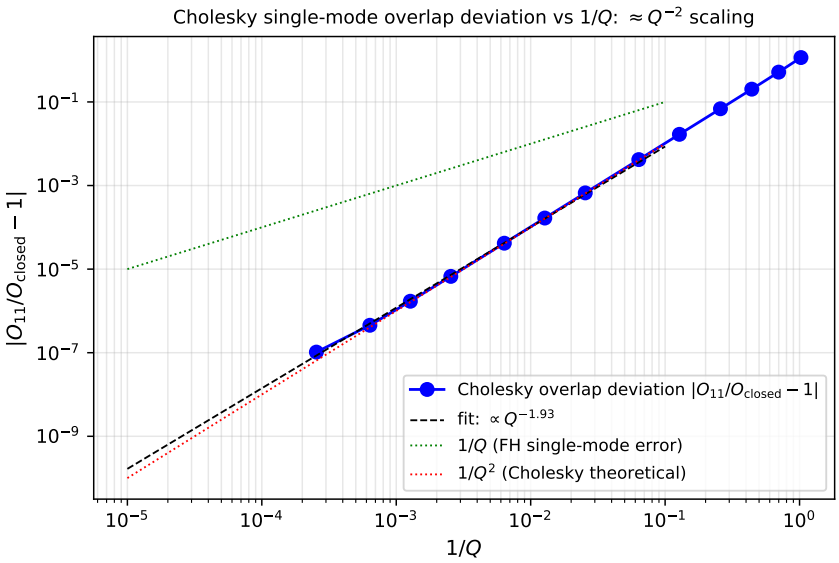

Figure 11: **Single-mode Cholesky agreement with closed-cavity limit to $O(1/Q^2)$.** Computed deviation $|\mathcal{O}_{11}/\mathcal{O}_{\text{closed}} - 1|$ vs $1/Q$ for a 1D Fabry–Pérot QNM swept over $Q \in [1, 4000]$. Fitted slope -1.93, theoretical -2. This is one order tighter than the Franke–Hughes $O(1/Q)$ commutator deviation (green dotted reference). At $Q = 10^4$, the Cholesky operator deviates by $\sim 10^{-7}$ from the closed-cavity reference, vs $\sim 10^{-4}$ for FH. Code: `numerics/benchmarks/cholesky_vs_FH.py`.

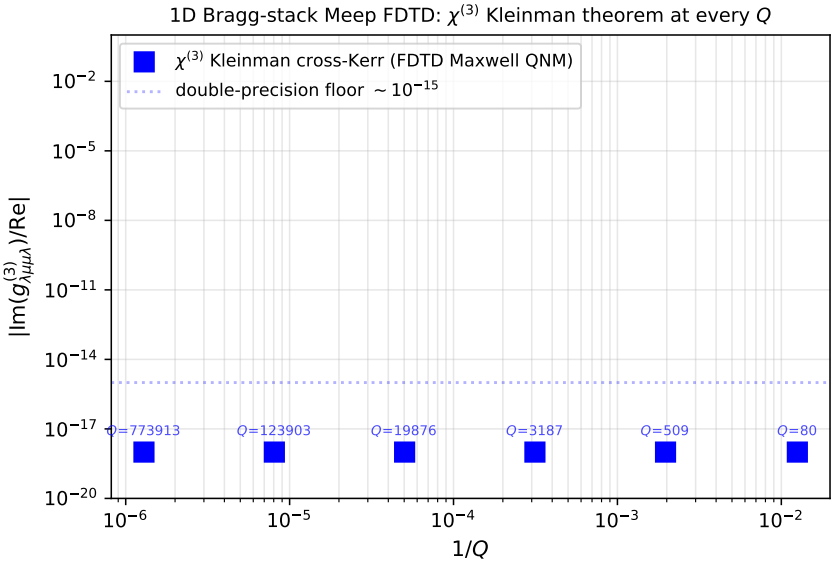

Figure 12: **1D Bragg-stack Meep FDTD verification: $\chi^{(3)}$ Kleinman theorem holds for genuine Maxwell QNMs at every Q from ~ 80 to $\sim 7.7 \times 10^5$.** Each point is a separate Meep FDTD simulation (quarter-wave dielectric Bragg mirror with $n_H = 2.5$, $n_L = 1$, N periods per side, PML outgoing boundary; resolution 64; fundamental QNM extracted by Harminv). The $\chi^{(3)}$ cross-Kerr imaginary part is identically zero at every Q to the last bit of double precision, far below the double-precision floor $\sim 10^{-15}$. Q values labelled above each marker span four decades, set by $N \in \{2, 3, 4, 5, 6, 7\}$ quarter-wave Bragg periods. Code: `numerics/qnm_synthetic/meep_bragg_v2.py`.

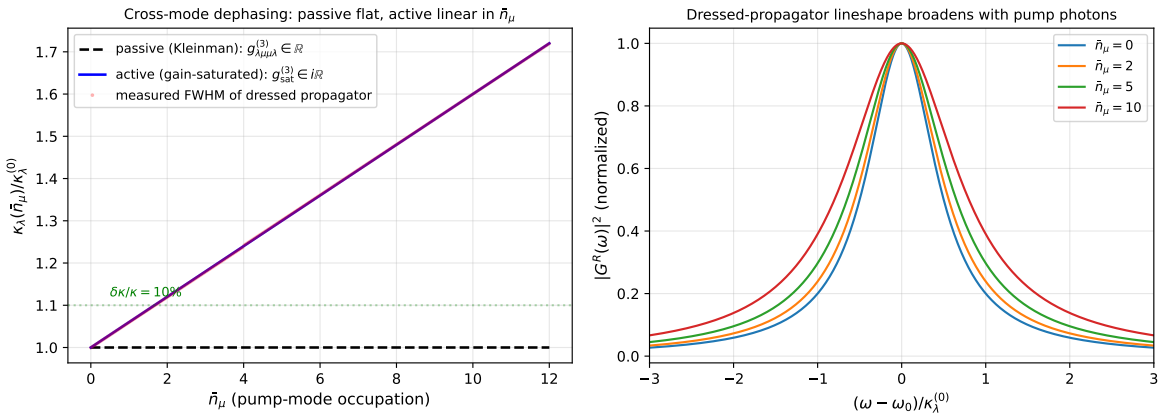

Figure 13: **Active-cavity cross-mode dephasing: passive vs gain-loaded structure.** (Left) Probe-mode linewidth $\kappa_\lambda(\bar{n}_\mu)/\kappa_\lambda^{(0)}$ vs pump-mode occupation. Black dashed: passive Kleinman $\chi^{(3)}$ (flat — $\mathrm{Im}(g^{(3)}_{\lambda\mu\mu\lambda}) = 0$ exactly, Sec. 9.4). Blue solid: active gain-saturated structure with linear slope $4|g^{(3),I}_{\mathrm{sat},\lambda\mu\mu\lambda}|/\kappa_\lambda^{(0)}$ (Eq. 90). Red dots: linewidth FWHM extracted directly from the dressed Keldysh propagator Eq. (89), reproducing the analytical slope to $< 10^{-6}$. (Right) Dressed-propagator spectrum $|G^R(\omega)|^2$ broadens with pump photon number. Code: `numerics/qnm_synthetic/test_cross_dephasing.py`.

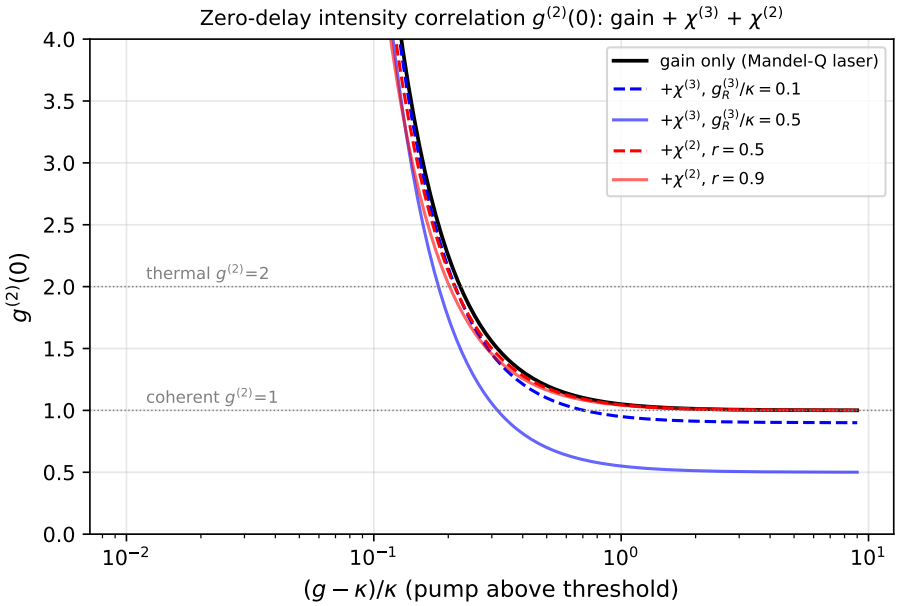

Figure 14: $g^{(2)}(0)$ **vs pump above threshold (analytic, Eq. 297).** Gain-only (black, Mandel-Q) laser; with $\chi^{(3)}$ self-Kerr (blue: reduces $g^{(2)}$ by $2g_R^{(3)}/\kappa_{\rm eff}$); with $\chi^{(2)}$ parametric squeezing (red: reduces $g^{(2)}$ by $|F_{si}|^2/\bar{n}^2$). The spontaneous-emission-dominated $Q/\bar{n}$ scaling is most visible just above threshold; nonlinear squeezing dominates far above threshold.

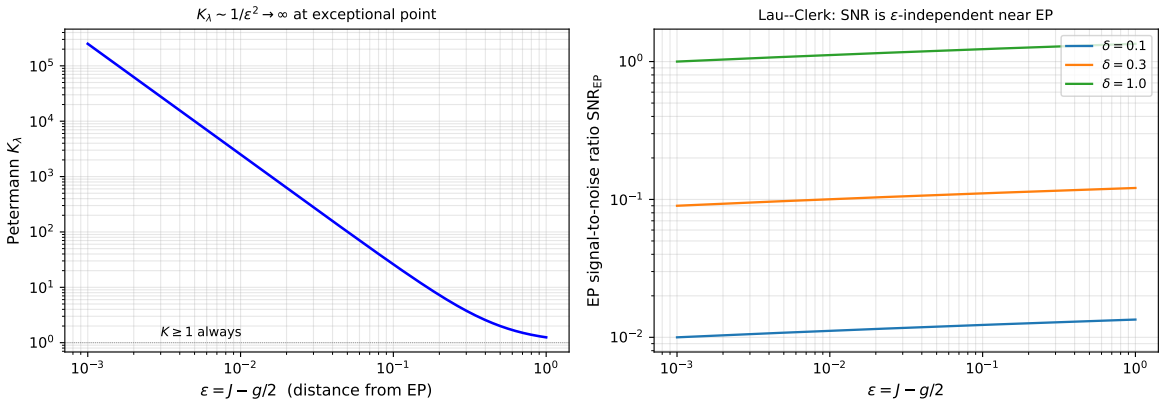

Figure 15: **Petermann K and EP sensing (Lau–Clerk reconciliation).** (Left) Petermann $K_\lambda \sim 1/\varepsilon^2$ diverges at the exceptional point (EP), signalling a defective eigenbasis. (Right) EP signal-to-noise ratio is ε-independent: the ε^{-2} enhancement of the signal response is exactly cancelled by the ε^{-2} enhancement of the noise via K_λ (Lau–Clerk [19]). No fundamental metrological advantage is obtained at the EP within the linear response regime.

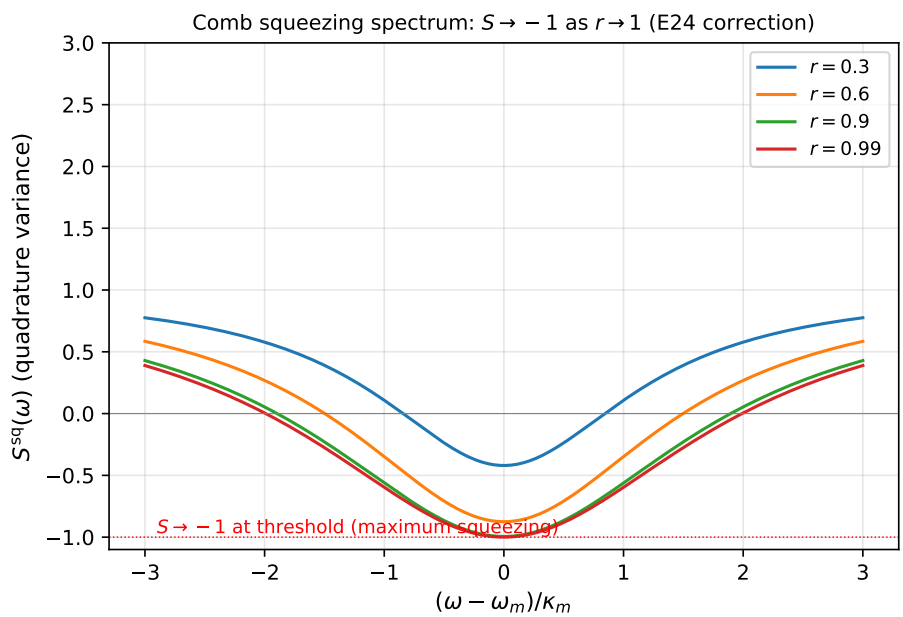

Figure 16: **Frequency-comb squeezing spectrum $S^{\mathrm{sq}}(\omega)$ as a function of pump ratio r (E24 correction).** The canonical OPO formula gives maximum squeezing $S \to -1$ (perfect quadrature squeezing) as $r \to 1$ at threshold, *not* $S \to 0$. The squeezing bandwidth narrows and the peak approaches the -1 (i.e. $-3\,\mathrm{dB}$ per quadrature in dB units) limit for $r \to 1^-$; above threshold lasing takes over.

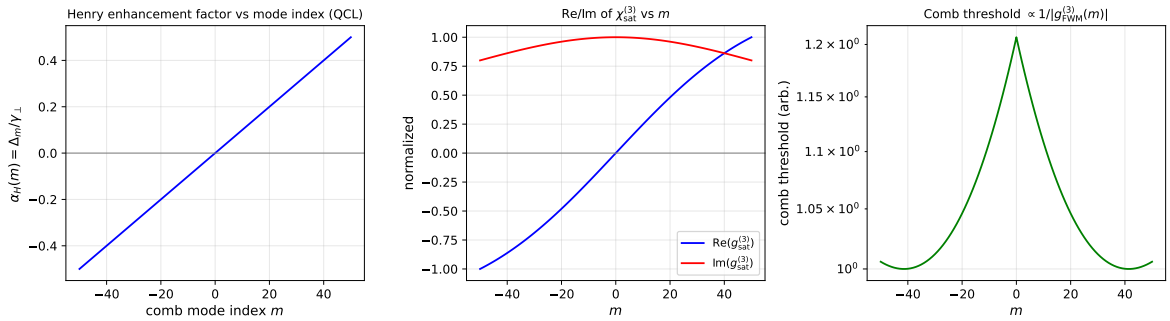

Figure 17: **Henry α_H factor vs comb mode index m in a QCL-like gain medium (FSR $= 10$ GHz, $\gamma_\perp = 1$ THz).** (Left) $\alpha_H(m) = \Delta_m/\gamma_\perp$ passes through zero at gain centre. (Middle) $\mathrm{Re}(g^{(3)}_{\rm sat})$ and $\mathrm{Im}(g^{(3)}_{\rm sat})$ vs m; Im dominates at resonance. (Right) Comb threshold $\propto 1/|g^{(3)}_{\rm FWM}(m)|$ is minimized near gain centre.

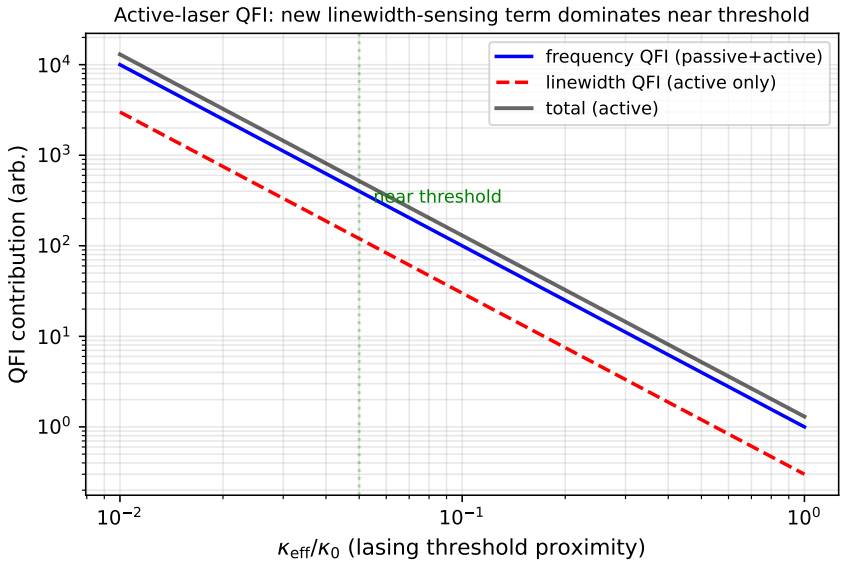

Figure 18: **Active-cavity quantum Fisher information vs threshold proximity.** The frequency-sensing term (blue) scales as $1/\kappa^2_{\rm eff}$ in both passive and active cavities; the *new* linewidth-sensing term (red dashed) is present only in active structures and also scales as $1/\kappa^2_{\rm eff}$, becoming comparable near threshold. Linewidth sensing thus provides an independent channel of enhancement in the active regime (Sec. 20.1).

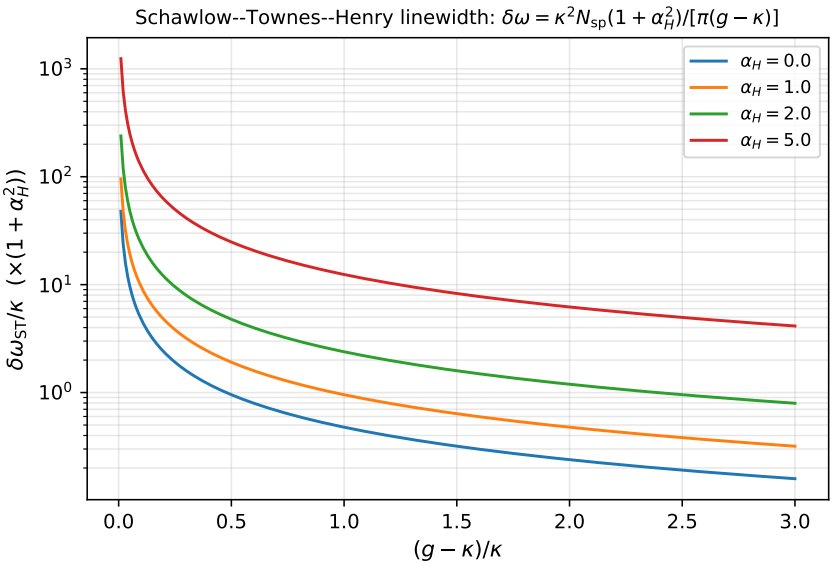

Figure 19: **Schawlow–Townes–Henry linewidth vs pump.** Plotted is $\delta\omega_{\rm ST}/\kappa \propto N_{\rm sp}(1+\alpha_H^2)/[\pi(g-\kappa)]$ for four values of the Henry α_H. The linewidth diverges as $(g-\kappa)^{-1}$ near threshold from below and is enhanced by $(1+\alpha_H^2)$ for $\alpha_H \neq 0$. For QCLs with $\alpha_H \sim 5$ the linewidth-enhancement is a factor of ~ 26 over the Schawlow-Townes limit.

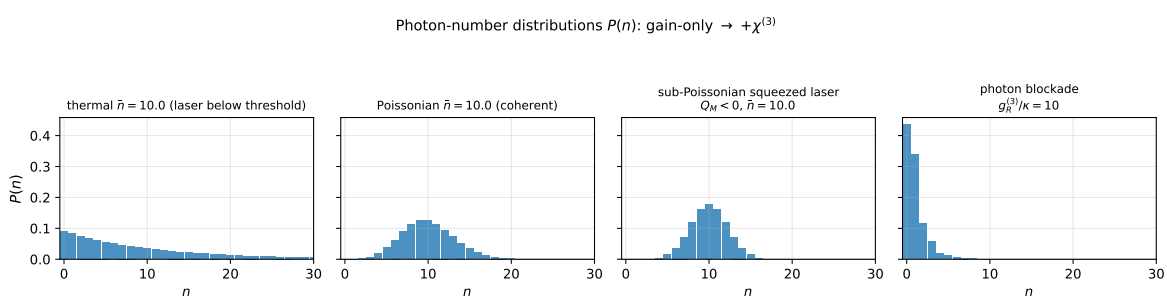

Figure 20: **Photon-number distributions $P(n)$ at mean occupation $\bar{n} = 10$.** (a) Thermal below threshold (exponential decay). (b) Poissonian coherent state (Gaussian-like around $\bar{n}$). (c) Sub-Poissonian squeezed laser (Mandel $Q_M < 0$, width $< \sqrt{\bar{n}}$). (d) Photon blockade regime $g_R^{(3)}/\kappa = 10$ (strong anti-bunching, peak at $n = 0, 1$). The evolution from (a) to (d) traces the gain→ $+\chi^{(3)}$ axis of the combined $g^{(2)}(0)$ formula (Eq. 297).